Selection of Optimal First-line Immunotherapy Based on Specific Pathological Characters for Patients With Advanced Driver-gene Wild-type Non-small Cell Lung Cancer: a Systematic Review and Network

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Supplementary materials

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Table S1. Checklist of the PRISMA Extension for Network Meta-analysis

Section/topic	Item#	Checklist item*	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review *incorporating a network meta- analysis (or related form of meta-analysis).*	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: • Background: main objectives; • Methods: data sources; study eligibility criteria, participants, and interventions; study appraisal and *synthesis methods, such as network meta- analysis.* • Results: number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; *treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.* • Discussion/Conclusions: limitations; conclusions and implications of findings. • Other: primary source of funding; systematic review registration number with registry name.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known, *including mention of why a network meta-analysis has been conducted.*	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. *Clearly describe eligible treatments included in the treatment network, and note whether any have been clustered or merged into the same node (with justification).*	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Table S2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7

Geometry of the network	S1	Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means). *Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses.*	8-9
Synthesis of results	14	Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to: • *Handling of multi-arm trials;* • *Selection of variance structure;* • *Selection of prior distributions in Bayesian analyses; and Assessment of model fit.*	8-9
Assessment of Inconsistency	S2	Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found.	9
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses, if done, indicating which were pre- specified. This may include, but not be limited to the following: • Sensitivity or subgroup analyses; • Meta-regression analyses; • *Alternative formulations of the treatment network; and Use of alternative prior distributions for Bayesian analyses (if applicable).*	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Presentation of network structure	S3	Provide a network graph of the included studies to enable visualization of the geometry of the treatment network	10
Summary of network geometry	S4	Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure.	10

Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	10 (Table 1)
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment.	Figure S1
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study:1) simple summary data for each intervention group, and 2) effect estimates and confidence/credible intervals. *Modified approaches may be needed to deal with information from larger networks.*	10-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence/credible intervals. *In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise comparisons.* If additional summary measures were explored (such as treatment rankings), these should also be presented.	10-14
Exploration for inconsistency	S5	Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, *P* values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network.	15
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies.	10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression, *alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses,* and so forth.	15
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). *Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons).*	19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19-20
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	20

PICOS = population, intervention, comparators, outcomes, study design.

*Text in italics indicates wording specific to reporting of network meta-analyses that has been added to guidance from the PRISMA statement.

† Authors may wish to plan for use of appendices to present all relevant information in full detail for items in this section.

Table S2. Literature Search Criteria
1. Search strategy on PubMed

(((((("Carcinoma, Non-Small-Cell Lung"[Mesh]) OR NSCLC[Title/Abstract]) OR "Non Small Cell"[Title/Abstract]) OR "Non-Small-Cell"[Title/Abstract]) OR "Non-Small Cell"[Title/Abstract])) AND (((chemotherapy [Title/Abstract]) OR Pemetrexed[Title/Abstract]) OR gemcitabine[Title/Abstract]) OR Paclitaxel[Title/Abstract]) OR Cisplatin[Title/Abstract]) OR Carboplatin[Title/Abstract]) OR nedaplatin[Title/Abstract]) OR lobaplatin[Title/Abstract]) OR oxaliplatin[Title/Abstract]) OR etoposide[Title/Abstract]) OR Docetaxel[Title/Abstract]) OR vinorelbine[Title/Abstract]) OR "pembrolizumab" [Supplementary Concept]) OR "lambrolizumab"[Title/Abstract]) OR "Keytruda"[Title/Abstract]) OR "MK-3475"[Title/Abstract]) OR “SCH-900475”[Title/Abstract]) OR "Nivolumab"[Mesh]) OR “Opdivo”[Title/Abstract]) OR “ONO-4538”[Title/Abstract]) OR “ONO 4538”[Title/Abstract]) OR “ONO4538”[Title/Abstract]) OR “MDX-1106”[Title/Abstract]) OR “MDX 1106”[Title/Abstract]) OR“MDX1106”[Title/Abstract]) OR “BMS-936558”[Title/Abstract]) OR “BMS 936558”[Title/Abstract]) OR “BMS936558”[Title/Abstract]) OR "atezolizumab" [Supplementary Concept]) OR “Immunoglobulin G1”[Title/Abstract]) OR “MPDL3280A”[Title/Abstract]) OR “MPDL-3280A”[Title/Abstract]) OR “Tecentriq” [Title/Abstract]) OR “RG7446”[Title/Abstract]) OR “RG-7446”[Title/Abstract]) OR "durvalumab" [Supplementary Concept]) OR “MEDI4736”[Title/Abstract]) OR “MEDI-4736”[Title/Abstract]) OR “Imfinzi”[Title/Abstract]) OR "avelumab" [Supplementary Concept]) OR “MSB-0010682”[Title/Abstract]) OR “MSB0010682”[Title/Abstract]) OR “bavencio”[Title/Abstract]) OR “MSB0010718C”[Title/Abstract]) OR “MSB-0010718C” [Title/Abstract]) OR "sintilimab" [Supplementary Concept]) OR “IBI 308”[Title/Abstract]) OR “IBI308”[Title/Abstract]) OR “IBI-308”[Title/Abstract]) OR "camrelizumab" [Supplementary Concept]) OR “SHR-1210”[Title/Abstract]) OR “SHR 1210”[Title/Abstract]) OR "toripalimab" [Supplementary Concept]) OR Tislelizumab [Title/Abstract]) OR "Immunotherapy"[Mesh]) OR “immune checkpoint”[Title/Abstract]) OR immunotherapy*[Title/Abstract]) OR ICB[Title/Abstract]) OR "Programmed Death-Ligand 1"[Title/Abstract]) OR "Programmed Death 1"[Title/Abstract]) OR "Programmed Cell Death 1 Receptor"[Title/Abstract]) OR "programmed cell death 1 ligand 1 protein"[Title/Abstract]) OR "antiPDL1"[Title/Abstract]) OR "antiPD1"[Title/Abstract]) OR "PD-1"[Title/Abstract]) OR "PD 1"[Title/Abstract]) OR "PD1"[Title/Abstract]) OR "PD-L1"[Title/Abstract]) OR "PD L1"[Title/Abstract]) OR "PDL1"[Title/Abstract]) OR VEGFR[Title/Abstract]) OR Antiangiogenesis[Title/Abstract]) OR Anlotinib[Title/Abstract]) OR Bevacizumab[Title/Abstract]) OR Endostar[Title/Abstract])) AND ((((((compare[Title/Abstract]) OR comparison[Title/Abstract]) OR comparative[Title/Abstract]) OR comparing[Title/Abstract]) OR versus[Title/Abstract]) OR vs[Title/Abstract]) Sort by: Best Match Filters: Randomized Controlled Trial; Publication date from 2005/01/01 to 2020/03/31; Humans

2. Search strategy on Embase

('non small cell lung cancer'/exp OR nsclc:ti,ab,kw OR 'non small cell':ti,ab,kw OR 'non-small-cell':ti,ab,kw OR 'non-small cell':ti,ab,kw) AND (chemotherapy:ti,ab,kw OR pemetrexed:ti,ab,kw OR gemcitabine:ti,ab,kw OR paclitaxel:ti,ab,kw OR cisplatin:ti,ab,kw OR carboplatin:ti,ab,kw OR nedaplatin:ti,ab,kw OR lobaplatin:ti,ab,kw OR oxaliplatin:ti,ab,kw OR etoposide:ti,ab,kw OR docetaxel:ti,ab,kw OR vinorelbine:ti,ab,kw OR 'pembrolizumab'/exp OR keytruda:ti,ab,kw OR lambrolizumab:ti,ab,kw OR 'mk 3475':ti,ab,kw OR mk3475:ti,ab,kw OR 'sch 900475':ti,ab,kw OR sch900475:ti,ab,kw OR 'nivolumab'/exp OR 'bms 936558':ti,ab,kw OR bms936558:ti,ab,kw OR 'cmab 819':ti,ab,kw OR cmab819:ti,ab,kw OR 'mdx 1106':ti,ab,kw OR mdx1106:ti,ab,kw OR 'ono 4538':ti,ab,kw OR ono4538:ti,ab,kw OR opdivo:ti,ab,kw OR 'atezolizumab'/exp OR 'monoclonal antibody mpdl 3280a':ti,ab,kw OR 'monoclonal antibody mpdl3280a':ti,ab,kw OR 'mpdl 3280a':ti,ab,kw OR mpdl3280a:ti,ab,kw OR 'rg 7446':ti,ab,kw OR rg7446:ti,ab,kw OR tecentriq:ti,ab,kw OR tecntriq:ti,ab,kw OR 'durvalumab'/exp OR imfinzi:ti,ab,kw OR 'medi 4736':ti,ab,kw OR medi4736:ti,ab,kw OR 'avelumab'/exp OR bavencio:ti,ab,kw OR 'msb 0010682':ti,ab,kw OR 'msb 0010718c':ti,ab,kw OR 'msb 10682':ti,ab,kw OR 'msb 10718c':ti,ab,kw OR msb0010682:ti,ab,kw OR msb0010718c:ti,ab,kw OR msb10682:ti,ab,kw OR msb10718c:ti,ab,kw OR 'pf 06834635':ti,ab,kw OR 'pf 6834635':ti,ab,kw OR pf06834635:ti,ab,kw OR pf6834635:ti,ab,kw OR 'sintilimab'/exp OR 'ibi 308':ti,ab,kw OR ibi308:ti,ab,kw OR tyvyt:ti,ab,kw OR 'camrelizumab'/exp OR 'toripalimab'/exp OR 'js 001':ti,ab,kw OR js001:ti,ab,kw OR 'tab 001':ti,ab,kw OR tab001:ti,ab,kw OR 'tislelizumab'/exp OR 'bgb a317':ti,ab,kw OR bgba317:ti,ab,kw OR 'immunotherapy'/exp OR immunotherap*:ti,ab,kw OR 'immune checkpoint':ti,ab,kw OR icb:ti,ab,kw OR 'programmed death 1 receptor'/exp OR pd1:ti,ab,kw OR 'pd 1':ti,ab,kw OR 'programmed death 1':ti,ab,kw OR 'programmed death 1 ligand 1'/exp OR pdl1:ti,ab,kw OR 'pd l1':ti,ab,kw OR 'programmed death ligand 1':ti,ab,kw OR 'angiogenesis inhibitor':ti,ab,kw OR antiangiogenesis:ti,ab,kw OR antiangiogenic:ti,ab,kw OR vegfr:ti,ab,kw) AND (compare:ti,ab,kw OR comparison:ti,ab,kw OR comparative:ti,ab,kw OR comparing:ti,ab,kw OR compared:ti,ab,kw OR versus:ti,ab,kw OR vs:ti,ab,kw) AND [randomized controlled trial]/lim AND [humans]/lim AND [1-1-2005]/sd NOT [1-4-2020]/sd AND [2005-2020]/py

3.Search strategy on Cochrane library

#1 MeSH descriptor: [Nivolumab] explode all trees
#2 "BMS 936558" OR BMS-936558 OR MDX-1106 OR "MDX 1106" OR ONO-4538 OR "ONO 4538" OR BMS936558 OR MDX1106 OR Opdivo OR ONO4538
#3 #1 OR #2
#4 pembrolizumab OR Tislelizumab OR toripalimab OR camrelizumab OR sintilimab OR avelumab OR durvalumab OR atezolizumab
#5 #3 OR #4
#6 " programmed death1" OR PD-1 OR PD1 OR "PD 1" OR "programmed death ligand 1" OR PD-L1 OR PDL1 OR "PD L1" OR anti-PD-1 OR anti-PD-L1 OR " immune checkpoint" OR Immunotherap* OR ICB
#7 MeSH descriptor: [Immunotherapy] explode all trees
#8 #6 OR #7 OR #5
#9 Paclitaxel OR Cisplatin OR Carboplatin OR nedaplatin OR oxaliplatin OR lobaplatin OR etoposide OR Docetaxel OR vinorelbine
#10 VEGFR OR Antiangiogenesis OR Anlotinib OR Bevacizumab OR ENDOSTAR
#11 compare OR comparison OR comparative OR comparing versus OR vs
#12 MeSH descriptor: [Carcinoma, Non-Small-Cell Lung] explode all trees
#13 "Non-Small-Cell Lung Carcinoma" OR "Carcinomas, Non-Small-Cell Lung" OR "Non Small Cell Lung Carcinoma" OR "Lung Carcinomas, Non-Small-Cell" OR "Carcinoma, Non-Small Cell Lung" OR "Carcinoma, Non Small Cell Lung" OR "Non-Small-Cell Lung Carcinomas" OR "Non-Small Cell Lung Cancer" OR "Lung Carcinoma, Non-Small-Cell" OR "Nonsmall Cell Lung Cancer"
#14 #12 OR #13
#15 #8 OR #9 OR #10
#16 #11 AND #14 AND #15 with Cochrane Library publication date from Jan 2005 to Mar 2020, in Trials

Table S3. Comparisons of the Fit of Consistency and Inconsistency Models Using Deviance Information Criteria (DIC)

		Model		
		Consistency Fixed	Consistency Random	Inconsistency
Overall	Overall survival	57.26	55.38	64.01
	Progression-free survival	**108.50**	**61.64**	**63.99**
	Objective response rate	**126.08**	**105.83**	**108.42**
	Grade ≥3 adverse events	92.23	88.77	92.36
	1-year overall survival	103.83	104.63	116.25
	1-year progression-free	**160.41**	**113.08**	**116.84**
	2-year overall survival	101.36	98.50	104.27
	2-year progression-free	70.30	66.18	66.73
Non-squamous subgroup	Overall survival	45.63	44.91	50.01
	Progression-free survival	**62.27**	**38.74**	**39.98**
Squamous subgroup	Overall survival	23.93	24.53	25.99
	Progression-free survival	17.60	16.22	16.00
PD-L1>=1 or TC/IC=1/2/3	Overall survival	27.35	28.41	30.01
	Progression-free survival	16.66	18.35	22.00
PD-L1<1 or TC/IC=0	Overall survival	21.30	22.50	23.98
	Progression-free survival	17.21	16.06	15.97
PD-L1>=50 or TC/IC=3	Overall survival	25.89	27.35	29.98
	Progression-free survival	25.56	21.95	24.00
PD-L1>=1, <=49 or TC/IC=1/2	Overall survival	18.80	19.23	20.00
	Progression-free survival	9.15	10.83	14.01
High TMB	Overall survival	14.18	15.02	15.98
	Progression-free survival	16.09	14.23	14.00
Low TMB	Overall survival	14.98	15.32	15.98
	Progression-free survival	13.21	13.57	14.00
Ever smoking	Overall survival	30.36	31.64	34.00
	Progression-free survival	25.15	24.00	23.99
Never smoking	Overall survival	32.73	33.03	35.98
	Progression-free survival	25.18	21.46	21.97
Male	Overall survival	40.28	35.12	35.99
	Progression-free survival	**38.75**	**30.16**	**30.01**
Female	Overall survival	30.68	32.51	38.02
	Progression-free survival	26.12	26.84	29.99
Age >= 65	Overall survival	36.90	38.17	43.99
	Progression-free survival	22.78	23.93	26.00
Age < 65	Overall survival	39.63	37.71	39.99
	Progression-free survival	**38.98**	**25.84**	**25.99**
ECOG PS=0	Overall survival	33.71	34.97	40.01
	Progression-free survival	26.30	26.78	27.99
ECOG PS=1	Overall survival	39.62	36.10	36.00
	Progression-free survival	33.90	28.14	28.01
Liver metastasis	Overall survival	16.01	16.94	17.98
	Progression-free survival	10.40	11.28	12.00
No liver metastasis	Overall survival	12.33	13.33	13.99
	Progression-free survival	7.09	7.69	7.99

CNS metastasis	Overall survival	13.98	13.98	13.98
	Progression-free survival	6.01	6.00	5.98
No CNS metastasis	Overall survival	14.01	14.01	13.98
	Progression-free survival	6.01	6.00	6.00
The first sensitive analysis by excluding phase II trials				
Overall	Overall survival	56.70	53.83	57.99
	Progression-free survival	106.56	57.29	57.97
	Objective response rate	110.60	94.57	96.28
	Grade ≥3 adverse events	87.19	82.91	84.10
Non-squamous subgroup	Overall survival	43.22	41.63	43.98
	Progression-free survival	59.83	33.91	34.01
Squamous subgroup	Overall survival	21.42	22.50	23.99
	Progression-free survival	14.01	13.98	14.00
The second sensitive analysis by excluding trials with highly selected populations				
Overall	Overall survival	54.98	53.64	61.99
	Progression-free survival	82.87	53.29	56.02
	Objective response rate	107.36	92.08	94.31
	Grade ≥3 adverse events	89.54	85.40	88.32
Non-squamous subgroup	Overall survival	37.84	38.05	44.03
	Progression-free survival	58.25	34.73	36.00
Squamous subgroup	Overall survival	19.94	19.67	20.01
	Progression-free survival	13.61	12.26	11.99

Number highlighted in bold indicates DIC value was higher using fixed consistency model than inconsistency model, where random model was used.
PD-L1=programmed-death ligand 1; TMB=tumor mutation burden; ECOG PS= Eastern Cooperative Oncology Group Performance Status; CNS=central nervous system.

Table S4. Node-splitting Analysis of Inconsistency

Nodes	Direct effect	Indirect effect	Overall	P
Overall survival				
PEM/CT, CT	0.71 (0.58, 0.87)	0.49 (0.39, 0.62)	0.60 (0.52, 0.71)	**0.02**
ATE/CT, CT	0.84 (0.73, 0.97)	0.74 (0.62, 0.87)	0.80 (0.72, 0.88)	0.22
CT/BEV, CT	0.87 (0.78, 0.96)	0.95 (0.79, 1.13)	0.89 (0.81, 0.97)	0.40
CT/Mpem, CT	0.78 (0.64, 0.96)	1.06 (0.90, 1.24)	0.94 (0.83, 1.07)	**0.02**
NIV/IPI, NIV	0.90 (0.65, 1.25)	0.63 (0.45, 0.88)	0.80 (0.66, 0.97)	0.13
NIV/CT, NIV/IPI	1.22 (0.77, 1.93)	1.02 (0.80, 1.31)	1.08 (0.89, 1.32)	0.52
CT/Mpem, PEM/CT	1.79 (1.48, 2.16)	1.23 (0.96, 1.58)	1.56 (1.34, 1.81)	**0.02**
CT/BEV, ATE/CT	1.19 (1.00, 1.41)	1.06 (0.92, 1.24)	1.12 (1.00, 1.25)	0.32
CT/Mpem, ATE/CT	1.23 (0.97, 1.55)	1.16 (0.98, 1.38)	1.18 (1.03, 1.36)	0.70
CT/Mpem, CT/BEV	1.07 (0.84, 1.37)	1.06 (0.90, 1.24)	1.06 (0.93, 1.22)	0.94
Progression-free survival				
PEM/CT, CT	0.57 (0.33, 0.99)	0.38 (0.21, 0.66)	0.47 (0.31, 0.69)	0.27
ATE/CT, CT	0.69 (0.47, 1.03)	0.52 (0.32, 0.82)	0.63 (0.46, 0.83)	0.31
CT/BEV, CT	0.63 (0.46, 0.84)	0.74 (0.44, 1.23)	0.65 (0.51, 0.83)	0.56
CT/Mpem, CT	0.59 (0.34, 1.01)	0.94 (0.65, 1.35)	0.81 (0.59, 1.11)	0.15
NIV/IPI, NIV	0.83 (0.43, 1.61)	0.63 (0.27, 1.48)	0.76 (0.47, 1.20)	0.59
NIV/CT, NIV/IPI	0.97 (0.45, 2.08)	0.73 (0.32, 1.67)	0.86 (0.53, 1.41)	0.60
CT/Mpem, PEM/CT	1.99 (1.28, 3.06)	1.31 (0.67, 2.51)	1.74 (1.21, 2.51)	0.27
CT/BEV, ATE/CT	1.07 (0.60, 1.90)	1.06 (0.69, 1.62)	1.06 (0.76, 1.47)	0.98
CT/Mpem, ATE/CT	1.68 (0.97, 2.90)	1.11 (0.70, 1.76)	1.32 (0.92, 1.89)	0.23
CT/Mpem, CT/BEV	1.06 (0.59, 1.90)	1.36 (0.88, 2.13)	1.24 (0.88, 1.76)	0.48
Objective response rate				
ATE/CT, CT	1.76 (1.07, 2.90)	2.48 (0.97, 6.37)	1.89 (1.24, 2.91)	0.48
CT/BEV, CT	2.38 (1.59, 3.60)	1.69 (0.63, 4.50)	2.25 (1.57, 3.27)	0.48
NIV/IPI, NIV	1.48 (0.64, 3.43)	0.69 (0.20, 2.38)	1.19 (0.59, 2.28)	0.28
NIV/CT, NIV/IPI	1.63 (0.61, 4.33)	1.59 (0.44, 5.57)	1.61 (0.80, 3.17)	0.98
CT/BEV, ATE/CT	0.96 (0.41, 2.24)	1.34 (0.71, 2.58)	1.19 (0.73, 1.96)	0.48
Grade ≥3 adverse events				
PEM/CT, CT	1.08 (0.75, 1.55)	1.15 (0.69, 1.92)	1.10 (0.82, 1.48)	0.84
ATE/CT, CT	1.75 (1.39, 2.20)	1.19 (0.79, 1.79)	1.59 (1.30, 1.95)	0.11
CT/BEV, CT	1.26 (0.77, 2.08)	2.24 (1.58, 3.20)	1.85 (1.39, 2.47)	0.06
NIV/IPI, NIV	2.29 (1.66, 3.18)	2.86 (1.62, 5.07)	2.52 (1.90, 3.37)	0.50
NIV/CT, NIV/IPI	3.43 (2.21, 5.38)	1.77 (1.18, 2.66)	2.41 (1.80, 3.24)	**0.03**
CT/Mpem, PEM/CT	0.74 (0.53, 1.03)	0.79 (0.46, 1.35)	0.75 (0.57, 0.99)	0.84
CT/BEV, ATE/CT	1.31 (0.99, 1.74)	0.74 (0.43, 1.27)	1.16 (0.91, 1.49)	0.07
CT/Mpem, ATE/CT	0.53 (0.38, 0.74)	0.50 (0.29, 0.84)	0.52 (0.39, 0.69)	0.84
1-year overall survival rate				
PEM/CT, CT	1.99 (1.42, 2.81)	2.67 (1.80, 3.97)	2.26 (1.75, 2.93)	0.27
ATE/CT, CT	1.13 (0.90, 1.41)	1.43 (1.08, 1.89)	1.24 (1.04, 1.47)	0.19
CT/BEV, CT	1.22 (1.02, 1.47)	1.08 (0.80, 1.46)	1.18 (1.01, 1.38)	0.48
CT/Mpem, CT	1.41 (0.97, 2.05)	0.95 (0.74, 1.22)	1.08 (0.87, 1.33)	0.08
NIV/IPI, NIV	1.26 (0.95, 1.68)	1.39 (0.84, 2.32)	1.29 (1.01, 1.65)	0.74
NIV/CT, NIV/IPI	0.95 (0.63, 1.45)	1.21 (0.81, 1.80)	1.08 (0.81, 1.45)	0.42
CT/Mpem, PEM/CT	0.43 (0.31, 0.59)	0.57 (0.38, 0.87)	0.48 (0.37, 0.61)	0.27
CT/BEV, ATE/CT	0.83 (0.60, 1.12)	1.05 (0.82, 1.35)	0.96 (0.79, 1.16)	0.22
CT/Mpem, ATE/CT	0.84 (0.60, 1.16)	0.90 (0.66, 1.21)	0.87 (0.70, 1.09)	0.76
CT/Mpem, CT/BEV	0.83 (0.55, 1.26)	0.95 (0.72, 1.24)	0.91 (0.73, 1.14)	0.59
1-year progression-free survival rate				
PEM/CT, CT	2.79 (0.79, 9.91)	6.19 (1.80, 22.2)	4.18 (1.76, 10.2)	0.34
ATE/CT, CT	2.47 (1.04, 5.88)	5.55 (2.03, 16.1)	3.45 (1.80, 6.84)	0.21
CT/BEV, CT	3.66 (1.90, 7.60)	1.94 (0.64, 6.01)	3.07 (1.76, 5.65)	0.31
CT/Mpem, CT	4.23 (1.19, 15.5)	1.86 (0.81, 4.48)	2.39 (1.18, 5.08)	0.27
NIV/IPI, NIV	1.40 (0.38, 5.36)	2.67 (0.41, 17.2)	1.73 (0.65, 4.64)	0.55
NIV/CT, NIV/IPI	0.78 (0.19, 3.08)	1.14 (0.18, 7.03)	0.90 (0.33, 2.42)	0.72

CT/Mpem, PEM/CT	0.46 (0.19, 1.18)	1.02 (0.23, 4.68)	0.57 (0.26, 1.26)	0.34
CT/BEV, ATE/CT	0.79 (0.22, 2.87)	0.96 (0.37, 2.66)	0.89 (0.43, 1.90)	0.79
CT/Mpem, ATE/CT	0.41 (0.12, 1.36)	1.04 (0.37, 2.98)	0.69 (0.31, 1.56)	0.22
CT/Mpem, CT/BEV	1.30 (0.35, 4.87)	0.59 (0.21, 1.56)	0.78 (0.35, 1.70)	0.32
2-year overall survival rate				
ATE/CT, CT	1.51 (1.19, 1.92)	1.73 (1.20, 2.51)	1.57 (1.28, 1.92)	0.54
CT/BEV, CT	1.36 (1.10, 1.67)	1.26 (0.90, 1.77)	1.33 (1.11, 1.59)	0.72
CT/Mpem, CT	1.50 (1.03, 2.22)	1.34 (0.76, 2.37)	1.45 (1.05, 2.01)	0.74
NIV/IPI, NIV	1.18 (0.88, 1.57)	3.37 (1.96, 5.81)	1.49 (1.15, 1.93)	**0.001**
NIV/CT, NIV/IPI	0.79 (0.51, 1.20)	1.00 (0.66, 1.49)	0.90 (0.67, 1.20)	0.43
CT/BEV, ATE/CT	0.79 (0.58, 1.08)	0.91 (0.66, 1.23)	0.85 (0.68, 1.06)	0.54
CT/Mpem, CT/BEV	1.02 (0.59, 1.76)	1.14 (0.75, 1.76)	1.09 (0.78, 1.53)	0.74
2-year progression-free survival rate				
ATE/CT, CT	2.86 (1.79, 4.79)	26.2 (7.53, 112)	4.11 (2.58, 6.78)	**0.001**
CT/BEV, CT	6.09 (2.24, 21.9)	0.68 (0.27, 1.60)	1.80 (1.01, 3.28)	**0.001**
NIV/CT, NIV/IPI	0.59 (0.31, 1.09)	0.64 (0.34, 1.20)	0.62 (0.40, 0.95)	0.84
CT/BEV, ATE/CT	0.24 (0.11, 0.47)	2.13 (0.69, 8.24)	0.44 (0.26, 0.74)	**0.001**
Overall survival for non-squamous subgroup				
ATE/CT, CT	0.79 (0.64, 0.98)	0.70 (0.59, 0.83)	0.73 (0.64, 0.84)	0.40
CT/BEV, CT	0.87 (0.78, 0.96)	0.84 (0.68, 1.03)	0.86 (0.78, 0.95)	0.77
CT/Mpem, CT	0.78 (0.65, 0.96)	0.93 (0.77, 1.14)	0.86 (0.74, 0.99)	0.21
CT/BEV, ATE/CT	1.19 (1.00, 1.41)	1.15 (0.95, 1.39)	1.17 (1.03, 1.33)	0.77
CT/Mpem, ATE/CT	1.23 (0.97, 1.55)	1.11 (0.91, 1.37)	1.16 (1.00, 1.36)	0.53
CT/Mpem, CT/BEV	1.07 (0.84, 1.37)	0.96 (0.80, 1.14)	0.99 (0.86, 1.15)	0.47
Progression-free survival for non-squamous subgroup				
ATE/CT, CT	0.64 (0.34, 1.22)	0.48 (0.27, 0.83)	0.55 (0.37, 0.80)	0.44
CT/BEV, CT	0.63 (0.44, 0.88)	0.62 (0.31, 1.22)	0.63 (0.47, 0.82)	0.97
CT/Mpem, CT	0.59 (0.31, 1.11)	0.82 (0.47, 1.39)	0.71 (0.47, 1.06)	0.38
CT/BEV, ATE/CT	1.06 (0.55, 2.08)	1.22 (0.67, 2.16)	1.15 (0.77, 1.70)	0.72
CT/Mpem, ATE/CT	1.67 (0.94, 2.97)	1.02 (0.57, 1.82)	1.31 (0.85, 2.01)	0.19
CT/Mpem, CT/BEV	1.06 (0.53, 2.10)	1.20 (0.67, 2.16)	1.14 (0.76, 1.71)	0.75
Overall survival for ever smoking				
ATE/CT, CT	0.85 (0.73, 0.98)	0.97 (0.65, 1.44)	0.86 (0.75, 0.98)	0.52
CT/BEV, CT	1.01 (0.84, 1.21)	0.88 (0.60, 1.28)	0.98 (0.83, 1.16)	0.51
CT/Mpem, ATE/CT	1.20 (0.94, 1.54)	1.39 (0.98, 1.95)	1.26 (1.03, 1.54)	0.51
CT/Mpem, CT/BEV	1.16 (0.90, 1.49)	1.01 (0.72, 1.42)	1.10 (0.90, 1.35)	0.52
Overall survival for never smoking				
ATE/CT, CT	0.70 (0.42, 1.16)	0.26 (0.05, 1.28)	0.64 (0.39, 1.03)	0.25
CT/BEV, CT	0.69 (0.51, 0.93)	1.86 (0.36, 9.64)	0.71 (0.53, 0.95)	0.25
CT/Mpem, ATE/CT	1.54 (0.76, 3.10)	0.57 (0.13, 2.61)	1.29 (0.68, 2.44)	0.25
CT/Mpem, CT/BEV	0.58 (0.14, 2.35)	1.56 (0.63, 3.89)	1.16 (0.54, 2.49)	0.25
Overall survival for male				
PEM/CT, CT	0.69 (0.36, 1.33)	0.69 (0.25, 1.89)	0.69 (0.42, 1.14)	1.00
CT/BEV, CT	0.83 (0.57, 1.20)	0.83 (0.26, 2.63)	0.83 (0.60, 1.13)	1.00
CT/Mpem, PEM/CT	1.43 (0.73, 2.81)	1.43 (0.51, 3.90)	1.43 (0.86, 2.37)	1.00
CT/Mpem, CT/BEV	1.19 (0.61, 2.32)	1.19 (0.43, 3.31)	1.19 (0.72, 1.97)	1.00
Progression-free survival for male				
PEM/CT, CT	0.58 (0.23, 1.47)	0.71 (0.16, 3.10)	0.62 (0.31, 1.20)	0.75
ATE/CT, CT	0.69 (0.36, 1.33)	0.56 (0.11, 2.82)	0.67 (0.40, 1.13)	0.75
CT/Mpem, PEM/CT	1.52 (0.59, 3.91)	1.86 (0.43, 8.07)	1.61 (0.82, 3.17)	0.76
CT/Mpem, ATE/CT	1.56 (0.62, 3.95)	1.27 (0.29, 5.58)	1.48 (0.76, 2.89)	0.75
Overall survival for female				
PEM/CT, CT	0.42 (0.22, 0.81)	0.23 (0.13, 0.42)	0.30 (0.20, 0.47)	0.18
CT/BEV, CT	0.88 (0.74, 1.06)	1.61 (0.67, 3.84)	0.90 (0.76, 1.08)	0.19
CT/Mpem, PEM/CT	3.45 (2.27, 5.25)	1.89 (0.86, 4.13)	3.02 (2.08, 4.36)	0.18

CT/Mpem, CT/BEV	0.90 (0.61, 1.33)	1.64 (0.74, 3.65)	1.01 (0.71, 1.44)	0.18
Progression-free survival for female				
PEM/CT, CT	0.49 (0.30, 0.80)	0.48 (0.29, 0.80)	0.49 (0.34, 0.69)	0.96
ATE/CT, CT	0.61 (0.49, 0.77)	0.63 (0.32, 1.23)	0.61 (0.50, 0.76)	0.96
CT/Mpem, PEM/CT	2.50 (1.83, 3.41)	2.45 (1.29, 4.64)	2.49 (1.88, 3.30)	0.95
CT/Mpem, ATE/CT	1.96 (1.40, 2.75)	2.00 (1.07, 3.74)	1.97 (1.46, 2.65)	0.95
Overall survival for ECOG PS=0				
PEM/CT, CT	0.54 (0.29, 0.99)	0.35 (0.19, 0.65)	0.44 (0.28, 0.67)	0.32
CT/BEV, CT	0.90 (0.74, 1.09)	1.40 (0.60, 3.27)	0.92 (0.76, 1.10)	0.32
CT/Mpem, PEM/CT	2.27 (1.43, 3.62)	1.46 (0.70, 3.05)	2.00 (1.35, 2.97)	0.32
CT/Mpem, CT/BEV	0.88 (0.61, 1.27)	1.37 (0.62, 3.02)	0.95 (0.68, 1.33)	0.32
Progression-free survival for ECOG PS=0				
PEM/CT, CT	0.45 (0.29, 0.69)	0.55 (0.34, 0.90)	0.49 (0.36, 0.70)	0.53
ATE/CT, CT	0.63 (0.52, 0.78)	0.51 (0.28, 0.95)	0.62 (0.51, 0.75)	0.53
CT/Mpem, PEM/CT	2.04 (1.47, 2.84)	2.51 (1.44, 4.38)	2.15 (1.62, 2.86)	0.53
CT/Mpem, ATE/CT	1.79 (1.33, 2.40)	1.45 (0.82, 2.58)	1.71 (1.31, 2.22)	0.53
Overall survival for ECOG PS=1				
PEM/CT, CT	0.66 (0.48, 0.90)	0.55 (0.34, 0.89)	0.63 (0.48, 0.81)	0.54
CT/BEV, CT	0.82 (0.72, 0.94)	0.98 (0.57, 1.70)	0.83 (0.73, 0.95)	0.54
CT/Mpem, PEM/CT	1.89 (1.38, 2.58)	1.58 (0.99, 2.54)	1.79 (1.38, 2.32)	0.54
CT/Mpem, CT/BEV	1.27 (0.92, 1.76)	1.51 (0.95, 2.41)	1.35 (1.03, 1.76)	0.55
Progression-free survival for ECOG PS=1				
PEM/CT, CT	0.61 (0.48, 0.78)	0.61 (0.42, 0.90)	0.61 (0.50, 0.75)	0.98
ATE/CT, CT	0.69 (0.59, 0.81)	0.69 (0.45, 1.05)	0.69 (0.60, 0.80)	0.98
CT/Mpem, PEM/CT	1.79 (1.38, 2.31)	1.80 (1.24, 2.61)	1.79 (1.45, 2.21)	0.98
CT/Mpem, ATE/CT	1.59 (1.25, 2.02)	1.58 (1.07, 2.32)	1.58 (1.29, 1.94)	0.98
Overall survival for age ≥ 65				
PEM/CT, CT	0.74 (0.51, 1.07)	0.69 (0.43, 1.12)	0.72 (0.54, 0.97)	0.82
ATE/CT, CT	0.80 (0.66, 0.97)	0.76 (0.48, 1.22)	0.79 (0.66, 0.95)	0.87
CT/BEV, CT	0.89 (0.73, 1.07)	0.97 (0.61, 1.54)	0.90 (0.75, 1.07)	0.72
CT/Mpem, PEM/CT	1.56 (1.05, 2.32)	1.45 (0.92, 2.31)	1.52 (1.12, 2.05)	0.82
CT/Mpem, ATE/CT	1.41 (0.99, 2.00)	1.35 (0.93, 1.95)	1.38 (1.07, 1.78)	0.87
CT/Mpem, CT/BEV	1.17 (0.84, 1.64)	1.28 (0.88, 1.86)	1.22 (0.95, 1.56)	0.72
Progression-free survival for age ≥ 65				
PEM/CT, CT	0.63 (0.47, 0.84)	0.89 (0.55, 1.44)	0.69 (0.54, 0.89)	0.24
ATE/CT, CT	0.65 (0.51, 0.83)	0.46 (0.28, 0.77)	0.61 (0.49, 0.76)	0.24
CT/Mpem, PEM/CT	1.33 (0.98, 1.82)	1.88 (1.17, 3.00)	1.48 (1.14, 1.91)	0.23
CT/Mpem, ATE/CT	1.82 (1.38, 2.40)	1.29 (0.79, 2.11)	1.68 (1.32, 2.13)	0.24
Overall survival for age < 65				
PEM/CT, CT	0.52 (0.34, 0.80)	0.38 (0.24, 0.58)	0.44 (0.33, 0.60)	0.30
ATE/CT, CT	0.85 (0.69, 1.04)	0.82 (0.52, 1.28)	0.84 (0.70, 1.01)	0.88
CT/BEV, CT	0.81 (0.71, 0.93)	1.05 (0.65, 1.69)	0.83 (0.72, 0.95)	0.31
CT/Mpem, PEM/CT	2.32 (1.66, 3.27)	1.68 (1.01, 2.79)	2.10 (1.59, 2.79)	0.30
CT/Mpem, ATE/CT	1.12 (0.82, 1.53)	1.08 (0.74, 1.58)	1.11 (0.87, 1.41)	0.88
CT/Mpem, CT/BEV	0.98 (0.68, 1.42)	1.27 (0.90, 1.77)	1.13 (0.88, 1.45)	0.32
Progression-free survival for age < 65				
PEM/CT, CT	0.50 (0.14, 1.86)	0.44 (0.05, 4.19)	0.48 (0.18, 1.31)	0.90
ATE/CT, CT	0.64 (0.18, 2.35)	0.73 (0.08, 6.93)	0.66 (0.25, 1.78)	0.90
CT/Mpem, PEM/CT	2.32 (0.63, 8.52)	2.03 (0.21, 19.3)	2.25 (0.83, 6.05)	0.90
CT/Mpem, ATE/CT	1.58 (0.43, 5.80)	1.82 (0.19, 17.3)	1.64 (0.61, 4.40)	0.89

Significant values are in bold and underlined, indicating a significant inconsistency between the direct effect and indirect
effect. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

Table S5. Sensitivity Analysis Description

The first sensitivity analysis (Remove Phase Ⅱ trials)					
Outcome	**No. Studies**	**No. patients**	**Removed studies**	**Changes in Ranking probabilities**	**Variations of HR/CrI from original analysis to sensitivity analysis**
OS	23	16545	KEYNOTE021G Lynch Niho	1. PEM/CT 2. NIV/IPI/CT 3. CAM/CT 4. ATE/CT/BEV 5. NIV/IPI 6. PEM 7. ATE/CT 8. NIV/CT 9. DUR/TRE/CT 10. ATE 11. CT/BEV/Mpem 12. CT/BEV 13. IPI/CT 14. NIV15. DUR/TRE 16. CT/Mpem 17. DUR 18. CT (No change with the original result)	ATE/CT/BEV vs IPI/CT: 0.79 (0.61,1.01) to 0.77(0.60,1.00)
PFS	23	16182	KEYNOTE021G Lynch Niho	1. ATE/CT/BEV 2. CT/BEV/Mpem(↑) 3. PEM/CT(↓) 4. CAM/CT 5. ATE/CT 6. CT/BEV 7. NIV/CT 8. NIV/IPI/CT 9. PEM 10. ATE 11. NIV/IPI 12. DUR/TRE/CT(↑) 13. CT/Mpem(↑) 14. DUR (↑)15. IPI/CT(↓) 16. CT 17. NIV 18. DUR/TRE	PEM vs ATE/CT/BEV: 1.98(1.01,3.82) to 1.96(0.92,4.12); NIV/IPI vs ATE/CT/BEV: 2.04(1.07,3.93) to 2.03(0.97,4.25) NIV vs ATE/CT: 1.70(1.05,2.76) to 1.70(0.99,2.94); ATE/CT/BEV vs IPI/CT: 0.49(0.25,0.96) to 0.45(0.19,1.05) DUR/TRE vs CAM/CT: 2.52(1.07,6.03) to 2.54(0.96,6.74) NIV vs CT/BEV: 1.60(1.02,2.53) to 1.59(0.94,2.70) DUR/TRE vs CT/BEV: 1.91(1.05,3.52) to 1.9(0.96,3.82); ATE/CT/BEV vs CT/BEV: 0.59(0.36,0.98) to 0.59(0.33,1.05) CT/BEV/Mpem vs CT/Mpem: 0.57(0.33,0.96) to 0.57(0.31,1.04) CAM/CT vs CT: 0.50(0.25,0.96) to 0.49(0.23,1.04)
ORR	20	13962	KEYNOTE021G Lynch Niho	1. ATE/CT/BEV 2. PEM/CT 3. NIV/CT 4. CAM/CT(↑) 5. CT/BEV(↓) 6. ATE/CT 7. NIV/IPI/CT 8. DUR/TRE/CT 9. NIV/IPI 10. PEM 11. NIV(↑) 12. CT(↑) 13. DUR 14. ATE(↑) 15. IPI/CT(↓) 16. DUR/TRE	ATE vs NIV/CT: 0.38(0.14,1.06) to 0.39(0.14,1.00) NIV/CT vs IPI/CT: 1.79(0.71,4.12) to 2.62(1.00,6.82) PEM/CT vs IPI/CT: 2.27(0.93,4.99) to 3.26(1.24,8.63) IPI/CT vs CT/BEV: 0.57(0.28,1.28) to 0.42(0.18,1.00)
≥3AE	17	12910	KEYNOTE021G Lynch	1. ATE/CT/BEV 2. IPI/CT 3. CT/BEV 4. NIV/CT 5. CAM/CT 6. ATE/CT 7. NIV/IPI/CT 8. PEM/CT 9. CT 10. CT/Mpem 11. NIV/IPI 12. DUR/TRE/CT 13. DUR/TRE 14. DUR(↑) 15. PEM(↓) 16. NIV 17. ATE	NIV/CT vs ATE/CT/BEV: 0.67(0.45,1.01) to 0.65(0.43,0.97) PEM/CT vs CT/Mpem: 1.35(1.02,1.78) to 1.27(0.94,1.71)

NSq OS	19	10946	KEYNOTE021G Lynch Niho	1. PEM/CT 2. CAM/CT 3. ATE 4. ATE/CT/BEV 5. NIV/IPI/CT 6. DUR 7. ATE/CT 8. NIV/IPI 9. PEM 10. CT/BEV/Mpem 11. NIV/CT 12. CT/Mpem 13. CT/BEV 14. CT 15. NIV	No change with the original result
NSq PFS	15	8651	KEYNOTE021G Lynch Niho	1. PEM/CT 2. ATE/CT/BEV 3. CT/BEV/Mpem 4. CAM/CT 5. ATE/CT 6. PEMI(↑) 7. NIV/IPI(↓) 8. CT/BEV 9. NIV/CT 10. CT/Mpem 11. CT 12. NIV	NIV vs ATE/CT: 2.36(1.14,4.97) to 2.38(0.98,5.95) ATE/CT/BEV vs CT/BEV: 0.57(0.33,1.00) to 0.57(0.28,1.14) NIV vs CAM/CT: 2.96(1.13,7.93) to 2.97(0.91,9.88) NIV vs CT/BEV: 2.06(1.05,4.11) to 2.07(0.90,4.89) PEM/CT vs CT/Mpem: 0.50(0.31,0.80) to 0.48(0.23,1.01) ATE/CT/BEV vs CT/Mpem: 0.5(0.26,0.95) to 0.50(0.23,1.10) CAM/CT vs CT: 0.44(0.20,0.91) to 0.43(0.17,1.08)
Sq OS	10	3582	Lynch	1. NIV/IPI/CT 2. NIV/IPI 3. ATE 4. NIV/CT 5. PEM/CT 6. PEM 7. NIV 8. DUR 9. ATE/CT 10. IPI/CT 11. CT (No change with the original result)	No change with the original result
Sq PFS	6	2488	Lynch	1. PEM 2. NIV/CT 3. PEM/CT 4. NIV/IPI 5. ATE/CT 6. NIV 7. IPI/CT 8. CT (No change with the original result)	No change with the original result
The second sensitivity analysis (Remove trials with highly-selected populations**)**					
Outcome	**No. Studies**	**No. patients**	**Removed studies**	**Changes in Ranking probabilities**	**Variations of HR/CrI from original analysis to sensitivity analysis**
OS	25	16672	KEYNOTE024	1. PEM/CT 2. NIV/IPI/CT 3. CAM/CT 4. ATE/CT/BEV 5. NIV/IPI 6. PEM 7. ATE/CT 8. NIV/CT 9. DUR/TRE/CT 10. ATE 11. CT/BEV/Mpem 12. CT/BEV 13. IPI/CT 14. NIV 15. DUR/TRE 16. CT/Mpem 17. DUR 18. CT (No change with the original result)	PEM vs DUR: 0.82(0.66, 1.00) to 0.85(0.69, 1.06) PEM vs CT/Mpem: 0.83(0.70, 0.99) to 0.87(0.72, 1.05)

PFS	23	15253	KEYNOTE024 MYSTIC CCTG BR.34	1. ATE/CT/BEV 2. PEM/CT 3. CT/BEV/Mpem 4. CAM/CT 5. ATE/CT 6. CT/BEV 7. NIV/CT 8. NIV/IPI/CT 9. ATE(↑) 10. NIV/IPI(↑) 11. IPI/CT(↑) 12. CT/Mpem(↑) 13. PEM(↓) 14. CT	PEM vs PEM/CT: 1.65(0.93, 2.86) to 2.25(1.26, 4.05) PEM vs ATE/CT: 1.25(0.75, 2.03) to 1.70(1.00, 2.91) PEM vs CAM/CT: 1.55(0.70, 3.34) to 2.11(1.00, 4.53) PEM vs CT/BEV/Mpem: 1.66(0.90, 3.06) to 2.25(1.23, 4.26) NIV/IPI vs PEM/CT: 1.69(0.98, 2.93) to 1.69(1.05, 2.74) NIV/IPI vs CT/BEV/Mpem: 1.71(0.95, 3.14) to 1.69 (1.01, 2.91) NIV/CT vs CT: 0.68(0.46, 1.01) to 0.68(0.48, 0.96)
ORR	20	13304	KEYNOTE024 MYSTIC CCTG BR.34	1. ATE/CT/BEV 2. PEM/CT 3. NIV/CT 4. CT/BEV 5. CAM/CT 6. ATE/CT 7. NIV/IPI/CT 8. NIV/IPI 9. IPI/CT(↑) 10. NIV (↑)11. PEM(↓) 12. CT 13. ATE	PEM vs PEM/CT: 0.48 (0.22, 1.07) to 0.36 (0.14, 0.90)
≥3AE	18	12859	KEYNOTE024	1. ATE/CT/BEV 2. IPI/CT 3. NIV/CT(↑) 4. CT/BEV(↓) 5. CAM/CT 6. ATE/CT 7. NIV/IPI/CT 8. PEM/CT 9.CT10.CT/Mpem 11. NIV/IPI 12. DUR/TRE//CT 13. DUR/TRE 14. DUR 15. PEM 16. NIV 17. ATE	No change with the original result
NSq OS	19	10717	KEYNOTE024 MYSTIC IMpower 110	1. PEM/CT 2. CAM/CT 3. ATE/CT/BEV 4. NIV/IPI/CT 5. ATE/CT 6. NIV/IPI 7. CT/BEV/Mpem(↑) 8. NIV/CT(↑) 9. CT/Mpem(↑) 10. PEM(↓) 11. CT/BEV 12. CT 13. IPI/CT 14. NIV	PEM vs CT: 0.79 (0.68,0.93) to 0.86(0.72, 1.03)
NSq PFS	16	8599	KEYNOTE024 CheckMate227 (High-TMB)	1. PEM/CT 2. ATE/CT/BEV 3. CT/BEV/Mpem 4. CAM/CT 5. ATE/CT 6. BEV/CT 7. NIV/CT 8. CT/Mpem 9. IPI/CT 10. CT 11. NIV (No change with the original result)	No change with the original result
Sq OS	8	3408	KEYNOTE024 MYSTIC IMpower 110	1.NIV/IPI/CT 2. NIV/IPI 3. NIV/CT 4. PEM/CT 5. PEM 6. NIV 7. ATE/CT 8. IPI/CT 9. CT (No change with the original result)	No change with the original result
Sq PFS	5	2368	KEYNOTE024 CheckMate227 (High-TMB)	1. NIV/CT 2. PEM/CT 3. ATE/CT 4. NIV 5. IPI/CT 6. CT (No change with the original result)	No change with the original result

OS=overall survival; PFS=progression-free survival; SQ=squamous NSCLC; NSQ=non-squamous NSCLC; ≥3AE=grade 3 or higher adverse event; PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; CAM=camrelizumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem= CT followed by maintenance of pemetrexed.

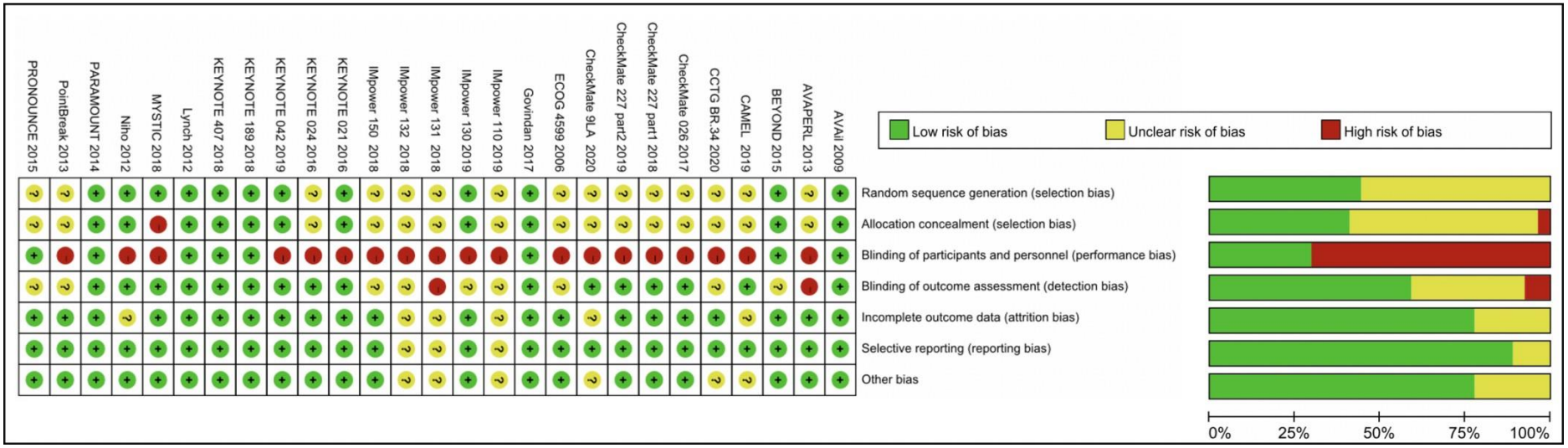


Figure S1. Summary of Results from Bias Risk Assessment of Studies Using the Cochrane Risk of Bias Tool

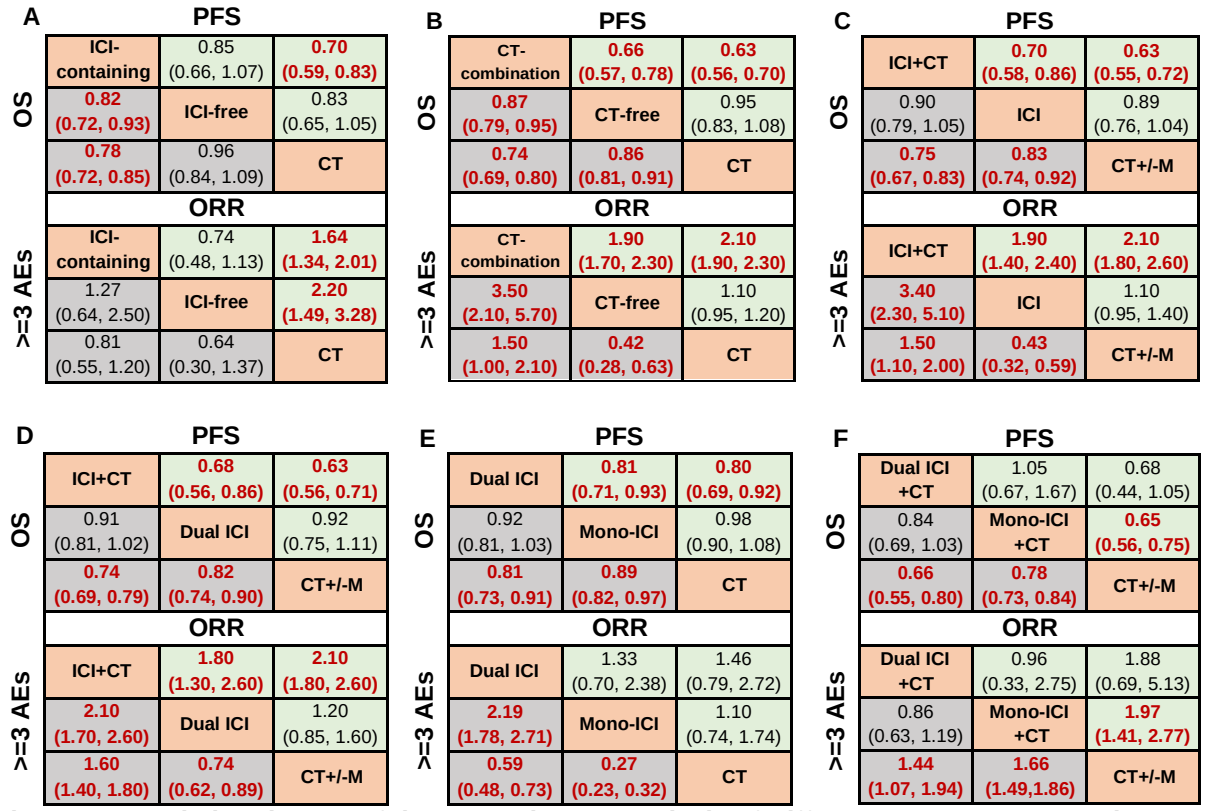
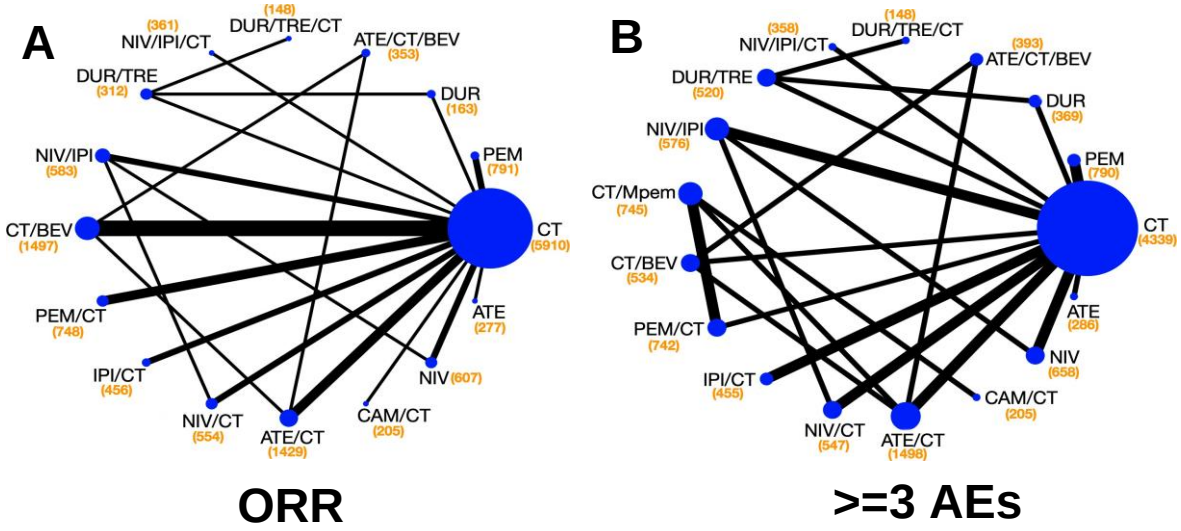


Figure S2. Pooled Estimates of the Network Meta-analysis of Different Treatment Strategies

(A) Pooled hazard/odds ratio (HR/OR) (95% CrIs (credible intervals)) for overall survival (OS), progression-free survival (PFS), adverse events of grade 3 or higher (>= 3 AEs), and objective response rate (ORR) in comparisons of immune checkpoints inhibitor (ICI)-containing, ICI-free and chemotherapy (CT). ICI-free group excludes patients receiving CT only. The treatment arm (ATE/CT/BEV) in the IMpower 150 trial was not included as ICI-containing in order to eliminate the additional impact of BEV while the treatment arm (ATE/CT) was included. (B) Pooled HR/OR (95% CrIs) for OS, PFS, >= 3 AEs, and ORR in comparisons of CT-combination, CT-free and CT. CT followed by maintenance of pemetrexed (CT/Mpem) was classified into CT. (D) Pooled HR/OR (95% CrIs) for OS, PFS, >= 3 AEs, and ORR in comparison of ICI plus CT, ICI only, and CT with or without maintenance (CT+/-M). (D) Pooled HR/OR (95% CrIs) for OS, PFS, >= 3 AEs, and ORR in comparison of ICI plus CT, dual ICI, and CT+/-M. (E) Pooled HR/OR (95% CrIs) for OS, PFS, >= 3 AEs, and ORR in comparison of dual ICI, mono-ICI, and CT after excluding highly selected population. (F) Pooled HR/OR (95% CrIs) for OS, PFS, >= 3 AEs, and ORR in comparison of dual ICI plus CT, mono-ICI plus CT, and CT+/-M. Data in each cell are HR or OR (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. HR less than 1 and OR more than 1 favour upper-row treatment. Significant results are highlighted in red and bold.


A
(361)
NIV/IPI/CT
(148)
DUR/TRE/CT
ATE/CT/BEV
(353)
DUR/TRE
(312)
DUR
(163)
NIV/IPI
(583)
PEM
(791)
CT/BEV
(1497)
CT
(5910)
PEM/CT
(748)
ATE
(277)
IPI/CT
(456)
NIV
(607)
NIV/CT
(554)
ATE/CT
(1429)
CAM/CT
(205)
ORR
B
(358)
NIV/IPI/CT
(148)
DUR/TRE/CT
(393)
ATE/CT/BEV
DUR/TRE
(520)
DUR
(369)
NIV/IPI
(576)
PEM
(790)
CT/Mpem
(745)
CT
(4339)
CT/BEV
(534)
ATE
(286)
PEM/CT
(742)
NIV
(658)
IPI/CT
(455)
CAM/CT
(205)
NIV/CT
(547)
ATE/CT
(1498)
>=3 AEs

C

>=3 AEs

ORR

PEM	1.10 (0.79, 1.53)	**1.80 (1.12, 2.92)**	0.97 (0.63, 1.49)	**0.57 (0.38, 0.85)**	0.47 (0.21, 1.04)	**0.43 (0.31, 0.60)**	**0.18 (0.13, 0.25)**	**0.23 (0.16, 0.33)**	**0.30 (0.21, 0.43)**	**0.12 (0.08, 0.17)**	**0.21 (0.15, 0.28)**	**0.17 (0.12, 0.24)**	**0.20 (0.11, 0.35)**	**0.18 (0.12, 0.26)**	**0.40 (0.27, 0.59)**	**0.33 (0.26, 0.41)**
1.15 (0.50, 2.67)	**NIV**	**1.64 (1.02, 2.69)**	0.88 (0.57, 1.37)	**0.52 (0.35, 0.78)**	**0.43 (0.19, 0.95)**	**0.40 (0.30, 0.53)**	**0.16 (0.12, 0.23)**	**0.21 (0.14, 0.31)**	**0.27 (0.19, 0.40)**	**0.11 (0.07, 0.16)**	**0.19 (0.14, 0.26)**	**0.15 (0.11, 0.22)**	**0.18 (0.10, 0.32)**	**0.16 (0.11, 0.24)**	**0.37 (0.25, 0.54)**	**0.30 (0.24, 0.38)**
1.58 (0.57, 4.56)	1.37 (0.50, 3.87)	**ATE**	**0.54 (0.31, 0.93)**	**0.32 (0.18, 0.53)**	**0.26 (0.11, 0.62)**	**0.24 (0.15, 0.39)**	**0.10 (0.06, 0.16)**	**0.13 (0.08, 0.21)**	**0.17 (0.10, 0.28)**	**0.07 (0.04, 0.11)**	**0.12 (0.07, 0.18)**	**0.09 (0.06, 0.15)**	**0.11 (0.06, 0.21)**	**0.10 (0.06, 0.16)**	**0.22 (0.13, 0.37)**	**0.18 (0.12, 0.28)**
1.53 (0.54, 4.48)	1.33 (0.47, 3.86)	0.97 (0.28, 3.25)	**DUR**	**0.59 (0.40, 0.85)**	0.49 (0.22, 1.06)	**0.45 (0.29, 0.69)**	**0.19 (0.12, 0.29)**	**0.24 (0.15, 0.38)**	**0.31 (0.20, 0.49)**	**0.12 (0.07, 0.20)**	**0.21 (0.14, 0.32)**	**0.17 (0.11, 0.27)**	**0.21 (0.11, 0.39)**	**0.18 (0.12, 0.29)**	**0.41 (0.26, 0.66)**	**0.34 (0.24, 0.49)**
1.62 (0.57, 4.74)	1.40 (0.49, 4.06)	1.02 (0.30, 3.42)	1.05 (0.43, 2.57)	**DUR/TRE**	0.83 (0.42, 1.64)	0.76 (0.51, 1.14)	**0.32 (0.21, 0.47)**	**0.41 (0.26, 0.63)**	**0.53 (0.34, 0.82)**	**0.21 (0.13, 0.33)**	**0.37 (0.25, 0.54)**	**0.29 (0.19, 0.45)**	**0.35 (0.19, 0.65)**	**0.31 (0.20, 0.49)**	0.70 (0.45, 1.10)	**0.58 (0.42, 0.80)**
0.85 (0.20, 3.57)	0.74 (0.18, 3.15)	0.54 (0.11, 2.50)	0.56 (0.15, 2.03)	0.53 (0.20, 1.37)	**DUR/TRE/CT**	0.92 (0.42, 2.05)	**0.38 (0.17, 0.85)**	0.49 (0.22, 1.11)	0.64 (0.28, 1.44)	**0.25 (0.11, 0.57)**	**0.44 (0.20, 0.97)**	**0.36 (0.16, 0.81)**	0.42 (0.17, 1.06)	**0.38 (0.17, 0.86)**	0.85 (0.38, 1.94)	0.70 (0.33, 1.51)
0.97 (0.44, 2.24)	0.84 (0.43, 1.68)	0.61 (0.22, 1.68)	0.63 (0.22, 1.81)	0.60 (0.21, 1.71)	1.14 (0.28, 4.71)	**NIV/IPI**	**0.41 (0.31, 0.56)**	**0.53 (0.36, 0.77)**	**0.69 (0.48, 1.00)**	**0.27 (0.18, 0.40)**	**0.48 (0.35, 0.65)**	**0.39 (0.27, 0.55)**	**0.46 (0.26, 0.80)**	**0.41 (0.28, 0.59)**	0.92 (0.63, 1.35)	**0.76 (0.60, 0.96)**
0.60 (0.27, 1.41)	0.52 (0.24, 1.17)	0.38 (0.14, 1.05)	0.39 (0.14, 1.13)	0.37 (0.13, 1.07)	0.71 (0.17, 2.97)	0.62 (0.31, 1.24)	**NIV/CT**	1.28 (0.88, 1.87)	**1.67 (1.15, 2.43)**	**0.65 (0.44, 0.97)**	1.15 (0.84, 1.57)	0.93 (0.65, 1.33)	1.11 (0.64, 1.95)	0.99 (0.68, 1.44)	**2.22 (1.51, 3.28)**	**1.84 (1.45, 2.33)**
0.75 (0.27, 2.10)	0.65 (0.24, 1.80)	0.47 (0.15, 1.54)	0.49 (0.15, 1.62)	0.46 (0.14, 1.54)	0.88 (0.19, 4.11)	0.77 (0.28, 2.07)	1.23 (0.46, 3.38)	**NIV/IPI/CT**	1.31 (0.86, 2.00)	**0.51 (0.33, 0.80)**	0.90 (0.63, 1.30)	0.73 (0.49, 1.09)	0.87 (0.49, 1.57)	0.78 (0.51, 1.18)	**1.74 (1.13, 2.68)**	**1.44 (1.07, 1.94)**
0.48 (0.22, 1.07)	**0.41 (0.19, 0.90)**	**0.30 (0.11, 0.81)**	**0.31 (0.11, 0.87)**	**0.30 (0.11, 0.81)**	0.57 (0.14, 2.28)	0.49 (0.23, 1.04)	0.79 (0.36, 1.71)	0.64 (0.24, 1.69)	**PEM/CT**	**0.39 (0.26, 0.59)**	**0.69 (0.50, 0.94)**	**0.56 (0.37, 0.83)**	0.67 (0.41, 1.09)	**0.59 (0.41, 0.87)**	**1.33 (1.01, 1.76)**	1.10 (0.82, 1.48)
0.37 (0.14, 0.98)	**0.32 (0.12, 0.84)**	**0.23 (0.07, 0.72)**	**0.24 (0.07, 0.77)**	**0.23 (0.07, 0.73)**	0.43 (0.10, 1.96)	**0.38 (0.14, 0.97)**	0.61 (0.23, 1.57)	0.49 (0.16, 1.50)	0.77 (0.31, 1.93)	**ATE/CT/BEV**	**1.76 (1.34, 2.33)**	1.43 (0.94, 2.18)	1.71 (0.97, 2.97)	**1.52 (1.15, 2.00)**	**3.40 (2.31, 5.03)**	**2.81 (2.03, 3.90)**
0.74 (0.36, 1.57)	0.64 (0.31, 1.33)	0.47 (0.18, 1.18)	0.48 (0.18, 1.28)	0.46 (0.17, 1.21)	0.87 (0.22, 3.43)	0.76 (0.37, 1.53)	1.22 (0.59, 2.51)	0.99 (0.39, 2.46)	1.55 (0.79, 2.99)	2.00 (0.96, 4.28)	**ATE/CT**	0.81 (0.58, 1.13)	0.97 (0.59, 1.58)	0.86 (0.67, 1.10)	**1.93 (1.45, 2.56)**	**1.59 (1.30, 1.95)**
1.09 (0.43, 2.53)	0.94 (0.37, 2.16)	0.68 (0.22, 1.89)	0.71 (0.22, 2.02)	0.67 (0.21, 1.91)	1.27 (0.29, 5.22)	1.11 (0.44, 2.51)	1.79 (0.70, 4.11)	1.45 (0.47, 3.91)	2.26 (0.94, 5.00)	**2.94 (1.02, 7.74)**	1.47 (0.64, 3.07)	**IPI/CT**	1.20 (0.67, 2.12)	1.07 (0.72, 1.58)	**2.39 (1.59, 3.60)**	**1.97 (1.51, 2.58)**
0.60 (0.22, 1.69)	0.52 (0.19, 1.46)	0.38 (0.11, 1.25)	0.39 (0.11, 1.33)	0.37 (0.11, 1.25)	0.70 (0.15, 3.35)	0.61 (0.22, 1.69)	0.99 (0.35, 2.75)	0.80 (0.25, 2.58)	1.25 (0.47, 3.39)	1.62 (0.51, 5.16)	0.81 (0.31, 2.09)	0.55 (0.20, 1.73)	**CAM/CT**	0.89 (0.51, 1.54)	**1.99 (1.34, 2.99)**	1.65 (0.99, 2.75)
0.62 (0.31, 1.26)	0.54 (0.27, 1.08)	**0.39 (0.15, 0.97)**	0.41 (0.16, 1.05)	**0.38 (0.15, 0.99)**	0.73 (0.19, 2.82)	0.64 (0.33, 1.24)	1.03 (0.52, 2.03)	0.84 (0.34, 2.03)	1.30 (0.69, 2.41)	1.69 (0.80, 3.54)	0.84 (0.51, 1.38)	0.57 (0.28, 1.29)	1.04 (0.41, 2.62)	**CT/BEV**	**2.24 (1.55, 3.24)**	**1.85 (1.39, 2.47)**
															CT/Mpem	0.83 (0.61, 1.12)
1.40 (0.78, 2.60)	1.21 (0.69, 2.20)	0.89 (0.38, 2.05)	0.91 (0.38, 2.21)	0.87 (0.36, 2.08)	1.65 (0.45, 6.06)	1.44 (0.82, 2.52)	**2.32 (1.30, 4.14)**	1.88 (0.82, 4.27)	**2.93 (1.76, 4.91)**	**3.80 (1.79, 8.26)**	**1.90 (1.25, 2.92)**	1.30 (0.70, 2.64)	**2.34 (1.01, 5.49)**	**2.25 (1.57, 3.27)**		**CT**

Figure S3. Network Meta-analysis of Comparisons on Objective Response Rate and Adverse Events of Grade 3 or Higher in patients with advanced wild-type NSCLC

(A) Network diagrams of comparisons on objective response rate (ORR). (B) Network diagrams of comparisons on adverse events of grade 3 or higher (>= 3 AEs). Each circular node represents a type of treatment. Each line represents a type of head-to-head comparison. The size of the nodes and the thickness of the lines are weighted according to the number of studies evaluating each treatment and direct comparison, respectively. The total number of patients receiving a treatment was shown in brackets. (C) Pooled estimates of the network meta-analysis of ORR and >= 3 AEs. Data in each cell are OR (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. OR more than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab.

	NO. of studies	Sample size	PEM	NIV	DUR	DUR/TRE	DUR/TRE/ CT	NIV/IPI	NIV/CT	NIV/IPI/ CT	PEM/CT	ATE/CT/ BEV	ATE/CT	IPI/CT	CAM/CT	CT/BEV	CT/BEV/ Mpem	CT/Mpem	CT
Neutropenia	23	14148	1.3	0.0	2.7	1.2	35.9	0.0	9.7	6.7	17.5	13.7	32.1	12.5	38.1	32.4	22.2	18.8	15.2
Febrile neutropenia	11	6400	0.0	NA	NA	NA	NA	NA	NA	3.9	1.7	9.2	NA	2.8	NA	4.5	1.2	0.0	1.9
Thrombocytopenia	22	12439	0.0	0.4	0.0	0.6	18.2	NA	NA	2.8	7.4	4.1	9.5	6.8	16.6	7.4	18.2	12.5	5.8
Anaemia	22	13971	0.0	0.4	0.0	1.7	23.6	1.4	12.4	5.9	16.7	6.1	29.2	11.4	18.5	4.4	12.2	14.6	12.2
Nausea	15	10000	0.0	0.4	0.0	0.6	4.1	0.5	1.5	NA	2.4	3.8	2.7	0.7	1.0	2.0	NA	2.1	1.4
Vomiting	16	10087	0.0	0.0	0.0	0.4	2.7	0.3	2.3	NA	2.4	1.5	1.9	0.9	1.0	4.5	NA	2.1	1.6
Diarrhoea	15	9862	0.0	1.1	0.5	3.8	4.7	1.7	1.1	NA	4.3	2.8	4.9	7.3	NA	0.6	NA	2.7	1.4
Constipation	13	9119	0.0	0.0	0.0	0.0	1.4	0.0	0.4	NA	0.8	0.0	0.2	NA	0.0	0.0	NA	0.4	0.3
Colitis	12	8104	0.0	NA	0.3	4.6	6.8	NA	NA	NA	1.6	1.8	0.9	2.3	NA	0.5	NA	0.0	0.2
Abnormal liver function	16	9352	0.0	2.6	0.3	2.3	4.1	NA	NA	NA	1.3	5.1	1.5	1.1	5.4	1.4	NA	0.8	0.3
Hypertension	8	4228	NA	NA	NA	NA	NA	NA	NA	NA	NA	6.4	NA	NA	NA	5.6	6.2	0.0	0.9
Fatigue	18	11307	0.0	1.1	1.6	1.5	4.7	1.7	3.5	NA	5.3	3.3	6.1	3.7	3.4	3.6	9.2	2.1	2.7
Asthenia	13	9767	0.0	0.0	0.3	0.0	NA	1.4	0.9	NA	4.8	1.3	1.9	2.0	NA	3.3	0.0	3.5	1.7
Alopecia	12	7695	0.0	0.0	0.0	0.0	0.0	NA	0.5	0.8	0.3	0.0	0.0	0.0	NA	0.0	NA	0.0	0.5
Rash	17	11030	0.0	0.8	0.8	0.6	4.7	1.6	0.9	NA	1.9	2.3	2.8	2.4	1.0	1.5	NA	1.5	0.3
Pruritus	11	6390	0.0	0.0	0.0	0.4	1.4	0.5	0.0	NA	0.0	NA	NA	0.9	0.0	NA	NA	0.0	0.1
Pneumonia	12	8104	0.0	NA	1.4	2.9	2.0	NA	NA	NA	2.7	1.8	1.3	1.5	NA	0.8	NA	1.3	0.5
Hypothyroidism	12	7560	0.0	NA	0.5	0.8	0.7	NA	NA	NA	0.4	0.3	0.5	NA	0.5	0.0	NA	0.0	0.0
Hyperthyroidism	12	8104	0.0	NA	0.0	0.2	0.0	NA	NA	NA	0.1	0.3	0.2	0.0	NA	0.0	NA	0.0	0.0
Thrombogenesis	6	3087	NA	NA	NA	NA	NA	NA	NA	NA	NA	1.8	NA	NA	NA	3.1	2.8	0.0	6.0
Peripheral neurotoxicity	11	6410	0.0	0.0	NA	1.3	0.0	NA	NA	0.0	1.1	2.8	NA	0.7	NA	3.7	0.0	0.0	0.7
Pancreatitis	8	6008	0.1	NA	0.3	1.2	1.4	NA	NA	NA	0.5	0.5	0.3	NA	NA	0.0	NA	0.2	0.0
Nephritis	6	4599	0.3	NA	0.0	0.0	NA	NA	NA	NA	1.2	0.3	NA	NA	NA	0.0	NA	0.0	0.1
Adrenal insufficiency	7	5888	0.3	NA	0.3	1.0	0.7	NA	NA	NA	0.2	0.3	0.0	0.5	NA	0.3	NA	0.5	0.0
Diabetes	6	4163	0.6	NA	0.0	0.3	NA	NA	NA	NA	0.5	0.0	0.6	NA	NA	0.0	NA	0.0	0.1
Hypophysitis	6	4599	0.5	NA	0.0	0.3	NA	NA	NA	NA	0.3	0.3	NA	NA	NA	0.0	NA	0.0	0.0

Figure S4. A Frequency Toxicity Profile in Relation to the Incidence (%) of Each Specific Grade >=3 Adverse Events Based on the Population of Each Treatment We Included

NA=not applicable; PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

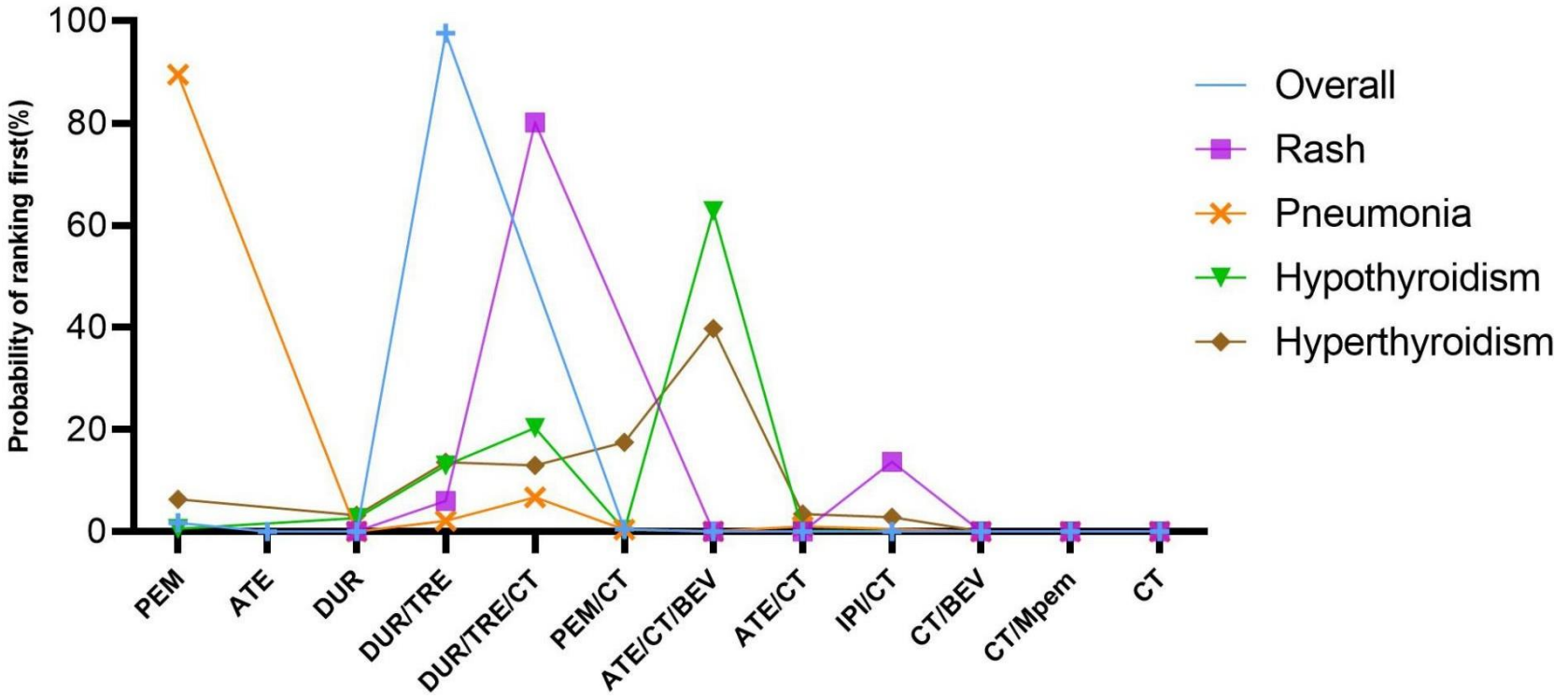


Figure S5. Relative Toxicity of Comparable Treatments on Overall and Four Common Immune-related Adverse Events (AEs) of Any Grade
PEM=pembrolizumab; ATE=atezolizumab; DUR=durvalumab; TRE=tremelimumab; IPI= ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

PFS

OS

PEM/ pem-CT	**0.52 (0.38, 0.72)**	0.81 (0.63, 1.05)	**0.39 (0.23, 0.67)**	0.80 (0.58, 1.11)	0.87 (0.62, 1.23)	**0.52 (0.38, 0.69)**	**0.36 (0.24, 0.55)**	**0.62 (0.45, 0.86)**	**0.49 (0.41, 0.58)**	**0.32 (0.23, 0.44)**	**0.29 (0.22, 0.38)**
0.67 (0.47, 0.96)	**ATE/ pac-CT**	**1.55 (1.11, 2.17)**	0.75 (0.48, 1.18)	**1.52 (1.03, 2.25)**	**1.67 (1.35, 2.07)**	0.99 (0.86, 1.13)	**0.69 (0.49, 0.96)**	1.19 (0.97, 1.45)	0.93 (0.71, 1.22)	**0.61 (0.53, 0.70)**	**0.55 (0.38, 0.78)**
0.69 (0.50, 0.94)	1.02 (0.71, 1.47)	**ATE/ pem-CT**	**0.48 (0.28, 0.83)**	0.98 (0.70, 1.38)	1.08 (0.76, 1.53)	**0.64 (0.47, 0.86)**	**0.44 (0.29, 0.68)**	0.77 (0.54, 1.08)	**0.60 (0.49, 0.73)**	**0.39 (0.28, 0.54)**	**0.35 (0.26, 0.48)**
0.42 (0.23, 0.74)	**0.62 (0.38, 1.00)**	0.61 (0.34, 1.09)	**IPI/ pac-CT**	**2.03 (1.14, 3.61)**	**2.22 (1.38, 3.59)**	1.31 (0.84, 2.05)	0.92 (0.53, 1.57)	1.58 (0.99, 2.54)	1.24 (0.75, 2.06)	0.81 (0.53, 1.25)	0.73 (0.42, 1.27)
0.78 (0.53, 1.15)	1.15 (0.75, 1.78)	1.13 (0.75, 1.69)	1.87 (0.99, 3.51)	**CAM/ pem-CT**	1.10 (0.73, 1.64)	**0.65 (0.45, 0.93)**	**0.45 (0.28, 0.73)**	0.78 (0.53, 1.16)	**0.61 (0.46, 0.80)**	**0.40 (0.27, 0.58)**	**0.36 (0.25, 0.51)**
0.75 (0.52, 1.08)	1.11 (0.89, 1.39)	1.09 (0.74, 1.59)	**1.80 (1.08, 2.98)**	0.96 (0.62, 1.51)	**ATE/pac-CT/ BEV**	**0.59 (0.50, 0.70)**	**0.41 (0.29, 0.58)**	**0.71 (0.57, 0.89)**	**0.56 (0.42, 0.74)**	**0.36 (0.30, 0.45)**	**0.33 (0.23, 0.47)**
0.60 (0.43, 0.83)	0.89 (0.77, 1.02)	0.87 (0.62, 1.22)	1.44 (0.89, 2.32)	0.77 (0.51, 1.16)	**0.80 (0.67, 0.95)**	**pac-CT/ BEV**	**0.70 (0.52, 0.94)**	**1.20 (1.04, 1.40)**	0.94 (0.74, 1.20)	**0.62 (0.55, 0.69)**	**0.56 (0.40, 0.77)**
0.53 (0.33, 0.85)	0.78 (0.53, 1.15)	0.77 (0.47, 1.25)	1.27 (0.70, 2.29)	0.68 (0.39, 1.17)	0.70 (0.47, 1.05)	0.88 (0.62, 1.26)	**pem-CT/ BEV**	**1.72 (1.33, 2.24)**	1.35 (0.92, 1.99)	0.88 (0.64, 1.22)	0.80 (0.51, 1.24)
0.60 (0.42, 0.85)	0.89 (0.72, 1.09)	0.87 (0.60, 1.26)	1.44 (0.87, 2.37)	0.77 (0.50, 1.20)	0.80 (0.64, 1.01)	1.00 (0.86, 1.16)	1.14 (0.82, 1.57)	**pem-CT/ BEV/Mpem**	0.78 (0.59, 1.04)	**0.51 (0.42, 0.62)**	**0.46 (0.32, 0.66)**
0.56 (0.46, 0.69)	0.83 (0.62, 1.10)	0.81 (0.64, 1.03)	1.34 (0.79, 2.30)	**0.72 (0.52, 1.00)**	0.75 (0.55, 1.01)	0.93 (0.73, 1.20)	1.06 (0.69, 1.64)	0.93 (0.70, 1.25)	**pem-CT/ Mpem**	**0.65 (0.50, 0.85)**	**0.59 (0.47, 0.74)**
0.49 (0.35, 0.69)	**0.72 (0.62, 0.84)**	0.71 (0.49, 1.01)	1.17 (0.74, 1.86)	**0.63 (0.41, 0.96)**	**0.65 (0.53, 0.80)**	**0.81 (0.72, 0.92)**	0.92 (0.64, 1.35)	**0.81 (0.67, 0.99)**	0.87 (0.66, 1.15)	**pac-CT**	0.90 (0.64, 1.28)
0.44 (0.33, 0.58)	**0.65 (0.46, 0.92)**	**0.63 (0.47, 0.86)**	1.05 (0.59, 1.86)	**0.56 (0.38, 0.83)**	**0.58 (0.41, 0.84)**	**0.73 (0.53, 1.00)**	0.83 (0.51, 1.34)	0.73 (0.51, 1.04)	**0.78 (0.64, 0.95)**	0.90 (0.64, 1.26)	**pem-CT**

Figure S6. Pooled Hazard Ratio (95% Credible Intervals) for Different Treatment Strategies Containing Either Paclitaxel- (pac-CT) or Pemetrexed-based (pem-CT) Chemotherapy in Non-Squamous Subgroup

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratios less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. OS=overall survival; PFS=progression-free survival; PEM=pembrolizumab; ATE=atezolizumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; Mpem=maintenance of pemetrexed.

A — PD-L1>=1% or TC/IC=1/2/3

PD-L1<1% or TC/IC=0

PEM	0.97 (0.79, 1.18)	**1.36 (1.08, 1.72)**	**1.28 (1.04, 1.58)**		**2.27 (1.81, 2.83)**	**1.70 (1.39, 2.07)**	**1.81 (1.22, 2.69)**	1.05 (0.93, 1.19)
	NIV	**1.41 (1.10, 1.82)**	**1.33 (1.05, 1.67)**		**2.35 (1.84, 2.99)**	**1.76 (1.41, 2.19)**	**1.87 (1.25, 2.82)**	1.09 (0.93, 1.27)
		ATE	0.94 (0.72, 1.22)		**1.66 (1.26, 2.18)**	1.25 (0.97, 1.60)	1.33 (0.87, 2.03)	**0.77 (0.63, 0.94)**
			NIV/IPI		**1.77 (1.38, 2.28)**	**1.33 (1.05, 1.67)**	1.41 (0.94, 2.14)	**0.82 (0.69, 0.97)**
			1.03 (0.72, 1.47)	**NIV/CT**				
			1.12 (0.81, 1.56)	1.09 (0.77, 1.54)	**PEM/CT**	**0.75 (0.59, 0.96)**	0.80 (0.53, 1.22)	**0.46 (0.39, 0.56)**
			1.08 (0.81, 1.44)	1.05 (0.77, 1.43)	0.96 (0.74, 1.26)	**ATE/CT**	1.07 (0.71, 1.60)	**0.62 (0.53, 0.72)**
			1.01 (0.57, 1.79)	0.99 (0.55, 1.76)	0.91 (0.52, 1.58)	0.94 (0.55, 1.60)	**CAM/CT**	**0.58 (0.40, 0.84)**
			0.75 (0.59, 0.96)	**0.73 (0.56, 0.95)**	**0.67 (0.54, 0.84)**	**0.70 (0.60, 0.81)**	0.74 (0.44, 1.24)	**CT**

PFS

B — PD-L1>=50% or TC/IC=3

PD-L1 1-49% or TC/IC=1/2

PEM	0.85 (0.66, 1.10)	1.15 (0.79, 1.66)	1.17 (0.88, 1.55)	**1.89 (1.40, 2.55)**	**1.52 (1.09, 2.11)**	1.85 (0.68, 5.04)	**0.72 (0.62, 0.85)**
	NIV	1.35 (0.91, 1.98)	**1.37 (1.01, 1.86)**	**2.22 (1.61, 3.05)**	**1.78 (1.26, 2.52)**	2.18 (0.80, 5.94)	0.85 (0.70, 1.03)
1.41 (1.05, 1.89)		**ATE**	1.02 (0.67, 1.53)	**1.65 (1.08, 2.51)**	1.32 (0.85, 2.05)	1.62 (0.57, 4.58)	**0.63 (0.45, 0.88)**
			NIV/IPI	**1.62 (1.14, 2.30)**	1.30 (0.90, 1.89)	1.59 (0.58, 4.39)	**0.62 (0.49, 0.79)**
2.42 (1.83, 3.21)		**1.72 (1.23, 2.39)**		**PEM/CT**	0.80 (0.55, 1.18)	0.98 (0.35, 2.72)	**0.38 (0.30, 0.49)**
1.83 (1.42, 2.34)		1.29 (0.96, 1.76)		0.75 (0.56, 1.01)	**ATE/CT**	1.22 (0.44, 3.41)	**0.48 (0.36, 0.64)**
						CAM/CT	0.39 (0.15, 1.05)
1.27 (1.08, 1.50)		0.90 (0.71, 1.15)		**0.52 (0.42, 0.66)**	**0.70 (0.58, 0.84)**		**CT**

PFS

C **PD-L1<1%**

PD-L1>=1%

PEM								
0.87 (0.72, 1.05)	**NIV**							
0.93 (0.74, 1.18)	1.07 (0.85, 1.35)	**DUR**	**1.62 (1.00, 2.61)**	**1.84 (1.25, 2.73)**	**1.53 (1.07, 2.20)**	**1.90 (1.22, 2.99)**	**1.86 (1.25, 2.78)**	1.18 (0.86, 1.62)
0.81 (0.64, 1.02)	0.93 (0.74, 1.17)	0.87 (0.67, 1.14)	**DUR/TRE**	1.14 (0.75, 1.74)	0.95 (0.64, 1.41)	1.18 (0.73, 1.90)	1.15 (0.75, 1.77)	0.73 (0.51, 1.04)
1.04 (0.84, 1.29)	1.20 (0.97, 1.47)	1.11 (0.87, 1.44)	**1.28 (1.00, 1.64)**	**NIV/IPI**	0.83 (0.62, 1.11)	1.03 (0.70, 1.53)	1.01 (0.72, 1.41)	**0.64 (0.51, 0.81)**
1.00 (0.76, 1.31)	1.15 (0.88, 1.5)	1.07 (0.79, 1.45)	1.23 (0.91, 1.66)	0.96 (0.72, 1.28)	**NIV/CT**	1.24 (0.87, 1.79)	1.22 (0.91, 1.64)	**0.77 (0.65, 0.92)**
1.28 (0.97, 1.70)	**1.47 (1.11, 1.95)**	**1.37 (1.01, 1.88)**	**1.58 (1.16, 2.15)**	1.23 (0.92, 1.66)	1.28 (0.91, 1.80)	**NIV/IPI/CT**	0.98 (0.66, 1.46)	**0.62 (0.45, 0.85)**
1.50 (1.14, 1.98)	**1.73 (1.31, 2.28)**	**1.61 (1.18, 2.20)**	**1.85 (1.36, 2.51)**	**1.45 (1.08, 1.94)**	**1.50 (1.07, 2.11)**	1.17 (0.83, 1.66)	**PEM/CT**	**0.63 (0.50, 0.81)**
0.82 (0.72, 0.94)	0.94 (0.83, 1.08)	0.88 (0.73, 1.07)	1.01 (0.84, 1.22)	**0.79 (0.67, 0.93)**	0.82 (0.65, 1.04)	**0.64 (0.50, 0.82)**	**0.55 (0.43, 0.70)**	**CT**

OS

D **PD-L1:1-49%**

PD-L1>=50%

PEM	1.17 (0.71, 1.91)		0.96 (0.61, 1.52)		0.97 (0.73, 1.29)		**1.49 (1.03, 2.15)**	**1.46 (1.08, 1.99)**	0.91 (0.76, 1.08)
0.83 (0.65, 1.07)	**NIV**								
0.90 (0.63, 1.28)	1.08 (0.74, 1.57)	**DUR**	0.82 (0.44, 1.54)		0.83 (0.50, 1.39)		1.28 (0.73, 2.25)	1.26 (0.74, 2.12)	0.78 (0.49, 1.24)
0.89 (0.62, 1.27)	1.07 (0.73, 1.56)	0.99 (0.63, 1.56)	**DUR/TRE**		1.01 (0.62, 1.64)		1.56 (0.91, 2.65)	1.53 (0.93, 2.51)	0.95 (0.62, 1.45)
1.38 (0.76, 2.52)	1.66 (0.90, 3.07)	1.54 (0.80, 2.98)	1.56 (0.97, 2.51)	**DUR/TRE/CT**					
0.97 (0.73, 1.31)	1.17 (0.86, 1.61)	1.09 (0.73, 1.63)	1.10 (0.73, 1.65)	0.71 (0.38, 1.32)	**NIV/IPI**		**1.54 (1.04, 2.29)**	**1.51 (1.08, 2.12)**	0.94 (0.75, 1.18)
1.02 (0.60, 1.72)	1.22 (0.72, 2.09)	1.13 (0.63, 2.05)	1.15 (0.63, 2.08)	0.74 (0.34, 1.58)	1.04 (0.60, 1.82)	**NIV/CT**			
1.03 (0.67, 1.60)	1.24 (0.79, 1.95)	1.15 (0.69, 1.93)	1.17 (0.69, 1.96)	0.75 (0.37, 1.51)	1.06 (0.66, 1.70)	1.02 (0.53, 1.93)	**NIV/IPI/CT**	0.98 (0.65, 1.48)	**0.61 (0.44, 0.84)**
1.01 (0.73, 1.40)	1.22 (0.86, 1.72)	1.13 (0.74, 1.73)	1.15 (0.74, 1.76)	0.73 (0.39, 1.40)	1.04 (0.71, 1.52)	1.00 (0.56, 1.77)	0.98 (0.60, 1.61)	**PEM/CT**	**0.62 (0.48, 0.80)**
0.68 (0.58, 0.80)	**0.82 (0.67, 1.00)**	0.76 (0.55, 1.05)	0.77 (0.56, 1.06)	**0.49 (0.28, 0.88)**	**0.70 (0.55, 0.90)**	0.67 (0.41, 1.10)	**0.66 (0.44, 0.99)**	**0.67 (0.51, 0.89)**	**CT**

OS

E **PD-L1<1%**

PD-L1>=1%

PEM						
0.99 (0.83, 1.19)	**NIV**					
1.28 (1.04, 1.58)	**1.29 (1.04, 1.60)**	**NIV/IPI**	1.03 (0.72, 1.47)	1.12 (0.80, 1.55)	1.01 (0.57, 1.78)	**0.75 (0.59, 0.96)**
			NIV/CT	1.09 (0.77, 1.54)	0.99 (0.55, 1.76)	**0.73 (0.56, 0.95)**
2.27 (1.81, 2.83)	**2.28 (1.82, 2.87)**	**1.77 (1.37, 2.28)**		**PEM/CT**	0.91 (0.52, 1.58)	**0.67 (0.54, 0.84)**
1.81 (1.22, 2.69)	**1.83 (1.23, 2.72)**	1.41 (0.94, 2.14)		0.80 (0.53, 1.22)	**CAM/CT**	0.74 (0.44, 1.24)
1.05 (0.93, 1.19)	1.06 (0.93, 1.20)	**0.82 (0.69, 0.97)**		**0.46 (0.38, 0.56)**	**0.58 (0.40, 0.85)**	**CT**

PFS

F **PD-L1:1-49%**

PD-L1>=50%

PEM			**2.42 (1.83, 3.20)**		**1.27 (1.08, 1.50)**
0.85 (0.66, 1.10)	**NIV**				
1.17 (0.88, 1.55)	**1.37 (1.01, 1.86)**	**NIV/IPI**			
1.89 (1.40, 2.55)	**2.22 (1.61, 3.05)**	**1.62 (1.14, 2.30)**	**PEM/CT**		**0.52 (0.42, 0.66)**
1.85 (0.68, 5.04)	2.17 (0.79, 5.94)	1.59 (0.57, 4.39)	0.98 (0.35, 2.73)	**CAM/CT**	
0.72 (0.62, 0.85)	0.85 (0.70, 1.03)	**0.62 (0.49, 0.79)**	**0.38 (0.30, 0.49)**	0.39 (0.15, 1.05)	**CT**

PFS

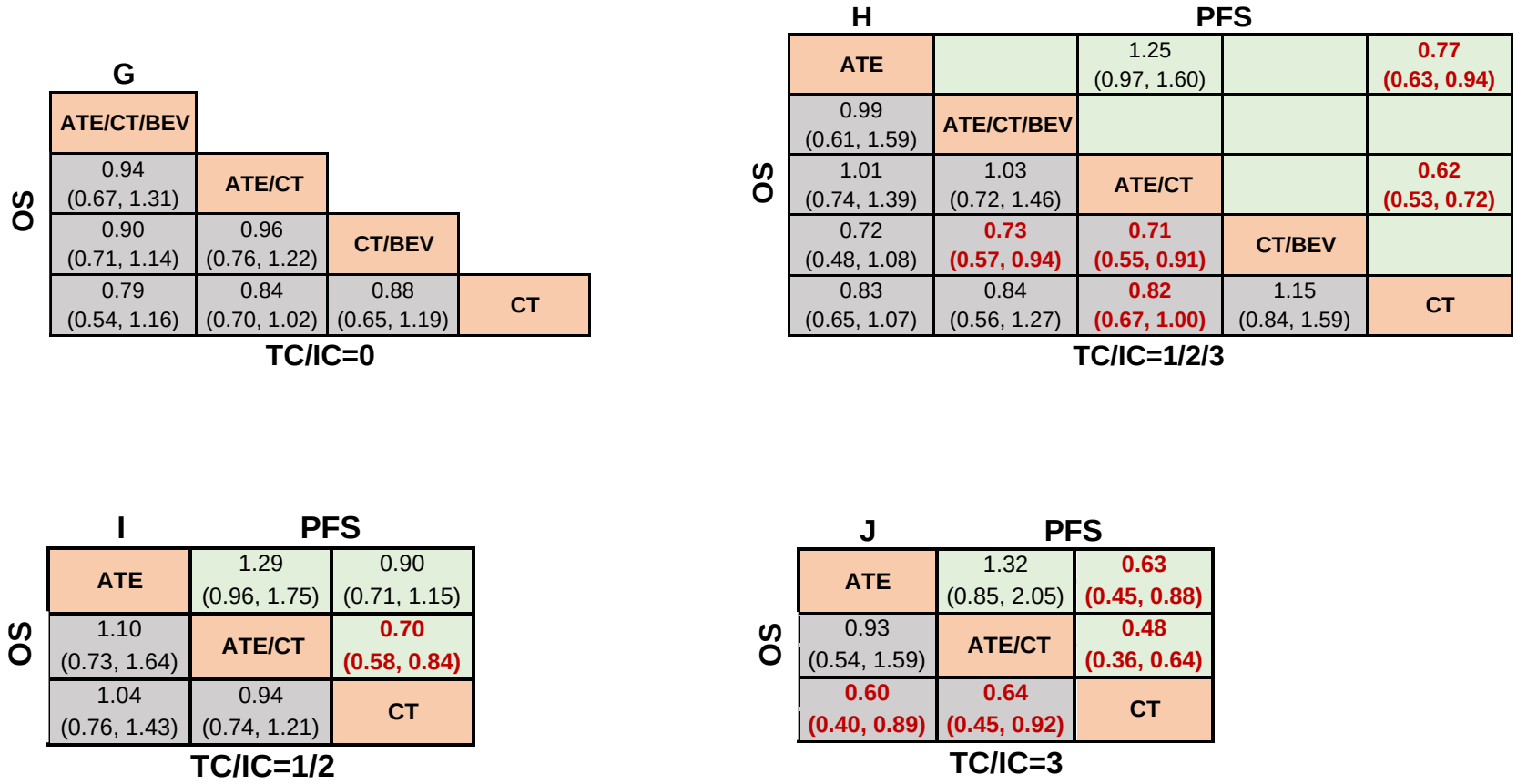


Figure S7. Pooled Estimates of the Network Meta-analysis According to Programmed-death Ligand 1 (PD-L1) or TC/IC Levels

(A and B) Pooled hazard ratio (HR) (95% CrIs (credible intervals)) for progression-free survival (PFS) in patients with PD-L1<1% (or TC/IC=0), PD-L1 >=1% (or TC/IC=1/2/3), PD-L1 1-49% (or TC/IC=1/2), or PD-L1>=50% (or TC/IC=3). (C, D, E, and F) Pooled HR (95% CrIs) for overall survival (OS) and PFS in patients with PD-L1<1%, PD-L1 >=1%, PD-L1 1-49%, or PD-L1>=50%. (G, H, I, and J) Pooled HR (95% CrIs) for OS and PFS in patients with TC/IC=0, TC/IC=1/2/3, TC/IC=1/2, or TC/IC=3. Data in each cell are HR (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. HR less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. TC=tumor cells; IC=tumor-infiltrating immune cells. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; CAM=camrelizumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy. CT followed by maintenance of pemetrexed (CT/Mpem) was also classified into CT.

A

Low TMB

High TMB \ OS	PEM	NIV	DUR	DUR/TRE	DUR/TRE/CT	NIV/IPI	PEM/CT	CT
PEM		1.10 (0.74, 1.65)	1.17 (0.86, 1.60)	0.94 (0.69, 1.28)	1.15 (0.70, 1.89)	**1.45 (1.06, 2.00)**	**1.46 (1.02, 2.10)**	1.09 (0.88, 1.35)
NIV	0.56 (0.31, 1.02)		1.06 (0.71, 1.60)	0.85 (0.57, 1.28)	1.04 (0.59, 1.83)	1.32 (0.87, 1.99)	1.33 (0.85, 2.07)	0.99 (0.71, 1.39)
DUR	0.86 (0.55, 1.35)	1.53 (0.79, 2.93)		0.80 (0.58, 1.10)	0.98 (0.59, 1.61)	1.24 (0.90, 1.71)	1.25 (0.86, 1.80)	0.93 (0.74, 1.16)
DUR/TRE	1.26 (0.78, 2.07)	**2.24 (1.13, 4.45)**	1.47 (0.84, 2.57)		1.22 (0.83, 1.80)	**1.55 (1.12, 2.13)**	**1.55 (1.08, 2.24)**	1.16 (0.93, 1.45)
DUR/TRE/CT	1.21 (0.45, 3.23)	2.13 (0.71, 6.38)	1.40 (0.50, 3.90)	0.95 (0.40, 2.24)		1.27 (0.76, 2.10)	1.28 (0.75, 2.18)	0.95 (0.61, 1.49)
NIV/IPI	0.91 (0.62, 1.34)	1.62 (0.88, 2.98)	1.06 (0.66, 1.70)	0.72 (0.43, 1.20)	0.76 (0.28, 2.06)		1.00 (0.69, 1.46)	**0.75 (0.59, 0.95)**
PEM/CT	0.88 (0.59, 1.32)	1.56 (0.84, 2.91)	1.02 (0.63, 1.66)	0.70 (0.41, 1.17)	0.73 (0.27, 2.00)	0.97 (0.63, 1.48)		**0.75 (0.56, 1.00)**
CT	**0.62 (0.48, 0.80)**	1.10 (0.64, 1.88)	0.72 (0.50, 1.04)	**0.49 (0.32, 0.74)**	0.51 (0.20, 1.34)	**0.68 (0.51, 0.91)**	**0.70 (0.52, 0.96)**	

B

Low TMB

High TMB \ PFS	NIV	DUR	DUR/TRE	DUR/TRE/CT	NIV/IPI	PEM/CT	CT
NIV		**1.53 (1.01, 2.30)**	1.17 (0.78, 1.76)	**1.99 (1.17, 3.39)**	**1.70 (1.13, 2.57)**	**3.06 (2.00, 4.66)**	**1.82 (1.30, 2.55)**
DUR	0.80 (0.43, 1.50)		0.77 (0.55, 1.06)	1.30 (0.81, 2.09)	1.11 (0.80, 1.55)	**2.00 (1.41, 2.83)**	1.19 (0.94, 1.50)
DUR/TRE	1.17 (0.61, 2.24)	1.45 (0.81, 2.60)		**1.69 (1.20, 2.40)**	**1.45 (1.04, 2.01)**	**2.60 (1.85, 3.66)**	**1.55 (1.23, 1.95)**
DUR/TRE/CT	1.33 (0.55, 3.18)	1.65 (0.73, 3.76)	1.14 (0.63, 2.03)		0.85 (0.53, 1.38)	1.54 (0.94, 2.50)	0.91 (0.60, 1.39)
NIV/IPI	1.07 (0.59, 1.92)	1.33 (0.79, 2.22)	0.91 (0.53, 1.58)	0.80 (0.36, 1.78)		**1.80 (1.27, 2.54)**	1.07 (0.84, 1.36)
PEM/CT	1.35 (0.77, 2.35)	**1.68 (1.04, 2.69)**	1.15 (0.69, 1.92)	1.01 (0.47, 2.20)	1.26 (0.82, 1.95)		**0.60 (0.46, 0.77)**
CT	0.62 (0.38, 1.01)	0.77 (0.52, 1.14)	**0.53 (0.34, 0.82)**	**0.47 (0.23, 0.96)**	**0.58 (0.41, 0.82)**	**0.46 (0.35, 0.60)**	

C

Low tTMB

High tTMB \ OS	PEM	NIV	DUR	DUR/TRE	NIV/IPI	PEM/CT	CT
PEM		1.10 (0.74, 1.65)	0.86 (0.58, 1.30)	0.78 (0.53, 1.16)	**1.45 (1.06, 2.00)**	**1.46 (1.02, 2.10)**	1.09 (0.88, 1.35)
NIV	0.56 (0.31, 1.02)		0.79 (0.49, 1.27)	0.71 (0.45, 1.14)	1.32 (0.88, 1.99)	1.33 (0.85, 2.07)	0.99 (0.71, 1.39)
DUR	0.89 (0.55, 1.43)	1.57 (0.80, 3.08)		0.91 (0.57, 1.45)	**1.68 (1.11, 2.53)**	**1.69 (1.08, 2.64)**	1.26 (0.90, 1.77)
DUR/TRE	0.86 (0.53, 1.40)	1.53 (0.78, 3.01)	0.97 (0.55, 1.73)		**1.85 (1.24, 2.76)**	**1.86 (1.20, 2.88)**	**1.39 (1.00, 1.92)**
NIV/IPI	0.91 (0.62, 1.34)	1.62 (0.88, 2.98)	1.03 (0.63, 1.69)	1.06 (0.64, 1.75)		1.00 (0.69, 1.46)	**0.75 (0.59, 0.95)**
PEM/CT	0.88 (0.59, 1.32)	1.56 (0.84, 2.91)	1.00 (0.60, 1.66)	1.02 (0.61, 1.71)	0.97 (0.63, 1.48)		**0.75 (0.56, 1.00)**
CT	**0.62 (0.48, 0.80)**	1.10 (0.64, 1.88)	0.70 (0.47, 1.05)	0.72 (0.48, 1.08)	**0.68 (0.51, 0.91)**	**0.70 (0.52, 0.96)**	

D

Low tTMB

High tTMB \ PFS	NIV	DUR	DUR/TRE	NIV/IPI	PEM/CT	CT
NIV		1.22 (0.75, 1.99)	0.92 (0.57, 1.48)	**1.70 (1.13, 2.57)**	**3.06 (2.01, 4.66)**	**1.82 (1.30, 2.55)**
DUR	0.72 (0.37, 1.39)		0.75 (0.46, 1.22)	1.39 (0.91, 2.13)	**2.50 (1.62, 3.87)**	**1.49 (1.05, 2.12)**
DUR/TRE	0.64 (0.33, 1.22)	0.89 (0.48, 1.64)		**1.85 (1.23, 2.79)**	**3.32 (2.18, 5.07)**	**1.98 (1.42, 2.77)**
NIV/IPI	1.07 (0.59, 1.93)	1.48 (0.85, 2.58)	1.67 (0.97, 2.89)		**1.80 (1.27, 2.55)**	1.07 (0.84, 1.36)
PEM/CT	1.35 (0.77, 2.35)	**1.87 (1.11, 3.14)**	**2.11 (1.27, 3.51)**	1.26 (0.82, 1.95)		**0.60 (0.46, 0.77)**
CT	0.62 (0.38, 1.01)	0.86 (0.55, 1.34)	0.97 (0.63, 1.49)	**0.58 (0.41, 0.82)**	**0.46 (0.35, 0.60)**	

E

High bTMB \ Low bTMB				
DUR	0.80 (0.58, 1.10)	0.98 (0.59, 1.61)	1.45 (0.88, 2.38)	0.93 (0.74, 1.17)
1.47 (0.84, 2.57)	**DUR/TRE**	1.22 (0.83, 1.80)	**1.81 (1.11, 2.96)**	1.16 (0.93, 1.45)
1.40 (0.50, 3.88)	0.95 (0.41, 2.24)	**NIV/IPI**	1.48 (0.79, 2.78)	0.95 (0.61, 1.49)
1.18 (0.61, 2.27)	0.80 (0.41, 1.59)	0.84 (0.28, 2.52)	**PEM/CT**	**0.64 (0.41, 1.00)**
0.72 (0.50, 1.04)	**0.49 (0.32, 0.75)**	0.51 (0.20, 1.33)	0.61 (0.36, 1.05)	**CT**

OS

F

High bTMB \ Low bTMB				
DUR	0.77 (0.55, 1.06)	1.30 (0.81, 2.09)	**2.38 (1.52, 3.74)**	1.19 (0.94, 1.50)
1.45 (0.81, 2.60)	**DUR/TRE**	**1.70 (1.20, 2.40)**	**3.10 (1.98, 4.82)**	**1.55 (1.23, 1.95)**
1.65 (0.72, 3.77)	1.14 (0.63, 2.04)	**DUR/TRE/CT**	**1.82 (1.04, 3.21)**	0.91 (0.60, 1.38)
2.20 (1.17, 4.14)	1.51 (0.78, 2.93)	1.33 (0.55, 3.21)	**PEM/CT**	**0.50 (0.34, 0.74)**
0.77 (0.52, 1.14)	**0.53 (0.34, 0.82)**	**0.47 (0.23, 0.97)**	**0.35 (0.21, 0.58)**	CT

PFS

Figure S8. Pooled Estimates of the Network Meta-analysis According to Tissue (tTMB) or Blood (bTMB) Tumor Mutation Burden
(A) and (B) Pooled hazard ratio (HR) (95% CrIs (credible intervals)) for overall survival (OS) and progression-free survival (PFS) according to either tTMB or bTMB. (C and D) Pooled HR (95% CrIs) for OS and PFS in patients with high or low tTMB. (E and F) Pooled HR (95% CrIs) for OS and PFS in patients with high or low bTMB. Data in each cell are HR (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. HR less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CT=chemotherapy. CT followed by maintenance of pemetrexed (CT/Mpem) was classified into CT.

A

Never smoking

Ever smoking

PEM	0.98 (0.48, 1.99)	0.55 (0.18, 1.66)	1.66 (0.80, 3.48)	1.05 (0.63, 1.72)	0.88 (0.51, 1.55)	0.87 (0.46, 1.66)	**5.25 (1.64, 16.82)**	1.56 (0.88, 2.78)	0.84 (0.46, 1.53)	1.40 (0.91, 2.14)	1.94 (0.98, 3.84)	1.21 (0.54, 2.70)	1.00 (0.73, 1.36)
0.71 (0.54, 0.94)	**NIV**	0.56 (0.16, 1.93)	1.70 (0.68, 4.29)	1.07 (0.51, 2.26)	0.90 (0.41, 1.99)	0.89 (0.38, 2.09)	**5.36 (1.48, 19.58)**	1.60 (0.72, 3.56)	0.86 (0.38, 1.94)	1.43 (0.71, 2.89)	1.99 (0.83, 4.80)	1.24 (0.47, 3.29)	1.02 (0.54, 1.93)
1.45 (0.91, 2.33)	**2.04 (1.23, 3.37)**	**ATE**	3.06 (0.87, 10.71)	1.93 (0.62, 5.98)	1.62 (0.51, 5.18)	1.61 (0.48, 5.35)	**9.63 (2.06, 45.19)**	2.87 (0.89, 9.23)	1.54 (0.47, 5.02)	2.57 (0.85, 7.76)	**3.57 (1.05, 12.20)**	2.22 (0.61, 8.11)	1.83 (0.63, 5.32)
1.00 (0.72, 1.38)	1.40 (0.97, 2.02)	0.69 (0.41, 1.17)	**DUR**	0.63 (0.29, 1.36)	0.53 (0.24, 1.20)	0.53 (0.22, 1.26)	3.16 (0.86, 11.64)	0.94 (0.41, 2.14)	0.50 (0.22, 1.17)	0.84 (0.41, 1.74)	1.17 (0.47, 2.88)	0.73 (0.27, 1.97)	0.60 (0.31, 1.17)
1.07 (0.86, 1.33)	**1.50 (1.14, 1.97)**	0.74 (0.46, 1.18)	1.07 (0.78, 1.47)	**NIV/IPI**	0.84 (0.46, 1.54)	0.83 (0.42, 1.64)	**5.00 (1.53, 16.36)**	1.49 (0.80, 2.76)	0.80 (0.42, 1.52)	1.33 (0.83, 2.16)	1.85 (0.91, 3.80)	1.15 (0.50, 2.65)	0.95 (0.65, 1.39)
1.06 (0.82, 1.36)	**1.48 (1.09, 2.00)**	0.73 (0.45, 1.18)	1.06 (0.75, 1.49)	0.99 (0.77, 1.26)	**NIV/CT**	0.99 (0.48, 2.05)	**5.94 (1.76, 20.00)**	1.77 (0.90, 3.46)	0.95 (0.48, 1.90)	1.58 (0.92, 2.74)	**2.20 (1.03, 4.73)**	1.37 (0.57, 3.29)	1.13 (0.71, 1.80)
1.24 (0.95, 1.63)	**1.74 (1.27, 2.39)**	0.86 (0.52, 1.40)	1.24 (0.87, 1.77)	1.16 (0.89, 1.51)	1.18 (0.88, 1.58)	**NIV/IPI/CT**	**6.00 (1.71, 20.98)**	1.78 (0.85, 3.74)	0.96 (0.45, 2.05)	1.60 (0.85, 3.01)	2.22 (0.97, 5.07)	1.38 (0.55, 3.50)	1.14 (0.65, 1.99)
1.31 (0.90, 1.92)	**1.84 (1.22, 2.79)**	0.90 (0.52, 1.59)	1.31 (0.84, 2.05)	1.23 (0.84, 1.79)	1.25 (0.84, 1.85)	1.06 (0.70, 1.59)	**PEM/CT**	**0.30 (0.10, 0.85)**	**0.16 (0.05, 0.55)**	**0.27 (0.09, 0.83)**	0.37 (0.11, 1.30)	**0.23 (0.10, 0.54)**	**0.19 (0.06, 0.58)**
0.90 (0.73, 1.10)	1.26 (0.96, 1.64)	**0.62 (0.39, 0.98)**	0.90 (0.66, 1.23)	0.84 (0.69, 1.03)	0.85 (0.67, 1.08)	**0.72 (0.56, 0.94)**	**0.68 (0.49, 0.96)**	**ATE/CT**	0.54 (0.27, 1.09)	0.90 (0.52, 1.55)	1.25 (0.58, 2.68)	0.77 (0.41, 1.46)	0.64 (0.39, 1.04)
0.88 (0.69, 1.11)	1.23 (0.92, 1.64)	**0.60 (0.37, 0.98)**	0.88 (0.63, 1.22)	0.82 (0.65, 1.03)	0.83. (0.63, 1.09)	**0.71 (0.53, 0.94)**	**0.67 (0.45, 0.98)**	0.98 (0.78, 1.22)	**IPI/CT**	1.67 (0.92, 3.01)	**2.32 (1.05, 5.14)**	1.44 (0.59, 3.56)	1.19 (0.71, 1.99)
0.78 (0.62, 0.98)	1.10 (0.83, 1.46)	**0.54 (0.34, 0.87)**	0.78 (0.57, 1.09)	**0.73 (0.59, 0.92)**	**0.74 (0.57, 0.96)**	**0.63 (0.48, 0.83)**	**0.60 (0.43, 0.84)**	0.87 (0.72, 1.06)	0.90 (0.70, 1.15)	**CT/BEV**	1.39 (0.82, 2.36)	0.86 (0.40, 1.86)	**0.71 (0.53, 0.95)**
0.77 (0.58, 1.02)	1.08 (0.78, 1.49)	**0.53 (0.32, 0.87)**	0.77 (0.53, 1.11)	**0.72 (0.55, 0.95)**	**0.73 (0.54, 0.99)**	**0.62 (0.45, 0.85)**	**0.59 (0.40, 0.85)**	0.86 (0.67, 1.10)	0.88 (0.65, 1.18)	0.98 (0.84, 1.15)	**CT/BEV/ Mpem**	0.62 (0.24, 1.58)	**0.51 (0.28, 0.94)**
0.71 (0.55, 0.92)	1.00 (0.74, 1.36)	**0.49 (0.30, 0.80)**	0.71 (0.50, 1.01)	**0.66 (0.51, 0.86)**	**0.67 (0.50, 0.90)**	**0.57 (0.42, 0.77)**	**0.54 (0.41, 0.71)**	**0.79 (0.65, 0.97)**	0.81 (0.61, 1.07)	0.91 (0.74, 1.11)	0.92 (0.71, 1.20)	**CT/Mpem**	0.83 (0.39, 1.73)
0.77 (0.66, 0.90)	1.08 (0.86, 1.36)	**0.53 (0.34, 0.83)**	0.77 (0.58, 1.02)	**0.72 (0.62, 0.84)**	**0.73 (0.60, 0.89)**	**0.62 (0.50, 0.77)**	**0.59 (0.42, 0.83)**	**0.86 (0.75, 0.98)**	0.88 (0.73, 1.06)	0.98 (0.83, 1.16)	1.00 (0.80, 1.26)	1.08 (0.88, 1.34)	**CT**

OS

B

Never smoking

Ever smoking										
PEM	0.36 (0.04, 3.28)		1.48 (0.15, 15.02)	2.11 (0.24, 18.54)	1.30 (0.15, 11.17)	1.69 (0.20, 14.21)	3.51 (0.39, 31.43)	0.64 (0.07, 5.92)	0.90 (0.11, 7.47)
0.48 (0.33, 0.68)	**NIV**		**4.12 (1.31, 13.02)**	**5.87 (2.57, 13.44)**	**3.62 (1.67, 7.85)**	**4.70 (2.31, 9.53)**	**9.78 (4.09, 23.34)**	1.77 (0.68, 4.62)	**2.51 (1.31, 4.81)**
0.93 (0.61, 1.42)	**1.95 (1.33, 2.84)**	**NIV/IPI**							
0.89 (0.58, 1.37)	**1.86 (1.27, 2.74)**	0.96 (0.62, 1.49)	**PEM/CT**	1.42 (0.49, 4.15)	0.88 (0.38, 2.04)	1.14 (0.42, 3.05)	2.37 (0.78, 7.17)	**0.43 (0.23, 0.81)**	0.61 (0.24, 1.57)
1.32 (0.90, 1.95)	**2.77 (1.97, 3.90)**	1.42 (0.95, 2.13)	1.48 (0.98, 2.23)	**ATE/CT/ BEV**	0.62 (0.32, 1.19)	0.80 (0.52, 1.23)	1.67 (0.86, 3.23)	**0.30 (0.13, 0.72)**	**0.43 (0.26, 0.71)**
0.79 (0.58, 1.08)	**1.65 (1.28, 2.13)**	0.85 (0.61, 1.19)	0.89 (0.66, 1.18)	**0.60 (0.45, 0.80)**	**ATE/CT**	1.30 (0.78, 2.15)	**2.7 (1.32, 5.54)**	**0.49 (0.28, 0.86)**	0.69 (0.45, 1.06)
0.77 (0.55, 1.08)	**1.60 (1.21, 2.14)**	0.82 (0.58, 1.18)	0.86 (0.60, 1.24)	**0.58 (0.48, 0.70)**	0.97 (0.78, 1.21)	**CT/BEV**	**2.08 (1.26, 3.46)**	**0.38 (0.18, 0.81)**	**0.53 (0.40, 0.71)**
0.86 (0.59, 1.26)	**1.80 (1.30, 2.50)**	0.93 (0.63, 1.37)	0.97 (0.65, 1.44)	**0.65 (0.51, 0.83)**	1.09 (0.83, 1.44)	1.12 (0.96, 1.32)	**CT/BEV/ Mpem**	**0.18 (0.07, 0.45)**	**0.26 (0.14, 0.46)**
0.48 (0.33, 0.70)	1.01 (0.73, 1.39)	**0.52 (0.35, 0.76)**	**0.54 (0.44, 0.67)**	**0.36 (0.26, 0.52)**	**0.61 (0.50, 0.74)**	**0.63 (0.47, 0.84)**	**0.56 (0.40, 0.78)**	**CT/Mpem**	1.42 (0.70, 2.87)
0.53 (0.40, 0.71)	1.11 (0.89, 1.38)	**0.57 (0.42, 0.78)**	**0.59 (0.43, 0.82)**	**0.40 (0.31, 0.52)**	**0.67 (0.59, 0.76)**	**0.69 (0.58, 0.83)**	**0.62 (0.48, 0.78)**	1.10 (0.87, 1.39)	**CT**

PFS

Figure S9. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) in Patients With a History of Ever or Never Smoking

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; ATE=atezolizumab; DUR=durvalumab; IPI= ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

A

Female

Male

PEM	0.78 (0.51, 1.22)	1.31 (0.62, 2.75)	1.37 (0.81, 2.31)	1.01 (0.73, 1.40)	1.07 (0.71, 1.63)	1.33 (0.84, 2.09)	**2.97 (1.79, 4.92)**	1.35 (0.94, 1.94)	0.68 (0.40, 1.14)	1.00 (0.74, 1.35)	1.10 (0.76, 1.59)	0.98 (0.63, 1.54)	0.90 (0.71, 1.15)
0.78 (0.57, 1.06)	**NIV**	1.67 (0.75, 3.69)	1.74 (0.96, 3.15)	1.29 (0.84, 1.98)	1.37 (0.83, 2.27)	1.69 (0.99, 2.89)	**3.79 (2.13, 6.74)**	**1.72 (1.09, 2.73)**	0.87 (0.48, 1.56)	1.27 (0.85, 1.92)	1.40 (0.88, 2.23)	1.26 (0.74, 2.13)	1.15 (0.79, 1.66)
1.32 (0.79, 2.21)	1.70 (0.98, 2.96)	**ATE**	1.04 (0.45, 2.42)	0.77 (0.37, 1.62)	0.82 (0.38, 1.80)	1.01 (0.46, 2.27)	2.27 (0.99, 5.22)	1.03 (0.49, 2.20)	0.52 (0.22, 1.20)	0.76 (0.37, 1.58)	0.84 (0.39, 1.79)	0.75 (0.34, 1.68)	0.69 (0.34, 1.40)
0.93 (0.66, 1.32)	1.20 (0.79, 1.81)	0.70 (0.39, 1.26)	**DUR**	0.74 (0.45, 1.24)	0.79 (0.44, 1.40)	0.97 (0.53, 1.78)	**2.17 (1.14, 4.13)**	0.99 (0.58, 1.69)	**0.50 (0.26, 0.95)**	0.73 (0.45, 1.20)	0.80 (0.47, 1.39)	0.72 (0.40, 1.31)	0.66 (0.42, 1.05)
1.11 (0.89, 1.39)	**1.43 (1.04, 1.96)**	0.84 (0.50, 1.41)	1.19 (0.83, 1.70)	**NIV/IPI**	1.06 (0.71, 1.59)	1.31 (0.85, 2.03)	**2.93 (1.80, 4.79)**	1.33 (0.94, 1.88)	0.67 (0.40, 1.11)	0.99 (0.75, 1.30)	1.08 (0.76, 1.54)	0.97 (0.63, 1.50)	0.89 (0.72, 1.10)
0.94 (0.73, 1.22)	1.21 (0.86, 1.70)	0.71 (0.42, 1.21)	1.01 (0.69, 1.48)	0.85 (0.65, 1.11)	**NIV/CT**	1.24 (0.74, 2.07)	**2.77 (1.58, 4.85)**	1.26 (0.81, 1.95)	0.63 (0.36, 1.12)	0.93 (0.63, 1.37)	1.02 (0.66, 1.60)	0.92 (0.55, 1.53)	0.84 (0.60, 1.18)
1.14 (0.87, 1.50)	**1.47 (1.04, 2.09)**	0.86 (0.51, 1.48)	1.23 (0.83, 1.81)	1.03 (0.78, 1.37)	1.21 (0.89, 1.65)	**NIV/IPI/CT**	**2.24 (1.24, 4.03)**	1.02 (0.63, 1.63)	**0.51 (0.28, 0.93)**	0.75 (0.49, 1.15)	0.83 (0.51, 1.34)	0.74 (0.43, 1.27)	**0.68 (0.46, 1.00)**
1.09 (0.81, 1.47)	1.40 (0.96, 2.03)	0.82 (0.47, 1.43)	1.17 (0.77, 1.76)	0.98 (0.72, 1.33)	1.15 (0.83, 1.61)	0.95 (0.67, 1.34)	**PEM/CT**	**0.46 (0.27, 0.76)**	**0.23 (0.12, 0.43)**	**0.34 (0.22, 0.52)**	**0.37 (0.23, 0.61)**	**0.33 (0.23, 0.48)**	**0.30 (0.20, 0.47)**
0.84 (0.68, 1.05)	1.08 (0.79, 1.48)	0.64 (0.38, 1.06)	0.90 (0.63, 1.29)	**0.76 (0.60, 0.96)**	0.89 (0.68, 1.16)	**0.74 (0.56, 0.97)**	0.77 (0.57, 1.05)	**ATE/CT**	**0.50 (0.29, 0.86)**	0.74 (0.54, 1.02)	0.81 (0.55, 1.20)	0.73 (0.46, 1.16)	**0.67 (0.51, 0.88)**
0.89 (0.70, 1.12)	1.14 (0.83, 1.58)	0.67 (0.40, 1.13)	0.95 (0.66, 1.37)	0.80 (0.63, 1.02)	0.94 (0.71, 1.24)	0.78 (0.58, 1.04)	0.82 (0.60, 1.12)	1.05 (0.83, 1.34)	**IPI/CT**	1.47 (0.90, 2.41)	1.62 (0.94, 2.78)	1.45 (0.80, 2.63)	1.33 (0.84, 2.11)
0.89 (0.73, 1.09)	1.15 (0.85, 1.55)	0.67 (0.41, 1.12)	0.96 (0.68, 1.35)	**0.80 (0.65, 1.00)**	0.94 (0.74, 1.21)	0.78 (0.60, 1.01)	0.82 (0.62, 1.08)	1.06 (0.86, 1.31)	1.00 (0.80, 1.26)	**CT/BEV**	1.10 (0.88, 1.37)	0.99 (0.69, 1.40)	0.90 (0.76, 1.08)
0.82 (0.62, 1.08)	1.05 (0.73, 1.50)	0.62 (0.36, 1.06)	0.88 (0.59, 1.30)	**0.74 (0.55, 0.99)**	0.87 (0.63, 1.19)	**0.71 (0.51, 0.99)**	0.75 (0.54, 1.05)	0.97 (0.73, 1.30)	0.92 (0.68, 1.24)	0.92 (0.75, 1.12)	**CT/BEV/ Mpem**	0.90 (0.59, 1.36)	0.82 (0.62, 1.09)
0.75 (0.55, 1.03)	0.97 (0.66, 1.42)	**0.57 (0.33, 1.00)**	0.81 (0.53, 1.23)	**0.68 (0.49, 0.94)**	0.80 (0.56, 1.13)	**0.66 (0.46, 0.94)**	**0.69 (0.53, 0.91)**	0.90 (0.65, 1.24)	0.85 (0.61, 1.18)	0.85 (0.65, 1.10)	0.92 (0.66, 1.28)	**CT/Mpem**	0.92 (0.63, 1.34)
0.76 (0.65, 0.88)	0.97 (0.74, 1.27)	**0.57 (0.35, 0.93)**	0.81 (0.59, 1.11)	**0.68 (0.57, 0.80)**	**0.80 (0.65, 0.99)**	**0.66 (0.53, 0.83)**	**0.69 (0.54, 0.90)**	0.90 (0.76, 1.05)	0.85 (0.71, 1.02)	**0.85 (0.74, 0.97)**	0.92 (0.73, 1.17)	1.00 (0.76, 1.32)	**CT**

OS

B

Female

Male \ PFS	PEM	NIV	NIV/IPI	PEM/CT	ATE/CT/ BEV	ATE/CT	CAM/CT	CT/BEV	CT/BEV/ Mpem	CT/Mpem	CT
PEM		**0.55 (0.31, 0.99)**	1.07 (0.52, 2.20)	1.54 (0.85, 2.81)	1.67 (0.93, 3.00)	1.22 (0.72, 2.07)	0.79 (0.35, 1.79)	1.22 (0.73, 2.04)	1.56 (0.90, 2.73)	0.62 (0.34, 1.11)	0.75 (0.46, 1.22)
NIV	0.37 (0.12, 1.20)		**1.94 (1.03, 3.64)**	**2.80 (1.72, 4.56)**	**3.03 (1.90, 4.83)**	**2.21 (1.49, 3.28)**	1.44 (0.70, 2.99)	**2.21 (1.53, 3.20)**	**2.84 (1.85, 4.34)**	1.12 (0.70, 1.80)	1.36 (0.98, 1.89)
NIV/IPI	0.75 (0.23, 2.47)	2.02 (0.64, 6.43)		1.44 (0.76, 2.74)	1.56 (0.83, 2.93)	1.14 (0.64, 2.03)	0.74 (0.32, 1.72)	1.14 (0.65, 2.00)	1.46 (0.80, 2.66)	0.58 (0.31, 1.09)	0.70 (0.41, 1.20)
PEM/CT	0.64 (0.22, 1.88)	1.71 (0.60, 4.87)	0.85 (0.29, 2.46)		1.08 (0.67, 1.76)	0.79 (0.56, 1.13)	**0.52 (0.28, 0.96)**	0.79 (0.54, 1.17)	1.01 (0.65, 1.58)	**0.40 (0.30, 0.53)**	**0.49 (0.34, 0.69)**
ATE/CT/ BEV	1.13 (0.34, 3.99)	3.04 (0.95, 10.42)	1.51 (0.46, 5.27)	1.78 (0.60, 5.60)		0.73 (0.49, 1.08)	**0.48 (0.23, 0.98)**	**0.73 (0.55, 0.97)**	0.94 (0.65, 1.34)	**0.37 (0.23, 0.59)**	**0.45 (0.32, 0.62)**
ATE/CT	0.58 (0.22, 1.58)	1.56 (0.60, 4.05)	0.77 (0.29, 2.06)	0.91 (0.44, 1.91)	0.51 (0.18, 1.39)		0.65 (0.35, 1.23)	1.00 (0.77, 1.30)	1.28 (0.91, 1.80)	**0.51 (0.38, 0.68)**	**0.61 (0.50, 0.76)**
CAM/CT	0.70 (0.17, 2.83)	1.89 (0.49, 7.40)	0.94 (0.24, 3.73)	1.10 (0.38, 3.22)	0.62 (0.15, 2.50)	1.21 (0.42, 3.50)		1.54 (0.79, 3.00)	1.97 (0.97, 3.97)	0.78 (0.45, 1.36)	0.94 (0.49, 1.80)
CT/BEV	0.62 (0.25, 1.66)	1.68 (0.70, 4.27)	0.83 (0.34, 2.17)	0.98 (0.46, 2.22)	0.55 (0.25, 1.21)	1.07 (0.57, 2.13)	0.89 (0.28, 2.95)		**1.28 (1.03, 1.59)**	**0.51 (0.35, 0.74)**	**0.61 (0.52, 0.72)**
CT/BEV/ Mpem	0.71 (0.21, 2.49)	1.91 (0.59, 6.49)	0.94 (0.29, 3.29)	1.11 (0.38, 3.52)	0.63 (0.21, 1.91)	1.22 (0.45, 3.50)	1.01 (0.25, 4.28)	1.14 (0.52, 2.50)		**0.40 (0.26, 0.61)**	**0.48 (0.37, 0.63)**
CT/Mpem	0.39 (0.13, 1.22)	1.06 (0.36, 3.15)	0.52 (0.17, 1.59)	0.62 (0.32, 1.22)	0.35 (0.11, 1.08)	0.68 (0.35, 1.33)	0.56 (0.25, 1.27)	0.63 (0.26, 1.42)	0.56 (0.17, 1.71)		1.21 (0.87, 1.69)
CT	**0.39 (0.17, 0.91)**	1.05 (0.47, 2.34)	0.52 (0.23, 1.20)	0.61 (0.31, 1.20)	**0.35 (0.14, 0.81)**	0.67 (0.40, 1.12)	0.56 (0.18, 1.66)	**0.63 (0.40, 0.92)**	0.55 (0.22, 1.29)	0.99 (0.47, 2.06)	

Figure S10. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) in Male or Female Patients

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; ATE=atezolizumab; DUR=durvalumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

A

ECOG PS=1

ECOG PS=0

PEM	0.76 (0.57, 1.03)	1.13 (0.69, 1.85)	0.90 (0.63, 1.27)		1.03 (0.83, 1.28)	0.92 (0.71, 1.18)	1.04 (0.79, 1.37)	1.25 (0.92, 1.68)	0.97 (0.78, 1.22)	0.91 (0.71, 1.16)	0.94 (0.77, 1.14)	0.85 (0.65, 1.13)	**0.70 (0.51, 0.95)**	**0.78 (0.67, 0.90)**
0.70 (0.43, 1.12)	**NIV**	1.48 (0.87, 2.53)	1.17 (0.78, 1.76)		1.35 (0.99, 1.83)	1.20 (0.86, 1.67)	1.36 (0.96, 1.92)	**1.63 (1.13, 2.35)**	1.27 (0.94, 1.73)	1.19 (0.86, 1.64)	1.23 (0.92, 1.64)	1.12 (0.79, 1.59)	0.91 (0.63, 1.33)	1.02 (0.79, 1.32)
1.84 (0.82, 4.11)	**2.64 (1.12, 6.26)**	**ATE**	0.79 (0.45, 1.40)		0.91 (0.55, 1.50)	0.81 (0.49, 1.36)	0.92 (0.55, 1.55)	1.10 (0.64, 1.89)	0.86 (0.52, 1.42)	0.80 (0.48, 1.34)	0.83 (0.51, 1.36)	0.76 (0.45, 1.28)	0.62 (0.36, 1.07)	0.69 (0.43, 1.10)
1.43 (0.85, 2.41)	**2.05 (1.12, 3.77)**	0.78 (0.32, 1.88)	**DUR**		1.15 (0.80, 1.64)	1.02 (0.70, 1.50)	1.16 (0.78, 1.72)	1.39 (0.92, 2.10)	1.09 (0.76, 1.56)	1.01 (0.70, 1.47)	1.05 (0.74, 1.48)	0.95 (0.64, 1.41)	0.78 (0.51, 1.19)	0.87 (0.63, 1.19)
1.02 (0.63, 1.65)	1.46 (0.82, 2.59)	0.55 (0.23, 1.31)	0.71 (0.39, 1.31)	**DUR/TRE**										
1.11 (0.77, 1.59)	1.59 (0.99, 2.57)	0.60 (0.27, 1.35)	0.78 (0.46, 1.31)	1.09 (0.67, 1.77)	**NIV/IPI**	0.89 (0.68, 1.16)	1.01 (0.76, 1.34)	1.21 (0.89, 1.65)	0.95 (0.75, 1.20)	0.88 (0.68, 1.14)	0.91 (0.74, 1.13)	0.83 (0.62, 1.10)	**0.68 (0.49, 0.94)**	**0.76 (0.64, 0.89)**
0.93 (0.60, 1.44)	1.34 (0.78, 2.29)	0.51 (0.22, 1.17)	0.65 (0.37, 1.16)	0.92 (0.53, 1.58)	0.84 (0.54, 1.30)	**NIV/CT**	1.13 (0.83, 1.55)	1.36 (0.97, 1.90)	1.06 (0.81, 1.39)	0.99 (0.74, 1.32)	1.02 (0.80, 1.31)	0.93 (0.68, 1.28)	0.76 (0.54, 1.08)	0.85 (0.69, 1.05)
1.61 (0.99, 2.61)	**2.31 (1.30, 4.11)**	0.87 (0.37, 2.07)	1.12 (0.61, 2.07)	1.59 (0.89, 2.83)	1.45 (0.90, 2.35)	**1.73 (1.01, 2.97)**	**NIV/IPI/CT**	1.20 (0.85, 1.70)	0.94 (0.70, 1.25)	0.87 (0.64, 1.19)	0.90 (0.69, 1.18)	0.82 (0.59, 1.14)	**0.67 (0.47, 0.96)**	**0.75 (0.60, 0.95)**
1.77 (1.07, 2.94)	**2.55 (1.41, 4.62)**	0.96 (0.40, 2.32)	1.24 (0.66, 2.32)	1.75 (0.96, 3.18)	1.60 (0.97, 2.65)	**1.91 (1.09, 3.34)**	1.10 (0.61, 2.00)	**PEM/CT**	0.78 (0.57, 1.07)	0.73 (0.52, 1.01)	**0.75 (0.57, 0.99)**	**0.69 (0.49, 0.96)**	**0.56 (0.43, 0.73)**	**0.63 (0.48, 0.81)**
0.85 (0.60, 1.21)	1.22 (0.76, 1.96)	0.46 (0.21, 1.03)	**0.60 (0.36, 0.99)**	0.84 (0.52, 1.35)	0.77 (0.54, 1.09)	0.91 (0.59, 1.41)	**0.53 (0.33, 0.85)**	**0.48 (0.29, 0.79)**	**ATE/CT**	0.93 (0.72, 1.21)	0.96 (0.78, 1.20)	0.88 (0.66, 1.17)	**0.72 (0.52, 0.99)**	**0.80 (0.68, 0.95)**
0.78 (0.53, 1.16)	1.12 (0.68, 1.85)	**0.42 (0.19, 0.96)**	**0.55 (0.32, 0.94)**	0.77 (0.46, 1.28)	0.70 (0.47, 1.04)	0.84 (0.53, 1.33)	**0.49 (0.29, 0.81)**	**0.44 (0.26, 0.75)**	0.92 (0.62, 1.35)	**IPI/CT**	1.04 (0.81, 1.32)	0.94 (0.69, 1.28)	0.77 (0.55, 1.08)	0.86 (0.70, 1.05)
0.84 (0.61, 1.16)	1.21 (0.78, 1.89)	0.46 (0.21, 1.01)	**0.59 (0.36, 0.96)**	0.83 (0.53, 1.31)	0.76 (0.55, 1.04)	0.91 (0.61, 1.35)	**0.53 (0.33, 0.82)**	**0.48 (0.31, 0.74)**	0.99 (0.73, 1.35)	1.08 (0.76, 1.54)	**CT/BEV**	0.91 (0.75, 1.10)	**0.74 (0.57, 0.97)**	**0.83 (0.73, 0.95)**
0.97 (0.66, 1.44)	1.39 (0.84, 2.30)	0.53 (0.23, 1.19)	0.68 (0.40, 1.16)	0.95 (0.58, 1.59)	0.87 (0.59, 1.29)	1.04 (0.66, 1.65)	**0.60 (0.36, 1.00)**	**0.55 (0.33, 0.89)**	1.14 (0.78, 1.67)	1.24 (0.82, 1.90)	1.15 (0.91, 1.45)	**CT/BEV/ Mpem**	0.82 (0.59, 1.14)	0.91 (0.72, 1.16)
0.89 (0.57, 1.38)	1.27 (0.74, 2.19)	0.48 (0.21, 1.12)	0.62 (0.35, 1.11)	0.87 (0.50, 1.51)	0.80 (0.51, 1.25)	0.95 (0.57, 1.58)	**0.55 (0.32, 0.95)**	**0.50 (0.34, 0.74)**	1.04 (0.67, 1.62)	1.14 (0.71, 1.82)	1.05 (0.75, 1.47)	0.91 (0.61, 1.37)	**CT/Mpem**	1.12 (0.85, 1.48)
0.77 (0.60, 1.00)	1.11 (0.74, 1.66)	**0.42 (0.20, 0.90)**	**0.54 (0.34, 0.85)**	0.76 (0.51, 1.15)	**0.70 (0.54, 0.90)**	0.83 (0.58, 1.18)	**0.48 (0.32, 0.72)**	**0.44 (0.28, 0.67)**	0.91 (0.71, 1.16)	0.99 (0.73, 1.34)	0.92 (0.76, 1.10)	0.80 (0.59, 1.07)	0.87 (0.61, 1.26)	**CT**

OS

B

ECOG PS=1

ECOG PS=0										
PEM	**0.51 (0.32, 0.79)**	0.93 (0.55, 1.57)	0.83 (0.55, 1.27)	1.29 (0.80, 2.06)	0.74 (0.50, 1.10)	0.72 (0.42, 1.22)	0.82 (0.54, 1.25)	0.90 (0.57, 1.44)	**0.47 (0.30, 0.72)**	**0.51 (0.35, 0.74)**
0.27 (0.14, 0.51)	**NIV**	**1.84 (1.17, 2.87)**	**1.65 (1.20, 2.28)**	**2.55 (1.74, 3.74)**	**1.46 (1.10, 1.95)**	1.42 (0.90, 2.24)	**1.63 (1.19, 2.23)**	**1.79 (1.23, 2.60)**	0.92 (0.66, 1.29)	1.01 (0.79, 1.30)
0.73 (0.35, 1.51)	**2.73 (1.48, 5.02)**	**NIV/IPI**	0.90 (0.59, 1.37)	1.39 (0.86, 2.23)	0.80 (0.53, 1.19)	0.77 (0.45, 1.32)	0.89 (0.58, 1.35)	0.98 (0.61, 1.55)	**0.50 (0.33, 0.77)**	**0.55 (0.38, 0.80)**
0.91 (0.49, 1.72)	**3.43 (2.12, 5.56)**	1.26 (0.70, 2.27)	**PEM/CT**	**1.54 (1.08, 2.20)**	0.89 (0.71, 1.11)	0.86 (0.59, 1.25)	0.99 (0.75, 1.31)	1.08 (0.77, 1.53)	**0.56 (0.45, 0.69)**	**0.61 (0.50, 0.75)**
1.30 (0.67, 2.53)	**4.89 (2.89, 8.24)**	1.79 (0.96, 3.34)	1.42 (0.86, 2.34)	**ATE/CT/ BEV**	**0.57 (0.41, 0.80)**	**0.56 (0.35, 0.90)**	**0.64 (0.51, 0.80)**	**0.70 (0.52, 0.95)**	**0.36 (0.25, 0.52)**	**0.40 (0.30, 0.53)**
0.73 (0.41, 1.29)	**2.73 (1.81, 4.10)**	1.00 (0.59, 1.70)	0.79 (0.57, 1.10)	**0.56 (0.36, 0.86)**	**ATE/CT**	0.97 (0.67, 1.41)	1.12 (0.88, 1.42)	1.22 (0.90, 1.67)	**0.63 (0.51, 0.77)**	**0.69 (0.60, 0.80)**
0.90 (0.37, 2.21)	**3.39 (1.53, 7.55)**	1.24 (0.52, 2.96)	0.99 (0.49, 2.00)	0.69 (0.31, 1.56)	1.24 (0.62, 2.50)	**CAM/CT**	1.15 (0.75, 1.76)	1.26 (0.79, 2.02)	**0.65 (0.48, 0.89)**	0.71 (0.49, 1.04)
0.71 (0.40, 1.30)	**2.69 (1.74, 4.14)**	0.99 (0.57, 1.71)	0.78 (0.52, 1.17)	**0.55 (0.41, 0.74)**	0.99 (0.72, 1.34)	0.79 (0.37, 1.69)	**CT/BEV**	1.10 (0.90, 1.34)	**0.57 (0.42, 0.76)**	**0.62 (0.51, 0.75)**
0.98 (0.52, 1.86)	**3.68 (2.25, 6.03)**	1.35 (0.74, 2.46)	1.07 (0.67, 1.71)	0.75 (0.52, 1.10)	1.35 (0.91, 1.99)	1.08 (0.49, 2.39)	**1.37 (1.08, 1.73)**	**CT/BEV/ Mpem**	**0.52 (0.36, 0.74)**	**0.56 (0.43, 0.74)**
0.42 (0.23, 0.79)	1.60 (1.00, 2.55)	0.58 (0.33, 1.04)	**0.46 (0.35, 0.62)**	**0.33 (0.20, 0.53)**	**0.58 (0.45, 0.76)**	**0.47 (0.25, 0.90)**	**0.59 (0.40, 0.87)**	**0.43 (0.28, 0.68)**	**CT/Mpem**	1.09 (0.88, 1.37)
0.45 (0.26, 0.78)	**1.69 (1.18, 2.42)**	0.62 (0.38, 1.02)	**0.49 (0.36, 0.68)**	**0.35 (0.24, 0.51)**	**0.62 (0.51, 0.75)**	0.50 (0.24, 1.02)	**0.63 (0.49, 0.80)**	**0.46 (0.33, 0.64)**	1.06 (0.79, 1.43)	**CT**

PFS

Figure S11. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) According to Eastern Cooperative Oncology Group Performance Status (ECOG PS)

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; ATE=atezolizumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

A

Age<65

Age>=65 \ OS	PEM	NIV	ATE	DUR	DUR/TRE	DUR/TRE/CT	NIV/IPI	NIV/CT	NIV/IPI/CT	PEM/CT	ATE/CT	IPI/CT	CT/BEV	CT/BEV/ Mpem	CT/Mpem	CT
PEM		**0.69 (0.48, 0.98)**	1.31 (0.73, 2.36)	0.90 (0.60, 1.35)			1.11 (0.86, 1.45)	0.99 (0.73, 1.35)	1.27 (0.92, 1.76)	**1.75 (1.23, 2.49)**	0.92 (0.72, 1.19)	0.95 (0.70, 1.28)	0.94 (0.75, 1.17)	0.90 (0.68, 1.19)	0.83 (0.62, 1.12)	**0.78 (0.65, 0.93)**
0.75 (0.52, 1.07)	NIV	**1.91 (1.01, 3.63)**	1.31 (0.82, 2.12)			**1.62 (1.13, 2.33)**	1.45 (0.97, 2.16)	**1.85 (1.22, 2.80)**	**2.55 (1.65, 3.93)**	1.34 (0.94, 1.92)	1.38 (0.93, 2.04)	1.37 (0.97, 1.91)	1.31 (0.90, 1.91)	1.21 (0.82, 1.80)	1.13 (0.83, 1.54)
1.16 (0.63, 2.16)	1.55 (0.80, 3.00)	ATE	0.69 (0.35, 1.33)			0.85 (0.47, 1.53)	0.76 (0.41, 1.40)	0.97 (0.52, 1.80)	1.33 (0.70, 2.52)	0.70 (0.39, 1.26)	0.72 (0.39, 1.32)	0.71 (0.40, 1.27)	0.69 (0.38, 1.25)	0.63 (0.35, 1.16)	0.59 (0.34, 1.03)
1.18 (0.78, 1.80)	1.58 (0.97, 2.55)	1.02 (0.51, 2.04)	DUR			1.24 (0.82, 1.86)	1.10 (0.71, 1.72)	1.41 (0.89, 2.22)	**1.94 (1.21, 3.11)**	1.02 (0.68, 1.54)	1.05 (0.68, 1.62)	1.04 (0.71, 1.53)	1.00 (0.66, 1.52)	0.92 (0.60, 1.43)	0.86 (0.60, 1.24)
1.08 (0.72, 1.62)	1.44 (0.91, 2.31)	0.93 (0.47, 1.85)	0.92 (0.55, 1.54)	DUR/TRE											
1.28 (0.73, 2.21)	1.70 (0.93, 3.10)	1.10 (0.50, 2.40)	1.08 (0.57, 2.05)	1.18 (0.81, 1.71)	DUR/TRE/CT										
0.99 (0.75, 1.30)	1.31 (0.92, 1.88)	0.85 (0.46, 1.58)	0.83 (0.55, 1.27)	0.91 (0.60, 1.37)	0.77 (0.44, 1.34)	NIV/IPI	0.89 (0.65, 1.23)	1.14 (0.82, 1.60)	**1.57 (1.10, 2.25)**	0.83 (0.64, 1.08)	0.85 (0.62, 1.16)	0.84 (0.67, 1.06)	0.81 (0.61, 1.08)	0.75 (0.55, 1.02)	**0.70 (0.58, 0.84)**
0.90 (0.65, 1.24)	1.19 (0.80, 1.78)	0.77 (0.41, 1.47)	0.76 (0.48, 1.20)	0.83 (0.53, 1.28)	0.70 (0.39, 1.25)	0.91 (0.66, 1.26)	NIV/CT	1.28 (0.88, 1.86)	**1.76 (1.18, 2.62)**	0.93 (0.68, 1.27)	0.95 (0.67, 1.35)	0.94 (0.71, 1.26)	0.91 (0.65, 1.27)	0.84 (0.59, 1.19)	0.78 (0.61, 1.01)
1.07 (0.76, 1.50)	1.42 (0.94, 2.15)	0.92 (0.48, 1.76)	0.90 (0.57, 1.44)	0.99 (0.63, 1.55)	0.84 (0.46, 1.51)	1.08 (0.77, 1.53)	1.19 (0.81, 1.75)	NIV/IPI/CT	1.38 (0.91, 2.08)	0.73 (0.52, 1.01)	0.74 (0.52, 1.07)	**0.74 (0.54, 1.00)**	0.71 (0.50, 1.01)	**0.66 (0.45, 0.95)**	**0.61 (0.46, 0.80)**
1.08 (0.76, 1.53)	1.44 (0.95, 2.20)	0.93 (0.48, 1.80)	0.92 (0.57, 1.47)	1.00 (0.63, 1.59)	0.85 (0.47, 1.54)	1.10 (0.77, 1.56)	1.21 (0.82, 1.79)	1.01 (0.68, 1.52)	PEM/CT	**0.53 (0.38, 0.73)**	**0.54 (0.37, 0.80)**	**0.54 (0.39, 0.74)**	**0.52 (0.36, 0.74)**	**0.48 (0.36, 0.63)**	**0.44 (0.33, 0.60)**
0.98 (0.76, 1.28)	1.31 (0.92, 1.86)	0.85 (0.46, 1.57)	0.83 (0.55, 1.26)	0.91 (0.61, 1.35)	0.77 (0.45, 1.34)	1.00 (0.77, 1.30)	1.10 (0.80, 1.51)	0.92 (0.66, 1.28)	0.91 (0.66, 1.26)	ATE/CT	1.03 (0.76, 1.39)	1.02 (0.82, 1.26)	0.98 (0.74, 1.29)	0.90 (0.71, 1.15)	0.84 (0.70, 1.01)
0.77 (0.57, 1.05)	1.03 (0.70, 1.51)	0.66 (0.35, 1.25)	0.65 (0.42, 1.02)	0.71 (0.46, 1.09)	0.61 (0.34, 1.07)	0.78 (0.58, 1.07)	0.86 (0.61, 1.23)	0.72 (0.50, 1.04)	0.71 (0.49, 1.04)	0.78 (0.58, 1.05)	IPI/CT	0.99 (0.75, 1.31)	0.95 (0.69, 1.32)	0.88 (0.63, 1.24)	0.82 (0.64, 1.05)
0.87 (0.67, 1.13)	1.16 (0.82, 1.64)	0.75 (0.40, 1.38)	0.74 (0.49, 1.11)	0.80 (0.54, 1.19)	0.68 (0.39, 1.18)	0.88 (0.68, 1.15)	0.97 (0.71, 1.33)	0.81 (0.59, 1.13)	0.80 (0.58, 1.11)	0.88 (0.70, 1.12)	1.13 (0.84, 1.51)	CT/BEV	0.96 (0.81, 1.14)	0.89 (0.69, 1.14)	**0.83 (0.72, 0.95)**
0.96 (0.65, 1.43)	1.29 (0.82, 2.03)	0.83 (0.42, 1.64)	0.82 (0.49, 1.36)	0.89 (0.54, 1.46)	0.76 (0.41, 1.41)	0.98 (0.66, 1.46)	1.08 (0.70, 1.66)	0.90 (0.58, 1.41)	0.89 (0.58, 1.38)	0.98 (0.67, 1.43)	1.25 (0.82, 1.90)	1.11 (0.83, 1.49)	CT/BEV/ Mpem	0.92 (0.68, 1.24)	0.86 (0.69, 1.06)
0.71 (0.52, 0.98)	0.95 (0.65, 1.40)	0.61 (0.32, 1.16)	**0.60 (0.39, 0.95)**	0.66 (0.43, 1.02)	**0.56 (0.32, 0.99)**	**0.72 (0.53, 0.99)**	0.80 (0.56, 1.14)	**0.67 (0.46, 0.97)**	**0.66 (0.49, 0.89)**	**0.73 (0.56, 0.93)**	0.92 (0.66, 1.30)	0.82 (0.64, 1.05)	0.74 (0.50, 1.09)	CT/Mpem	0.93 (0.73, 1.19)
0.78 (0.64, 0.94)	1.04 (0.77, 1.41)	0.67 (0.37, 1.21)	**0.66 (0.45, 0.96)**	0.72 (0.50, 1.03)	0.61 (0.36, 1.03)	**0.79 (0.65, 0.96)**	0.87 (0.67, 1.13)	**0.73 (0.55, 0.97)**	**0.72 (0.54, 0.97)**	**0.79 (0.66, 0.95)**	1.01 (0.80, 1.28)	0.90 (0.75, 1.07)	0.81 (0.57, 1.14)	1.09 (0.85, 1.40)	CT

B

Age<65

Age>=65										
PEM	0.52 (0.10, 2.71)	1.20 (0.22, 6.31)	1.27 (0.27, 5.90)	1.75 (0.30, 10.50)	0.92 (0.20, 4.27)	0.89 (0.12, 6.69)	1.14 (0.30, 4.53)	1.48 (0.26, 8.78)	0.56 (0.11, 2.92)	0.61 (0.19, 1.99)
0.37 (0.22, 0.63)	**NIV**	2.29 (0.44, 11.88)	2.42 (0.53, 11.14)	3.36 (0.59, 19.80)	1.77 (0.39, 8.12)	1.71 (0.23, 12.59)	2.18 (0.58, 8.48)	2.83 (0.50, 16.5)	1.08 (0.21, 5.49)	1.17 (0.37, 3.70)
0.73 (0.39, 1.36)	**1.95 (1.15, 3.32)**	**NIV/IPI**	1.06 (0.22, 4.97)	1.46 (0.25, 8.87)	0.77 (0.17, 3.61)	0.75 (0.10, 5.60)	0.95 (0.25, 3.78)	1.23 (0.21, 7.33)	0.47 (0.09, 2.43)	0.51 (0.16, 1.66)
0.65 (0.39, 1.08)	**1.76 (1.20, 2.57)**	0.90 (0.54, 1.49)	**PEM/CT**	1.38 (0.27, 7.45)	0.73 (0.23, 2.31)	0.71 (0.15, 3.26)	0.90 (0.27, 3.06)	1.17 (0.23, 6.19)	0.45 (0.17, 1.21)	0.48 (0.18, 1.31)
1.24 (0.71, 2.16)	**3.34 (2.15, 5.19)**	1.71 (0.98, 2.97)	**1.90 (1.26, 2.88)**	**ATE/CT/BEV**	0.53 (0.10, 2.71)	0.51 (0.06, 4.10)	0.65 (0.21, 2.04)	0.84 (0.17, 4.22)	0.32 (0.05, 1.82)	0.35 (0.09, 1.29)
0.74 (0.45, 1.21)	**1.99 (1.38, 2.87)**	1.02 (0.62, 1.67)	1.13 (0.86, 1.50)	**0.60 (0.40, 0.89)**	**ATE/CT**	0.97 (0.21, 4.42)	1.23 (0.38, 4.18)	1.60 (0.31, 8.45)	0.61 (0.23, 1.65)	0.66 (0.25, 1.78)
0.75 (0.34, 1.65)	2.01 (0.98, 4.13)	1.03 (0.47, 2.28)	1.15 (0.60, 2.19)	0.60 (0.29, 1.26)	1.01 (0.53, 1.92)	**CAM/CT**	1.27 (0.22, 7.55)	1.65 (0.21, 13.6)	0.63 (0.20, 2.00)	0.68 (0.13, 3.48)
0.67 (0.40, 1.13)	**1.80 (1.21, 2.69)**	0.92 (0.55, 1.56)	1.03 (0.71, 1.49)	**0.54 (0.45, 0.65)**	0.91 (0.64, 1.29)	0.90 (0.44, 1.83)	**CT/BEV**	1.30 (0.42, 4.02)	0.50 (0.13, 1.85)	0.54 (0.27, 1.03)
0.68 (0.38, 1.24)	**1.84 (1.12, 3.02)**	0.94 (0.52, 1.72)	1.05 (0.65, 1.68)	**0.55 (0.39, 0.78)**	0.93 (0.58, 1.46)	0.91 (0.42, 1.97)	1.02 (0.76, 1.37)	**CT/BEV/ Mpem**	0.38 (0.07, 2.15)	0.41 (0.11, 1.52)
0.44 (0.26, 0.74)	1.19 (0.79, 1.78)	0.61 (0.36, 1.03)	**0.68 (0.52, 0.88)**	**0.36 (0.23, 0.55)**	**0.60 (0.47, 0.76)**	0.59 (0.33, 1.07)	**0.66 (0.44, 0.97)**	0.64 (0.40, 1.05)	**CT/Mpem**	1.08 (0.34, 3.41)
0.45 (0.29, 0.70)	1.21 (0.91, 1.62)	**0.62 (0.40, 0.97)**	**0.69 (0.54, 0.88)**	**0.36 (0.26, 0.51)**	**0.61 (0.49, 0.76)**	0.60 (0.31, 1.16)	**0.67 (0.51, 0.88)**	**0.66 (0.44, 0.98)**	1.02 (0.77, 1.35)	**CT**

PFS

Figure S12. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) According to Age
Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; ATE=atezolizumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem= CT followed by maintenance of pemetrexed.

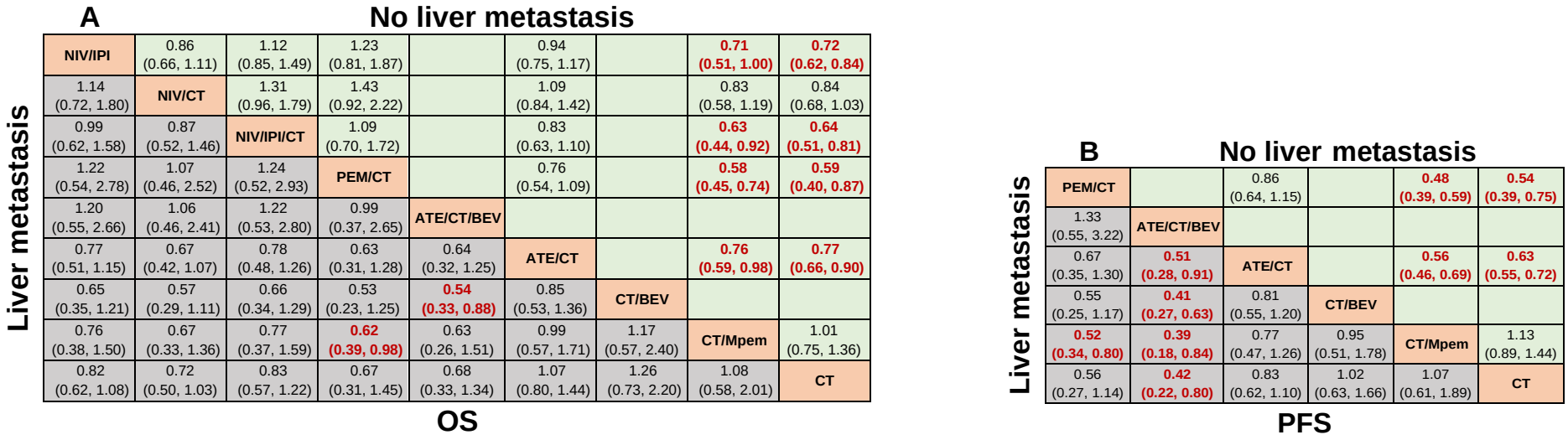


Figure S13. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) According to the Status of Liver Metastasis

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. Instead of using the recently updated overall survival data based on intention-to-treat population in the 2020 AACR meeting, early data presented in the 2018 ASCO annual meeting comparing ATE/CT/BEV with CT/BEV for wild-type population with liver metastasis was used in the network meta-analysis. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

A

CNS metastasis \ No CNS metastasis							
PEM	0.85 (0.60, 1.21)	0.80 (0.55, 1.16)	0.85 (0.58, 1.26)	1.09 (0.72, 1.63)	0.60 (0.29, 1.22)	0.61 (0.29, 1.27)	**0.64 (0.46, 0.88)**
1.14 (0.30, 4.41)	**NIV/IPI**	0.94 (0.74, 1.19)	1.00 (0.77, 1.29)	1.27 (0.96, 1.69)	0.70 (0.36, 1.35)	0.71 (0.36, 1.40)	**0.75 (0.65, 0.86)**
0.79 (0.20, 3.21)	0.70 (0.35, 1.39)	**NIV/CT**	1.07 (0.80, 1.42)	1.36 (0.99, 1.85)	0.74 (0.38, 1.46)	0.76 (0.38, 1.52)	**0.80 (0.66, 0.97)**
1.92 (0.49, 7.51)	1.68 (0.90, 3.16)	**2.42 (1.18, 4.98)**	**NIV/IPI/CT**	1.27 (0.92, 1.76)	0.70 (0.35, 1.37)	0.71 (0.35, 1.43)	**0.75 (0.61, 0.93)**
1.78 (0.45, 7.13)	1.56 (0.81, 3.03)	**2.25 (1.06, 4.75)**	0.93 (0.46, 1.86)	**PEM/CT**	0.55 (0.28, 1.09)	0.56 (0.28, 1.13)	**0.59 (0.46, 0.75)**
0.81 (0.22, 3.01)	0.71 (0.43, 1.17)	1.02 (0.55, 1.88)	**0.42 (0.25, 0.73)**	**0.45 (0.25, 0.81)**	**CT/BEV**	1.02 (0.86, 1.21)	1.08 (0.57, 2.05)
0.73 (0.18, 2.92)	0.64 (0.33, 1.25)	0.92 (0.43, 1.96)	**0.38 (0.19, 0.77)**	**0.41 (0.20, 0.85)**	0.90 (0.58, 1.41)	**CT/BEV/ Mpem**	1.05 (0.54, 2.05)
0.73 (0.20, 2.64)	**0.64 (0.42, 0.97)**	0.92 (0.53, 1.59)	**0.38 (0.24, 0.61)**	**0.41 (0.25, 0.68)**	0.90 (0.68, 1.19)	1.00 (0.59, 1.69)	**CT+/-M**

OS

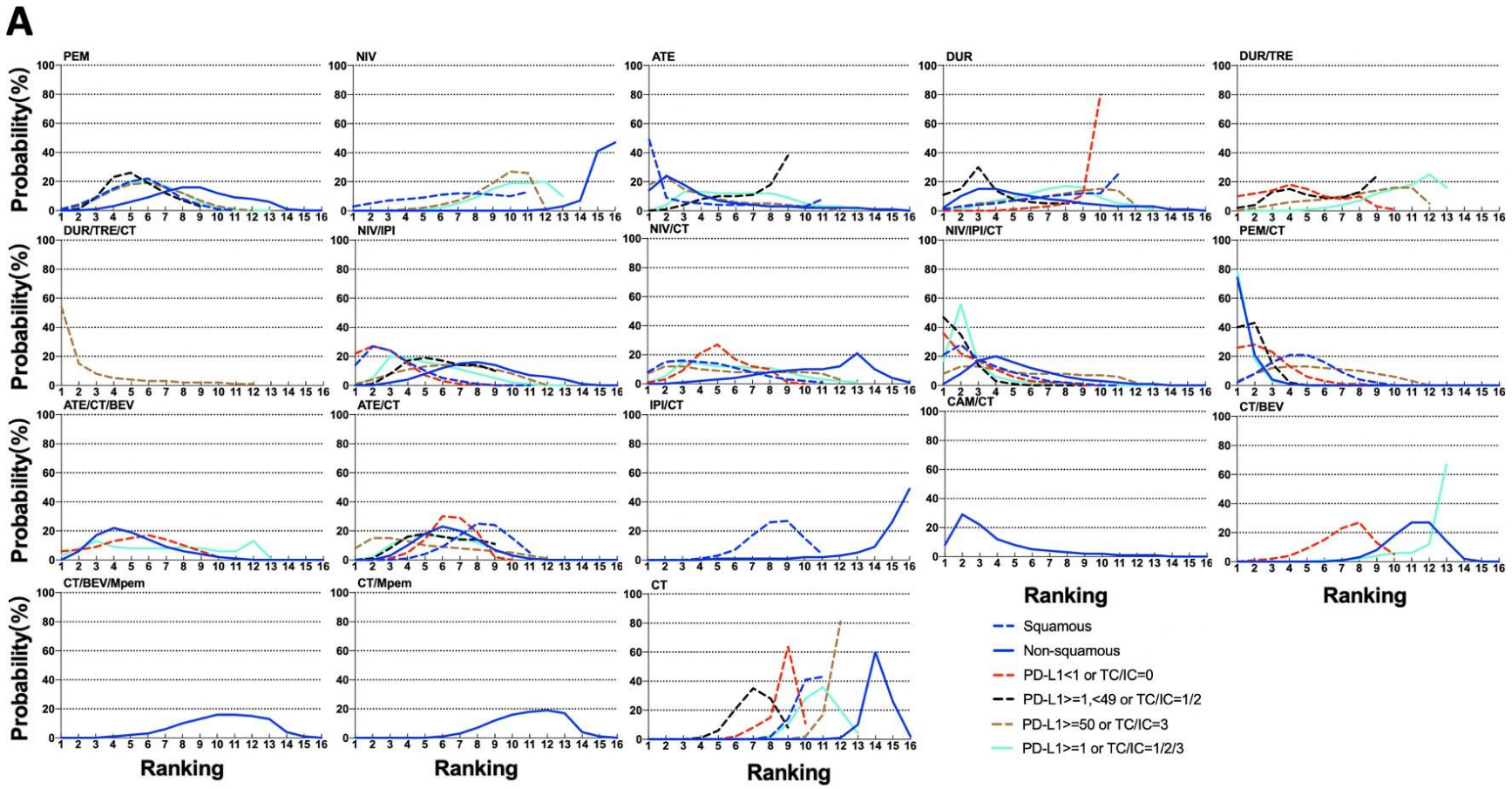
B

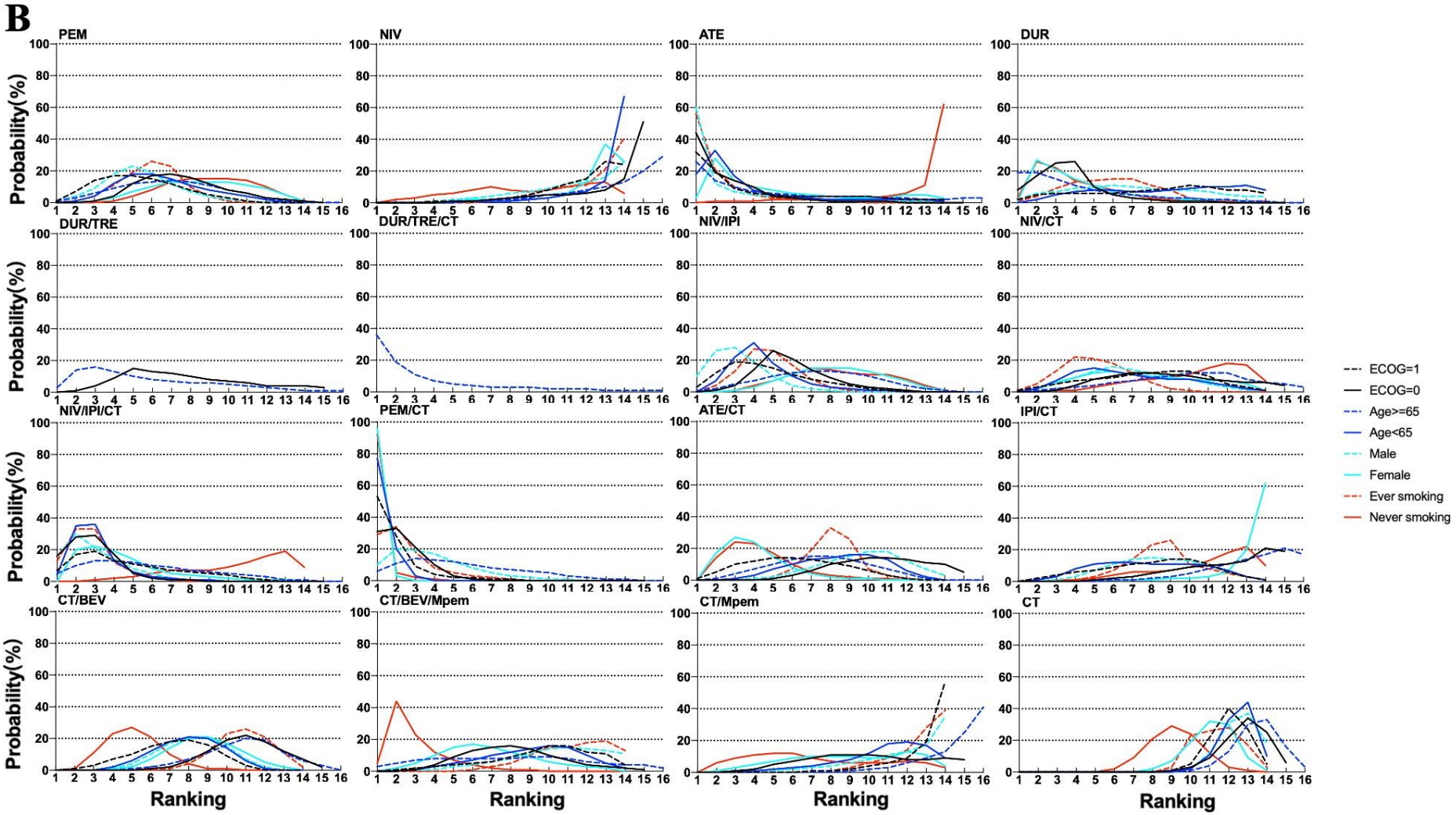
CNS metastasis \ No CNS metastasis			
PEM	1.04 (0.71, 1.52)	0.82 (0.54, 1.26)	**0.50 (0.36, 0.69)**
1.31 (0.43, 4.02)	**PEM/CT**	0.79 (0.55, 1.12)	**0.48 (0.39, 0.59)**
3.92 (0.33, 46.30)	2.99 (0.30, 29.40)	**CAM/CT**	**0.61 (0.46, 0.81)**
0.55 (0.20, 1.54)	**0.42 (0.27, 0.66)**	0.14 (0.01, 1.32)	**CT+/-M**

PFS

Figure S14. Pooled Estimates of the Network Meta-analysis for Overall Survival (OS) and Progression-Free Survival (PFS) According to the Status of Central Nervous System (CNS) Metastasis

Data in each cell are hazard ratio (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. Hazard ratio less than 1 favours upper-row treatment. Significant results are highlighted in red and bold. PEM=pembrolizumab; NIV=nivolumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; Mpem= maintenance of pemetrexed; CT+/-M=CT with or without maintenance of pemetrexed.


A
PEM
NIV
ATE
DUR
DUR/TRE
DUR/TRE/CT
NIV/IPI
NIV/CT
NIV/IPI/CT
PEM/CT
ATE/CT/BEV
ATE/CT
IPI/CT
CAM/CT
CT/BEV
CT/BEV/Mpem
CT/Mpem
CT
Probability(%)
Ranking
Squamous
Non-squamous
PD-L1<1 or TC/IC=0
PD-L1>=1,<49 or TC/IC=1/2
PD-L1>=50 or TC/IC=3
PD-L1>=1 or TC/IC=1/2/3


B
PEM
NIV
ATE
DUR
DUR/TRE
DUR/TRE/CT
NIV/IPI
NIV/CT
NIV/IPI/CT
PEM/CT
ATE/CT
IPI/CT
CT/BEV
CT/BEV/Mpem
CT/Mpem
CT
Probability(%)
Ranking
ECOG=1
ECOG=0
Age>=65
Age<65
Male
Female
Ever smoking
Never smoking

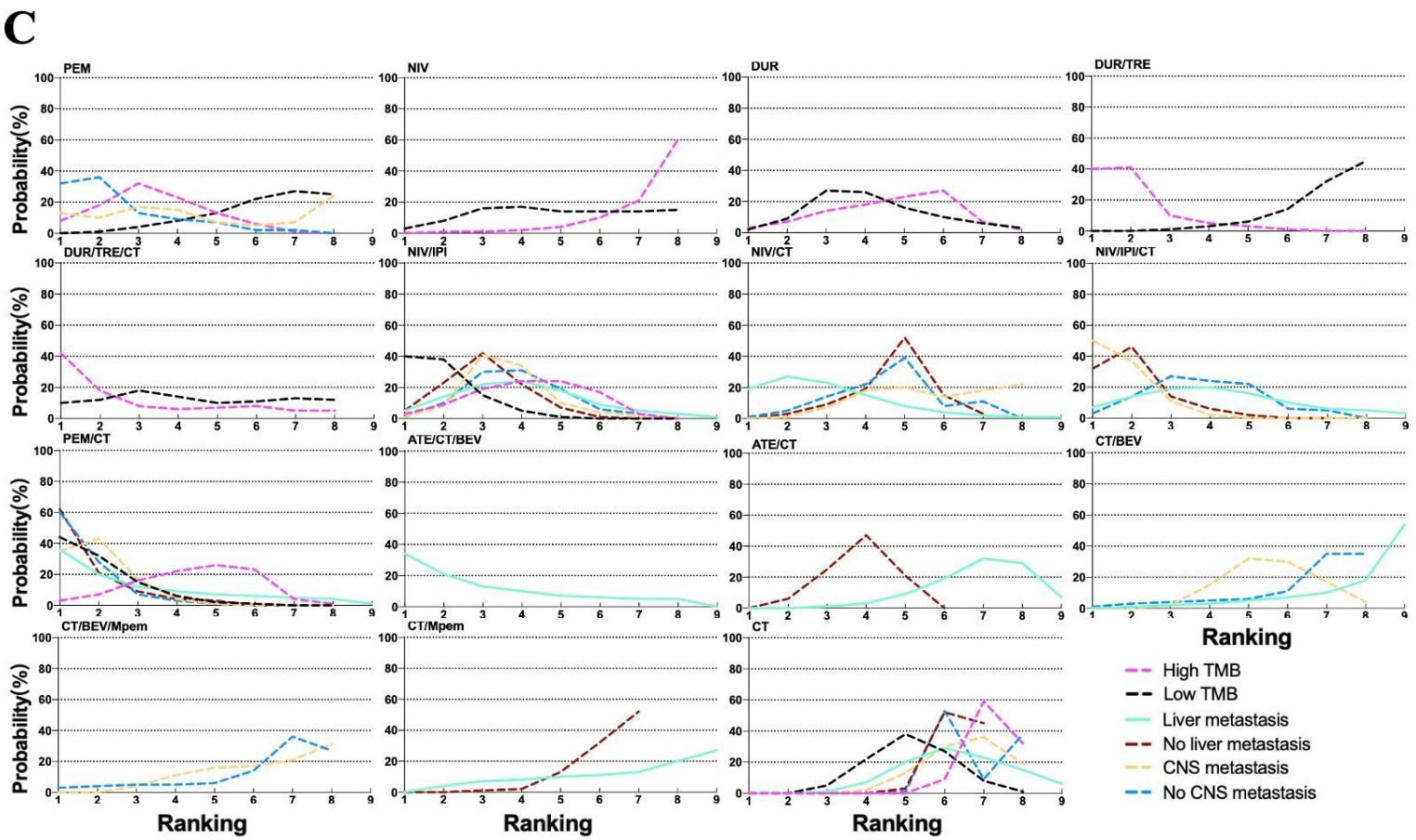


Figure S15. Bayesian Ranking Profiles of Comparable Treatments on Efficacy for Subgroup Patients With Advanced Wild-type Non-small Cell Lung Cancer (NSCLC)

(A) Profiles indicate the probability of each comparable treatment being ranked from first to last on overall survival (OS) in squamous, non-squamous subgroups or patients with different expression levels of programmed-death ligand 1 (PD-L1) or TC/IC. (B)Profiles indicate the probability of each comparable treatment being ranked from first to last on OS according to Eastern Cooperative Oncology Group (ECOG) performance status, age, gender, history of smoking. (C)Profiles indicate the probability of each comparable treatment being ranked from first to last on OS according to the status of tumor mutation burden (TMB), liver metastasis or central nervous system (CNS) metastasis. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI= ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.

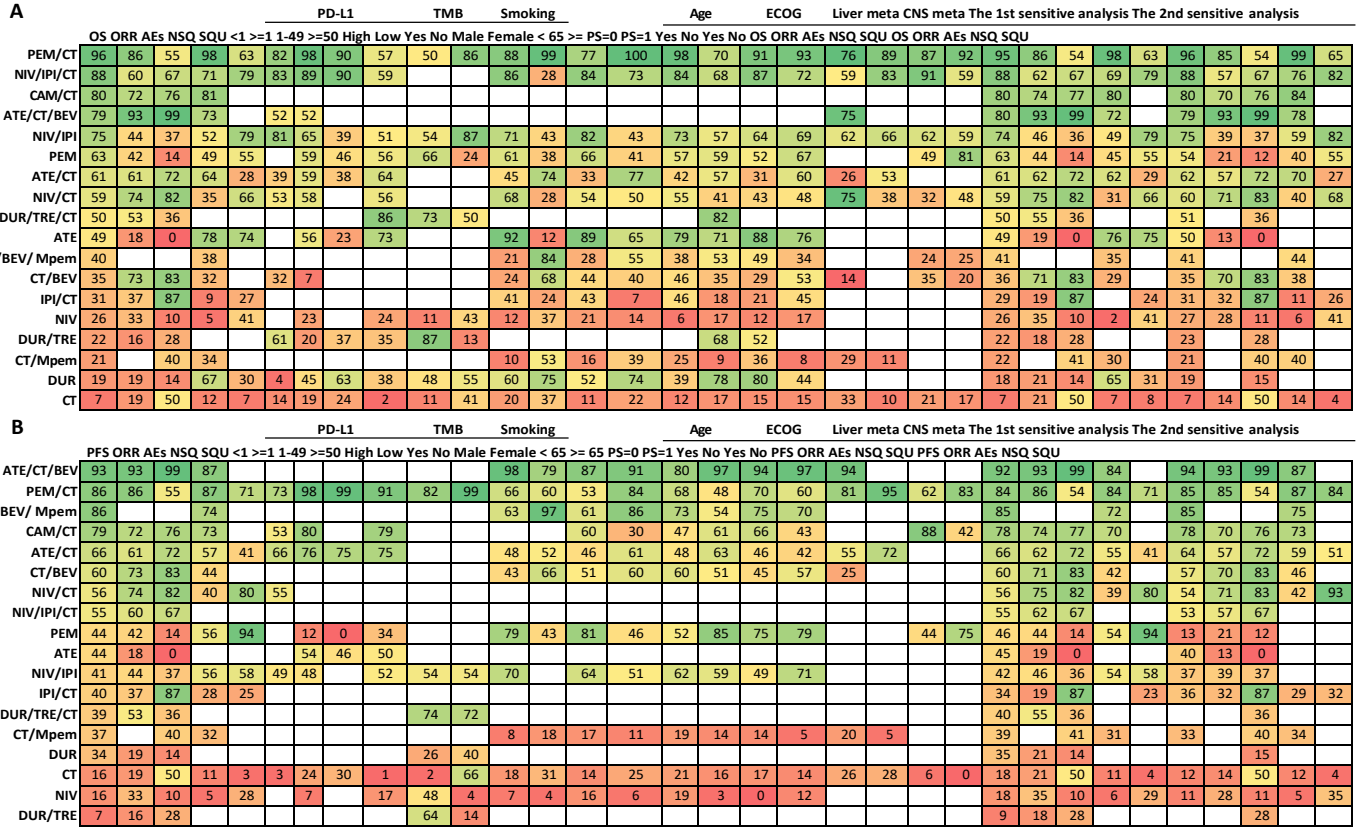
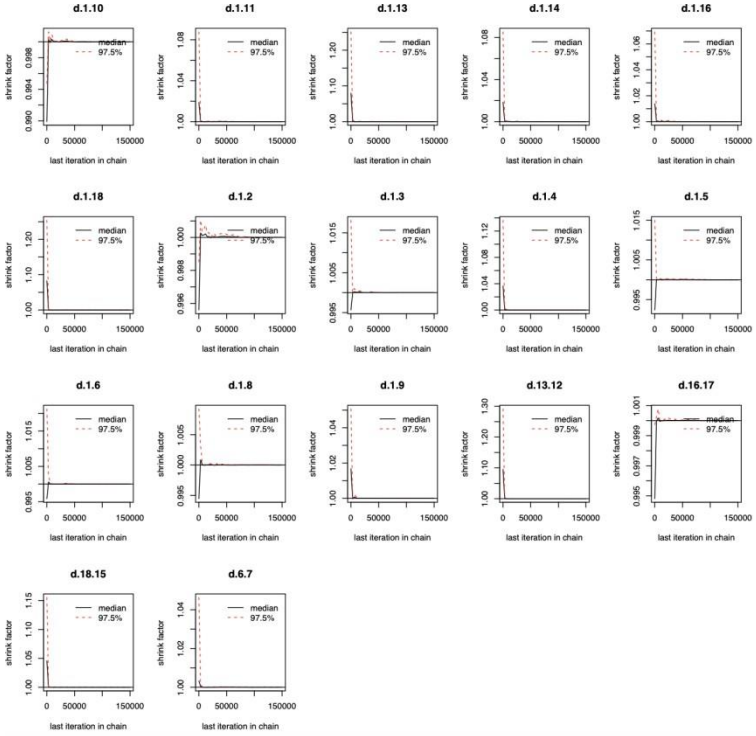


Figure S16. Bayesian Ranking Profiles of Comparable Treatments on Efficacy and Safety for Patients With Advanced Wild-type Non-small Cell Lung Cancer (NSCLC)

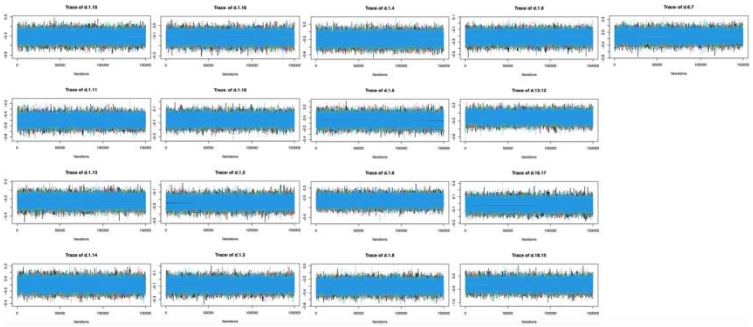
(A) Number (percentage of surface under the cumulative ranking curve (SUCRA)) in each cell indicates the probability of each treatment being ranked from first (high value) to last (low value) on overall overall survival (OS), OS for subgroups, objective response rate (ORR), and grade >=3 adverse events (>=3AEs). (B) Number (percentage of SUCRA) in each cell indicates the probability of each treatment being ranked from first (high value) to last (low value) on overall progression-free survival (PFS), PFS for subgroups, ORR and >=3AEs. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed; yr=year; NSQ=non-squamous; SQU=squamous; PD-L1=programmed-death ligand 1; TMB=tumor mutation burden; ECOG PS= Eastern Cooperative Oncology Group Performance Status; Meta=metastasis; CNS=central nervous system.

OS

A. Brooks-Gelman-Rubin diagnostic

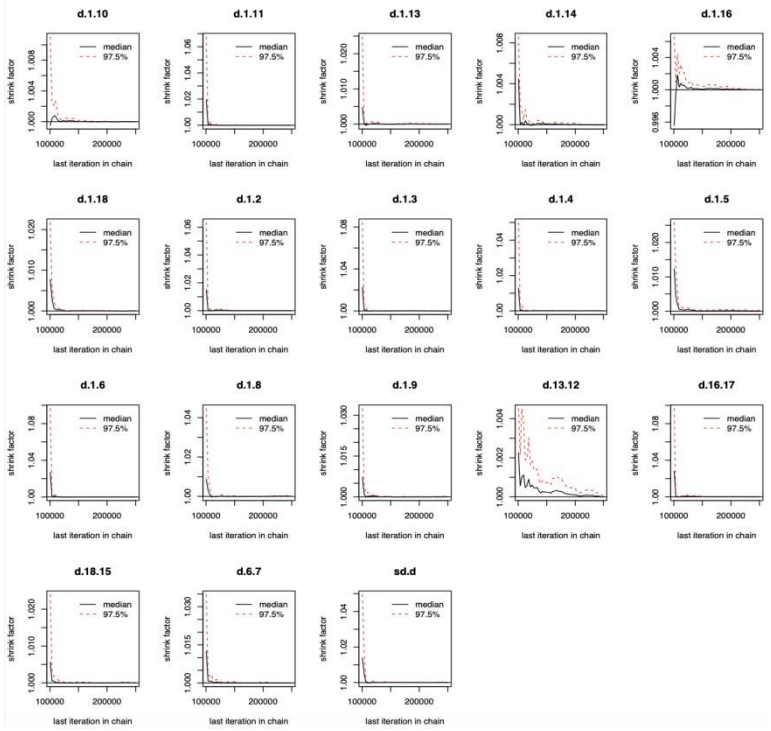


B. Trace plot

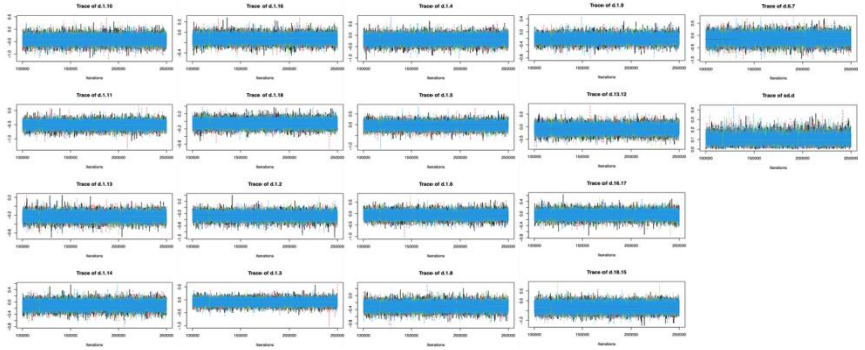


PFS

C. Brooks-Gelman-Rubin diagnostic

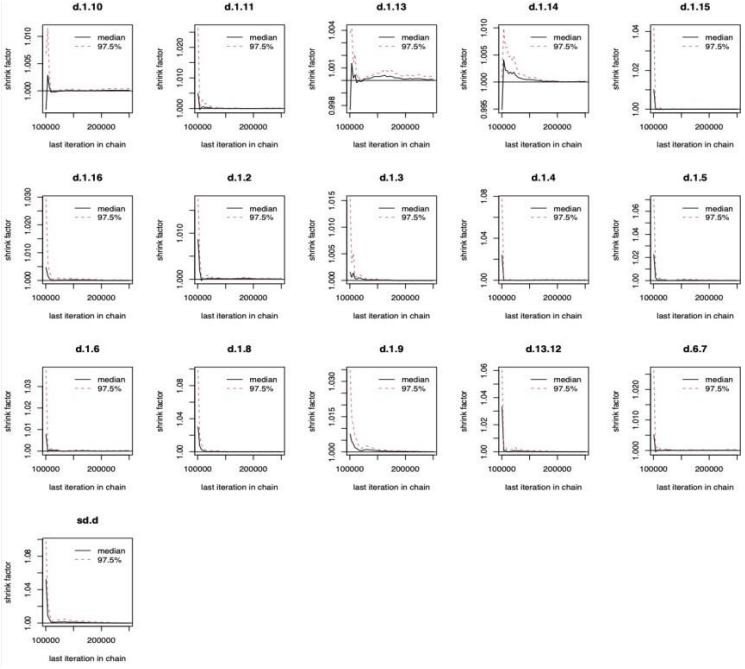


D. Trace plot

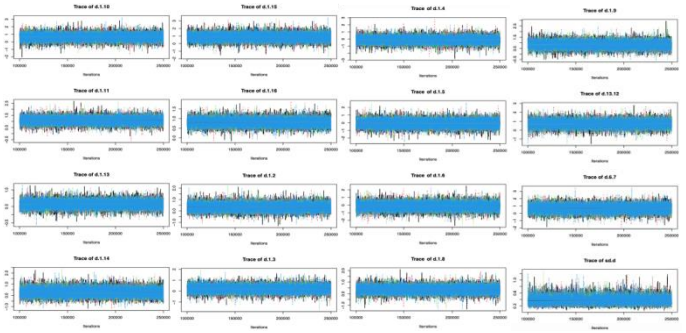


ORR

E. Brooks-Gelman-Rubin diagnostic

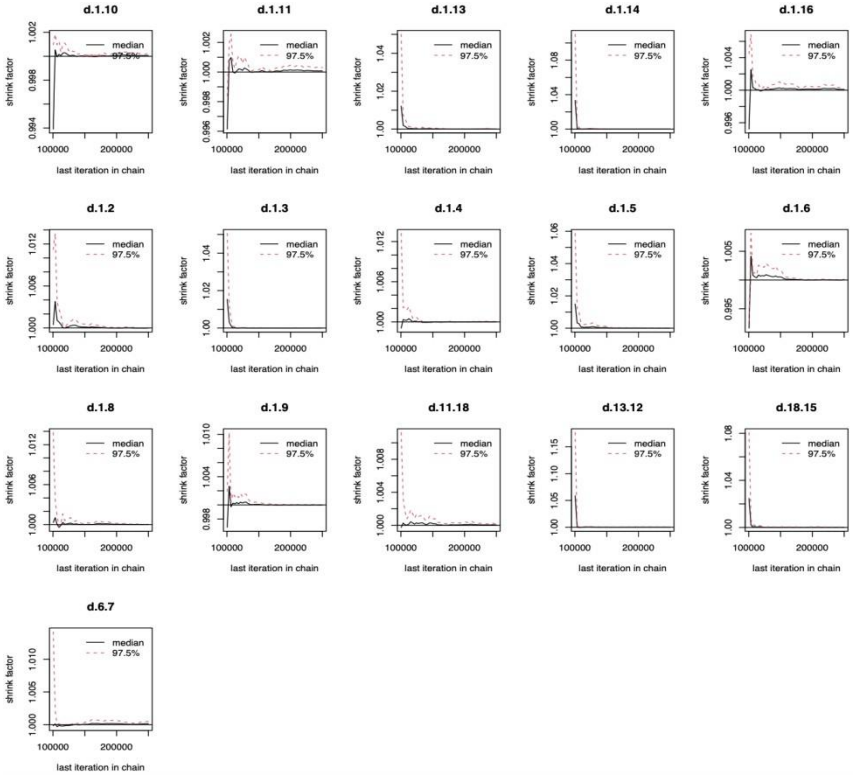


F. Trace plot

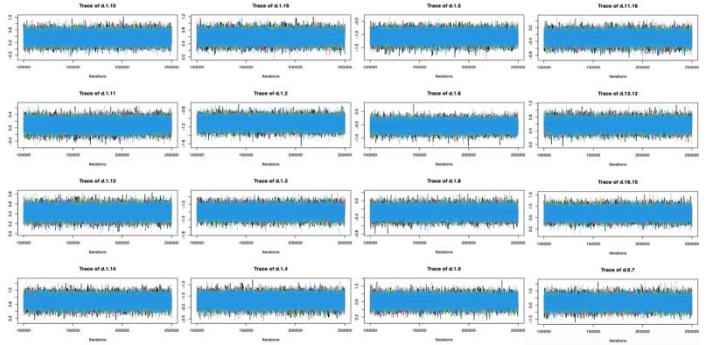


>=3 AEs

G. Brooks-Gelman-Rubin diagnostic

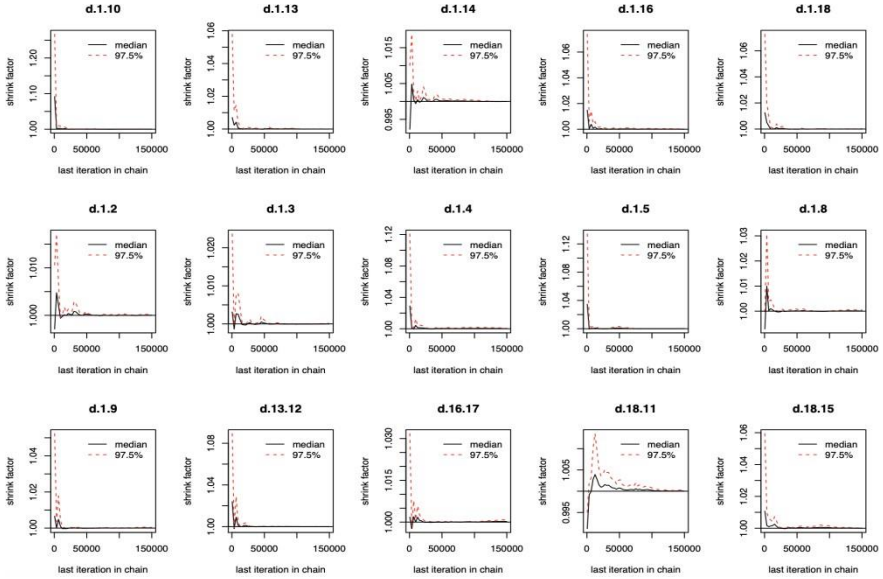


H. Trace plot

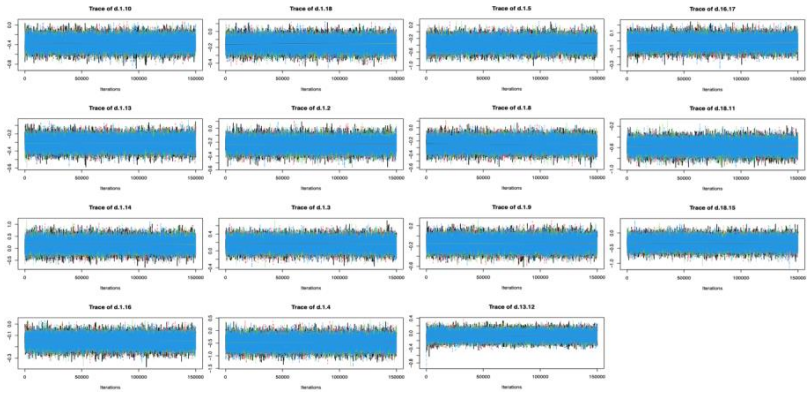


OS for non-squamous subgroup

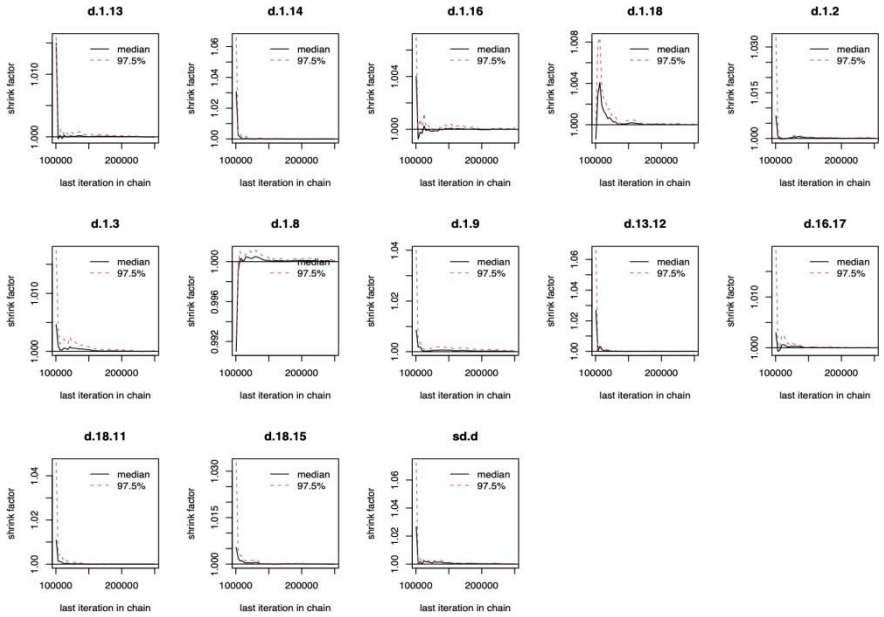
I. Brooks-Gelman-Rubin diagnostic



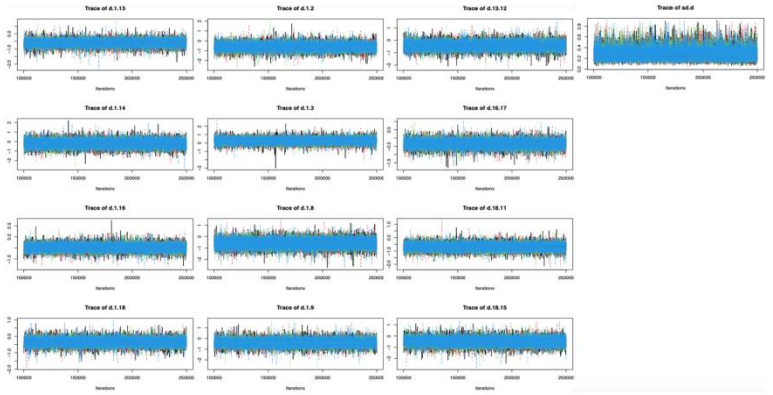
J. Trace plot



K. Brooks-Gelman-Rubin diagnostic

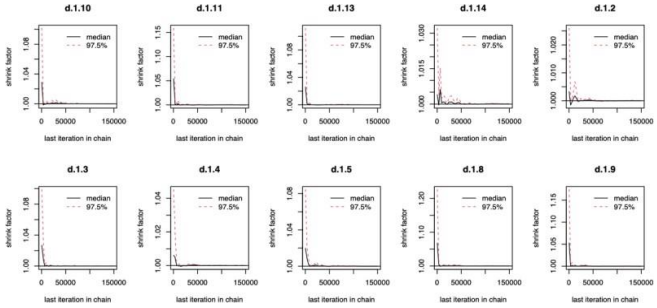


L. Trace plot

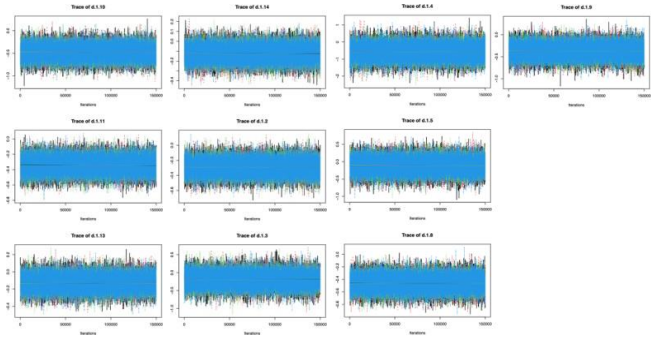


OS for squamous subgroup

M. Brooks-Gelman-Rubin diagnostic

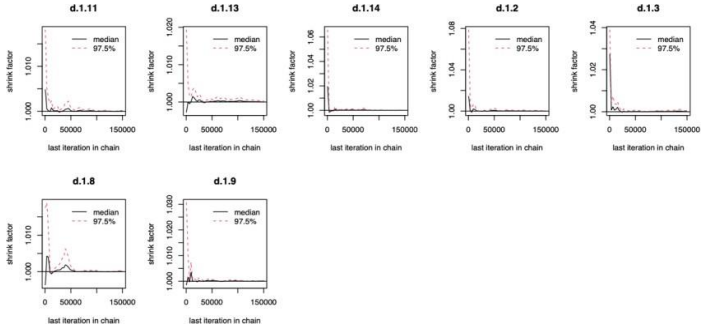


N. Trace plot



PFS for squamous subgroup

O. Brooks-Gelman-Rubin diagnostic



P. Trace plot

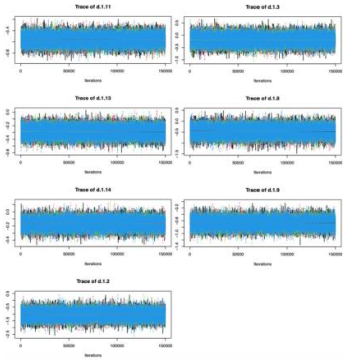
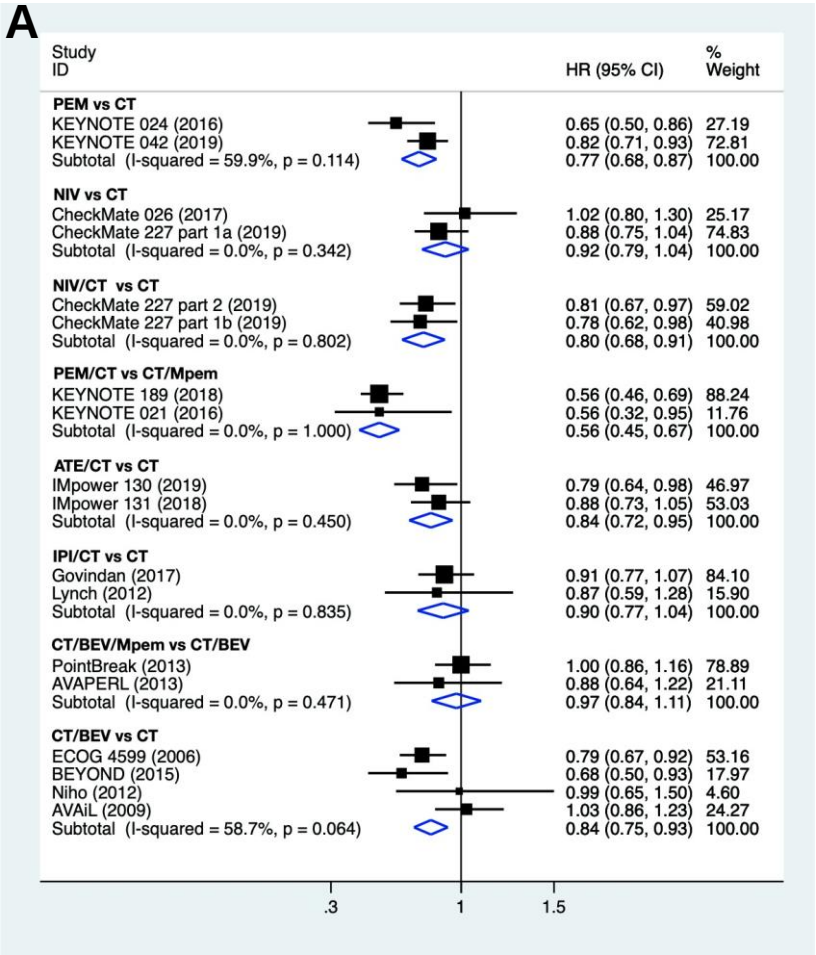


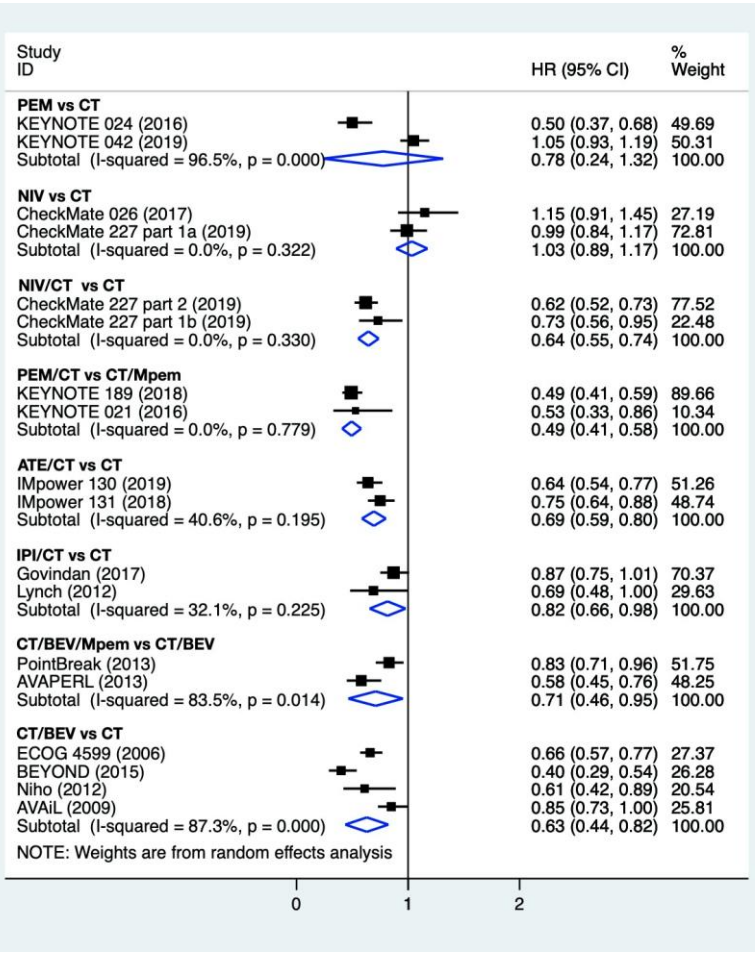
Figure S17. Convergence of the Four Chains Established by Inspection of the Brooks-Gelman-Rubin Diagnostic and the Trace Plot for Overall Survival (OS) (A and B), Progression-free Survival (PFS) (C and D), Objective Response Rate (ORR) (E and F), Grade >= 3 Adverse Events (>= 3 AEs) (G and H), OS for Non-squamous Subgroup (I and J), PFS for Non-squamous Subgroup (K and L), OS for Squamous Subgroup (M and N), PFS for Squamous Subgroup (O and P)

A


Study ID
HR (95% CI)
% Weight
PEM vs CT
KEYNOTE 024 (2016) 0.65 (0.50, 0.86) 27.19
KEYNOTE 042 (2019) 0.82 (0.71, 0.93) 72.81
Subtotal (I-squared = 59.9%, p = 0.114) 0.77 (0.68, 0.87) 100.00
NIV vs CT
CheckMate 026 (2017) 1.02 (0.80, 1.30) 25.17
CheckMate 227 part 1a (2019) 0.88 (0.75, 1.04) 74.83
Subtotal (I-squared = 0.0%, p = 0.342) 0.92 (0.79, 1.04) 100.00
NIV/CT vs CT
CheckMate 227 part 2 (2019) 0.81 (0.67, 0.97) 59.02
CheckMate 227 part 1b (2019) 0.78 (0.62, 0.98) 40.98
Subtotal (I-squared = 0.0%, p = 0.802) 0.80 (0.68, 0.91) 100.00
PEM/CT vs CT/Mpem
KEYNOTE 189 (2018) 0.56 (0.46, 0.69) 88.24
KEYNOTE 021 (2016) 0.56 (0.32, 0.95) 11.76
Subtotal (I-squared = 0.0%, p = 1.000) 0.56 (0.45, 0.67) 100.00
ATE/CT vs CT
IMpower 130 (2019) 0.79 (0.64, 0.98) 46.97
IMpower 131 (2018) 0.88 (0.73, 1.05) 53.03
Subtotal (I-squared = 0.0%, p = 0.450) 0.84 (0.72, 0.95) 100.00
IPI/CT vs CT
Govindan (2017) 0.91 (0.77, 1.07) 84.10
Lynch (2012) 0.87 (0.59, 1.28) 15.90
Subtotal (I-squared = 0.0%, p = 0.835) 0.90 (0.77, 1.04) 100.00
CT/BEV/Mpem vs CT/BEV
PointBreak (2013) 1.00 (0.86, 1.16) 78.89
AVAPERL (2013) 0.88 (0.64, 1.22) 21.11
Subtotal (I-squared = 0.0%, p = 0.471) 0.97 (0.84, 1.11) 100.00
CT/BEV vs CT
ECOG 4599 (2006) 0.79 (0.67, 0.92) 53.16
BEYOND (2015) 0.68 (0.50, 0.93) 17.97
Niho (2012) 0.99 (0.65, 1.50) 4.60
AVAiL (2009) 1.03 (0.86, 1.23) 24.27
Subtotal (I-squared = 58.7%, p = 0.064) 0.84 (0.75, 0.93) 100.00
.3
1
1.5

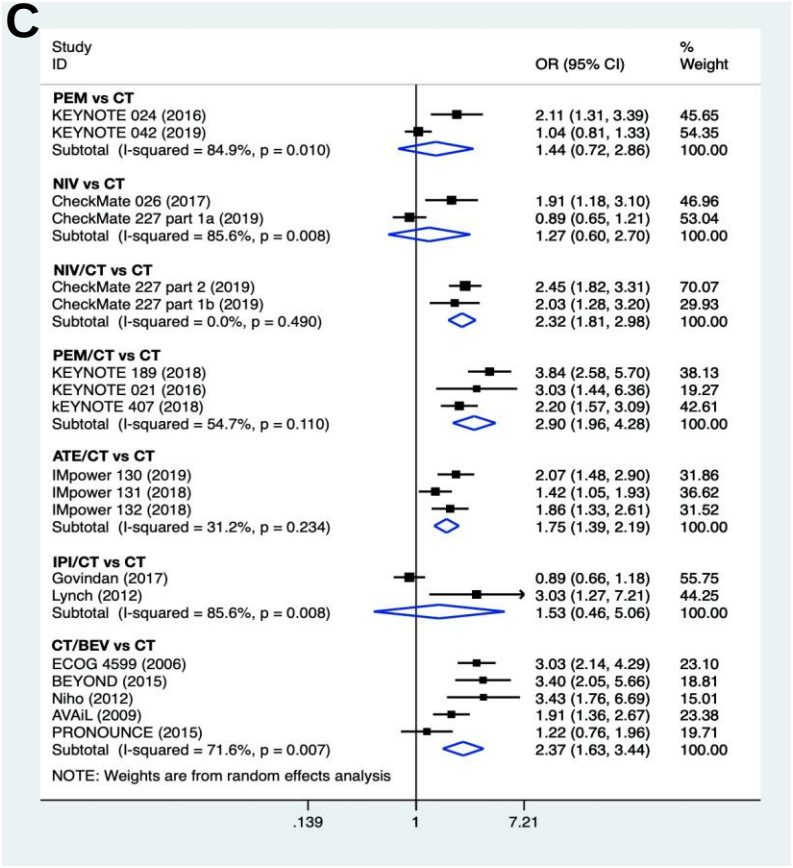
OS

B


Study ID
HR (95% CI)
% Weight
PEM vs CT
KEYNOTE 024 (2016) 0.50 (0.37, 0.68) 49.69
KEYNOTE 042 (2019) 1.05 (0.93, 1.19) 50.31
Subtotal (I-squared = 96.5%, p = 0.000) 0.78 (0.24, 1.32) 100.00
NIV vs CT
CheckMate 026 (2017) 1.15 (0.91, 1.45) 27.19
CheckMate 227 part 1a (2019) 0.99 (0.84, 1.17) 72.81
Subtotal (I-squared = 0.0%, p = 0.322) 1.03 (0.89, 1.17) 100.00
NIV/CT vs CT
CheckMate 227 part 2 (2019) 0.62 (0.52, 0.73) 77.52
CheckMate 227 part 1b (2019) 0.73 (0.56, 0.95) 22.48
Subtotal (I-squared = 0.0%, p = 0.330) 0.64 (0.55, 0.74) 100.00
PEM/CT vs CT/Mpem
KEYNOTE 189 (2018) 0.49 (0.41, 0.59) 89.66
KEYNOTE 021 (2016) 0.53 (0.33, 0.86) 10.34
Subtotal (I-squared = 0.0%, p = 0.779) 0.49 (0.41, 0.58) 100.00
ATE/CT vs CT
IMpower 130 (2019) 0.64 (0.54, 0.77) 51.26
IMpower 131 (2018) 0.75 (0.64, 0.88) 48.74
Subtotal (I-squared = 40.6%, p = 0.195) 0.69 (0.59, 0.80) 100.00
IPI/CT vs CT
Govindan (2017) 0.87 (0.75, 1.01) 70.37
Lynch (2012) 0.69 (0.48, 1.00) 29.63
Subtotal (I-squared = 32.1%, p = 0.225) 0.82 (0.66, 0.98) 100.00
CT/BEV/Mpem vs CT/BEV
PointBreak (2013) 0.83 (0.71, 0.96) 51.75
AVAPERL (2013) 0.58 (0.45, 0.76) 48.25
Subtotal (I-squared = 83.5%, p = 0.014) 0.71 (0.46, 0.95) 100.00
CT/BEV vs CT
ECOG 4599 (2006) 0.66 (0.57, 0.77) 27.37
BEYOND (2015) 0.40 (0.29, 0.54) 26.28
Niho (2012) 0.61 (0.42, 0.89) 20.54
AVAiL (2009) 0.85 (0.73, 1.00) 25.81
Subtotal (I-squared = 87.3%, p = 0.000) 0.63 (0.44, 0.82) 100.00
NOTE: Weights are from random effects analysis
0
1
2

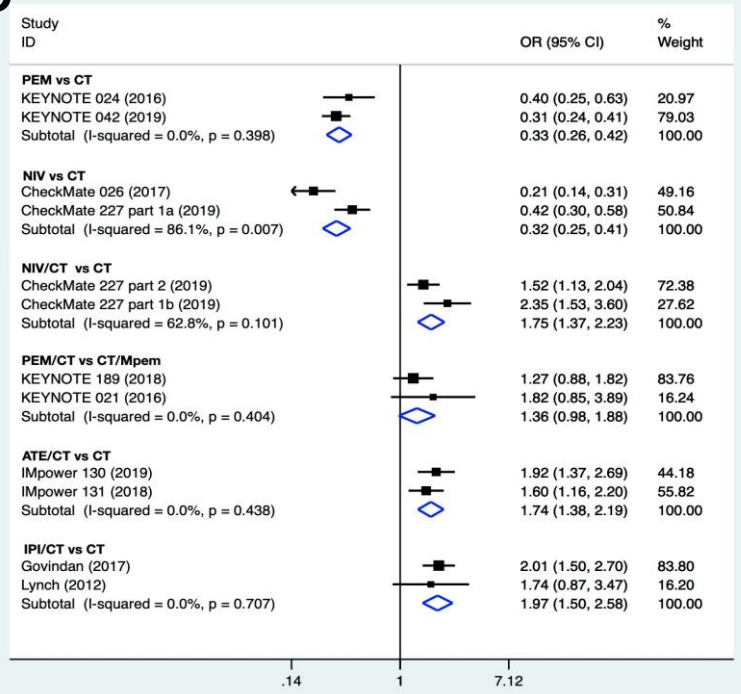
PFS

C


Study ID
OR (95% CI)
% Weight
PEM vs CT
KEYNOTE 024 (2016) 2.11 (1.31, 3.39) 45.65
KEYNOTE 042 (2019) 1.04 (0.81, 1.33) 54.35
Subtotal (I-squared = 84.9%, p = 0.010) 1.44 (0.72, 2.86) 100.00
NIV vs CT
CheckMate 026 (2017) 1.91 (1.18, 3.10) 46.96
CheckMate 227 part 1a (2019) 0.89 (0.65, 1.21) 53.04
Subtotal (I-squared = 85.6%, p = 0.008) 1.27 (0.60, 2.70) 100.00
NIV/CT vs CT
CheckMate 227 part 2 (2019) 2.45 (1.82, 3.31) 70.07
CheckMate 227 part 1b (2019) 2.03 (1.28, 3.20) 29.93
Subtotal (I-squared = 0.0%, p = 0.490) 2.32 (1.81, 2.98) 100.00
PEM/CT vs CT
KEYNOTE 189 (2018) 3.84 (2.58, 5.70) 38.13
KEYNOTE 021 (2016) 3.03 (1.44, 6.36) 19.27
kEYNOTE 407 (2018) 2.20 (1.57, 3.09) 42.61
Subtotal (I-squared = 54.7%, p = 0.110) 2.90 (1.96, 4.28) 100.00
ATE/CT vs CT
IMpower 130 (2019) 2.07 (1.48, 2.90) 31.86
IMpower 131 (2018) 1.42 (1.05, 1.93) 36.62
IMpower 132 (2018) 1.86 (1.33, 2.61) 31.52
Subtotal (I-squared = 31.2%, p = 0.234) 1.75 (1.39, 2.19) 100.00
IPI/CT vs CT
Govindan (2017) 0.89 (0.66, 1.18) 55.75
Lynch (2012) 3.03 (1.27, 7.21) 44.25
Subtotal (I-squared = 85.6%, p = 0.008) 1.53 (0.46, 5.06) 100.00
CT/BEV vs CT
ECOG 4599 (2006) 3.03 (2.14, 4.29) 23.10
BEYOND (2015) 3.40 (2.05, 5.66) 18.81
Niho (2012) 3.43 (1.76, 6.69) 15.01
AVAiL (2009) 1.91 (1.36, 2.67) 23.38
PRONOUNCE (2015) 1.22 (0.76, 1.96) 19.71
Subtotal (I-squared = 71.6%, p = 0.007) 2.37 (1.63, 3.44) 100.00
NOTE: Weights are from random effects analysis
.139
1
7.21

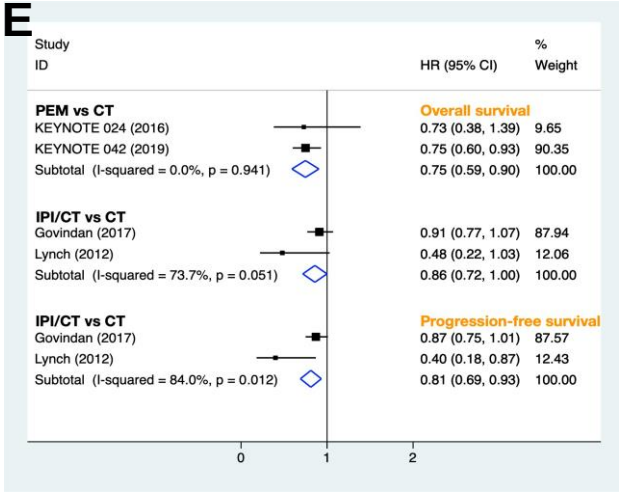
ORR

D


Study ID
OR (95% CI)
% Weight
PEM vs CT
KEYNOTE 024 (2016) 0.40 (0.25, 0.63) 20.97
KEYNOTE 042 (2019) 0.31 (0.24, 0.41) 79.03
Subtotal (I-squared = 0.0%, p = 0.398) 0.33 (0.26, 0.42) 100.00
NIV vs CT
CheckMate 026 (2017) 0.21 (0.14, 0.31) 49.16
CheckMate 227 part 1a (2019) 0.42 (0.30, 0.58) 50.84
Subtotal (I-squared = 86.1%, p = 0.007) 0.32 (0.25, 0.41) 100.00
NIV/CT vs CT
CheckMate 227 part 2 (2019) 1.52 (1.13, 2.04) 72.38
CheckMate 227 part 1b (2019) 2.35 (1.53, 3.60) 27.62
Subtotal (I-squared = 62.8%, p = 0.101) 1.75 (1.37, 2.23) 100.00
PEM/CT vs CT/Mpem
KEYNOTE 189 (2018) 1.27 (0.88, 1.82) 83.76
KEYNOTE 021 (2016) 1.82 (0.85, 3.89) 16.24
Subtotal (I-squared = 0.0%, p = 0.404) 1.36 (0.98, 1.88) 100.00
ATE/CT vs CT
IMpower 130 (2019) 1.92 (1.37, 2.69) 44.18
IMpower 131 (2018) 1.60 (1.16, 2.20) 55.82
Subtotal (I-squared = 0.0%, p = 0.438) 1.74 (1.38, 2.19) 100.00
IPI/CT vs CT
Govindan (2017) 2.01 (1.50, 2.70) 83.80
Lynch (2012) 1.74 (0.87, 3.47) 16.20
Subtotal (I-squared = 0.0%, p = 0.707) 1.97 (1.50, 2.58) 100.00
.14
1
7.12

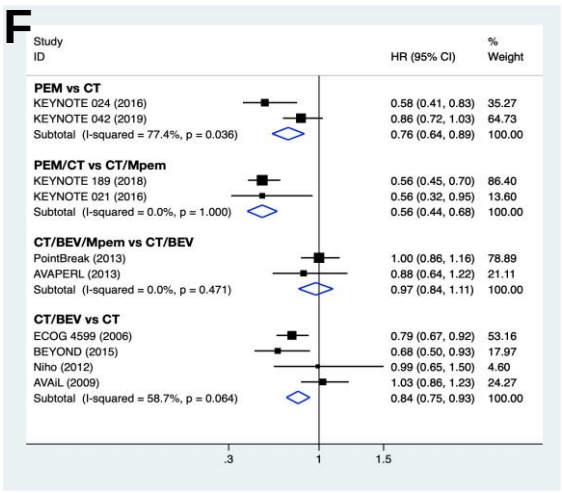
>=3 AEs

E



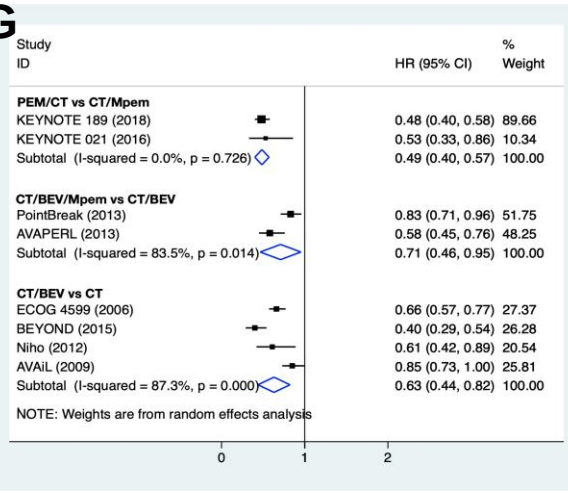
Squamous OS and PFS

F



Non-squamous OS

G



Non-squamous

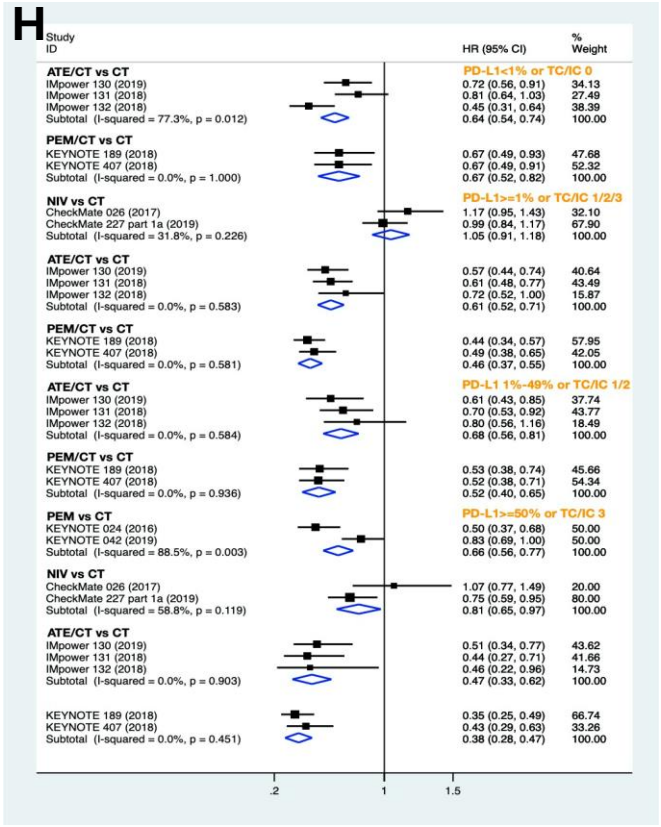
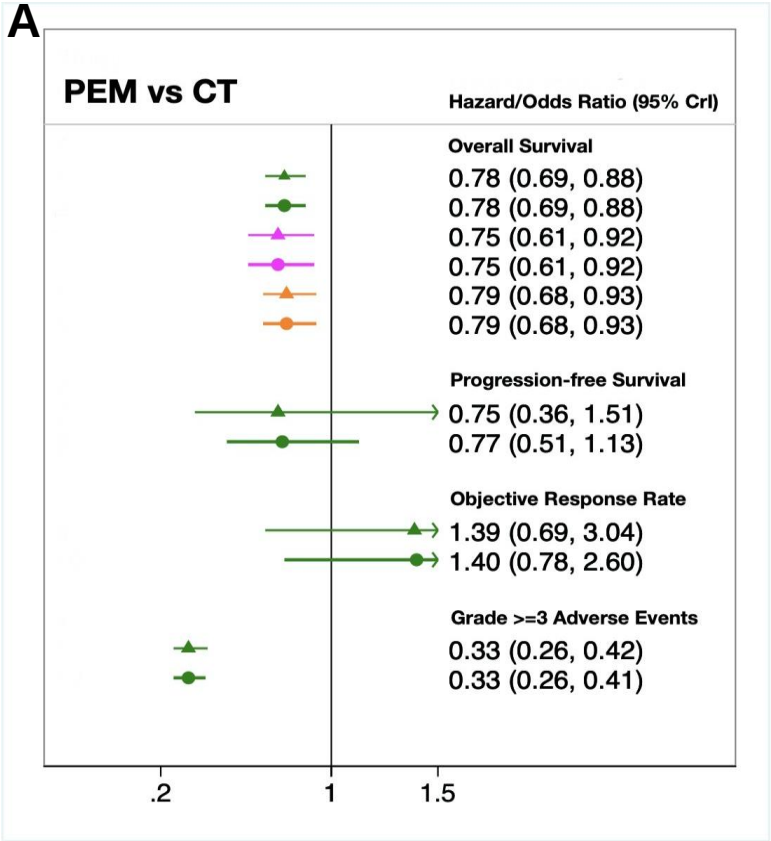
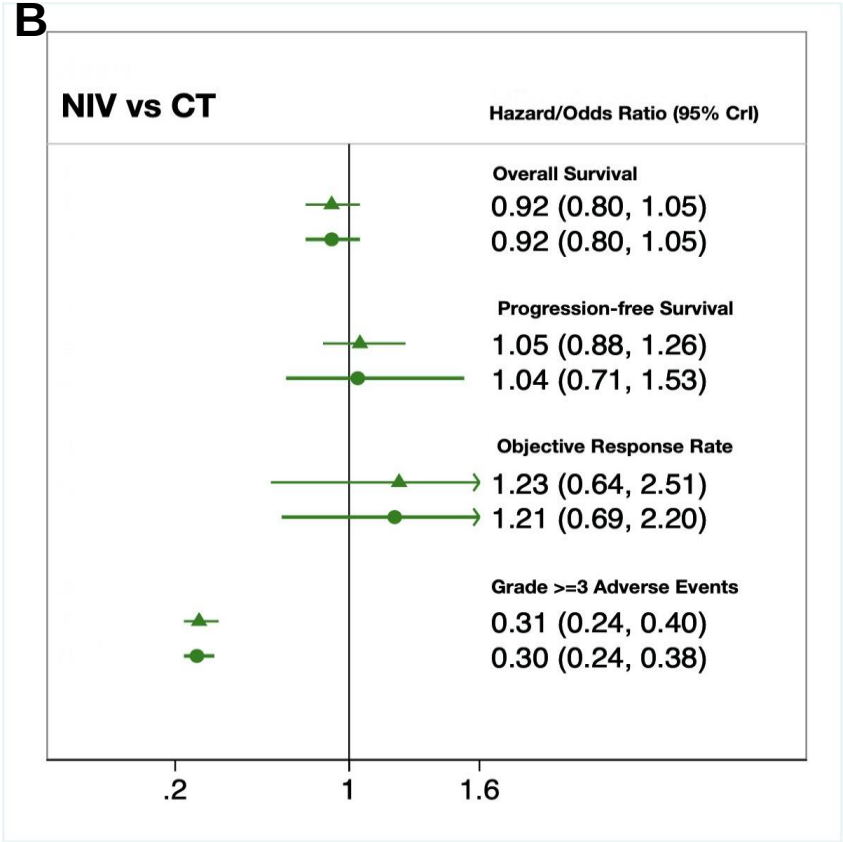
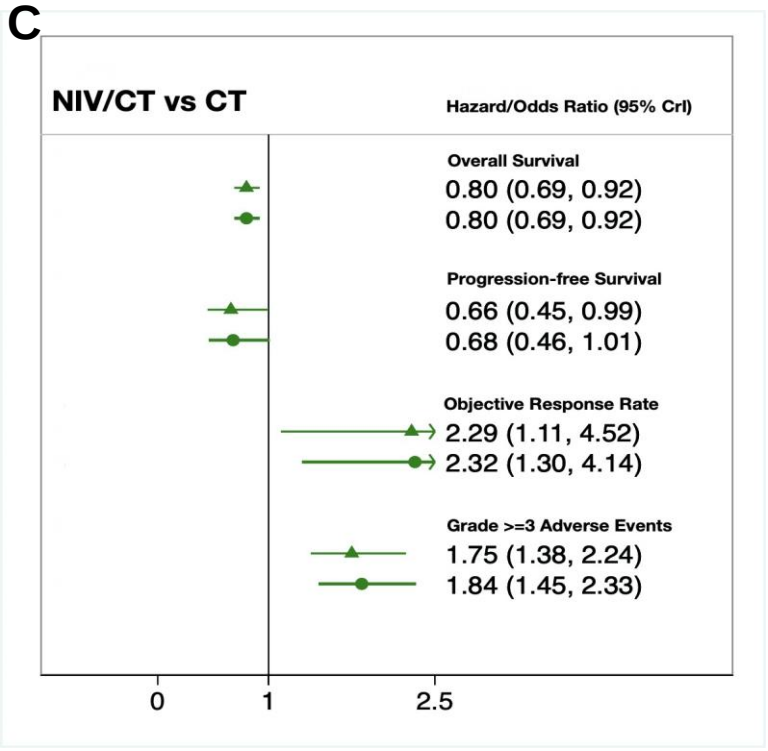


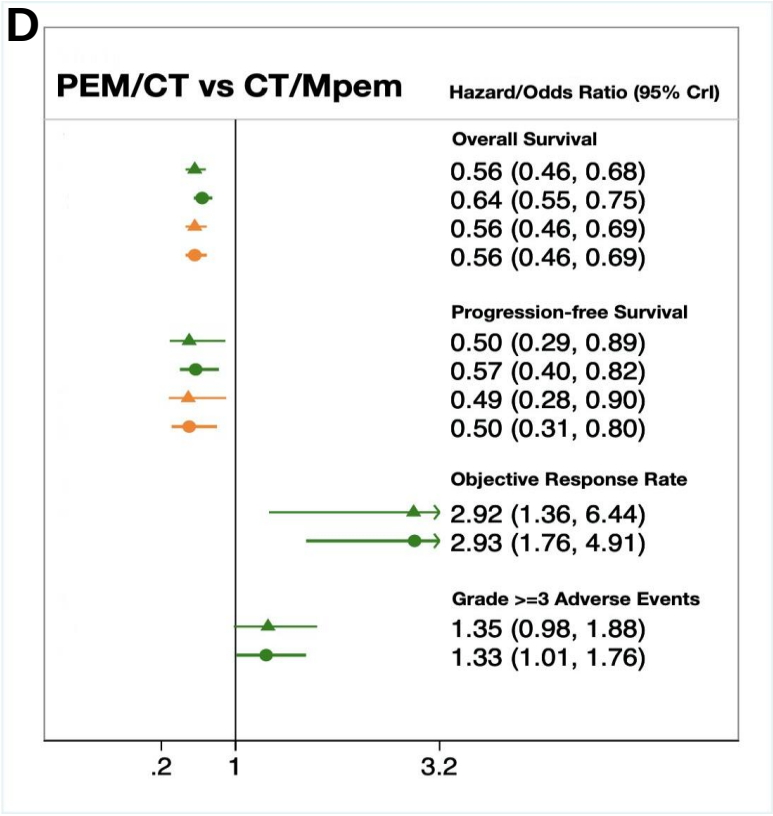
Figure S18. Forest Plots Depicting Results of Head-to-head Comparisons According to Frequentist Pairwise Meta-analyses on Different Outcomes in Advanced Wild-type Non-small Cell Lung Cancer (NSCLC)

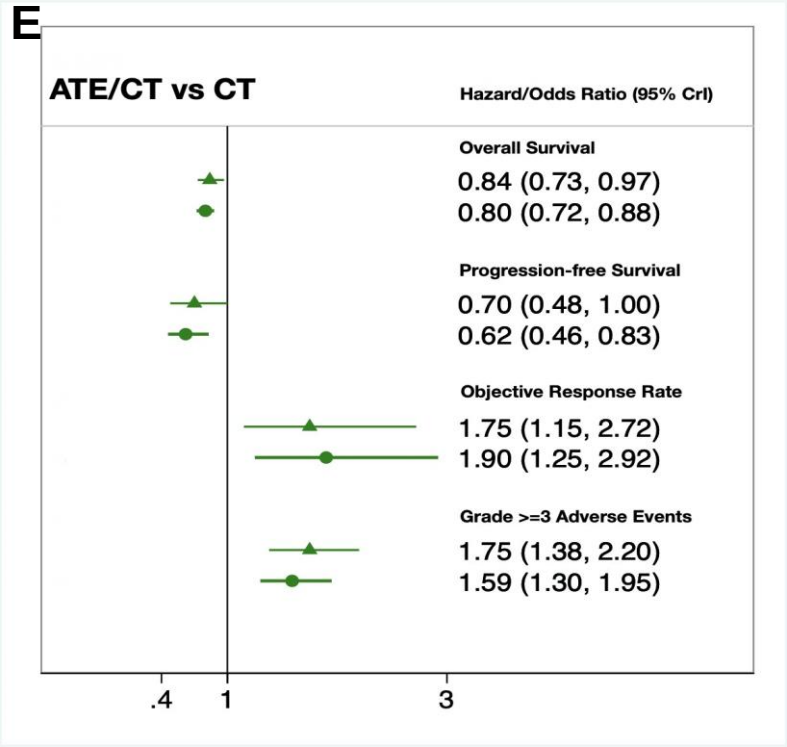
(A) Outcomes for overall survival (OS). (B) Outcomes for progression-free survival (PFS). (C) Outcomes for objective response rate (ORR). (D) Outcomes for adverse events of grade 3 or higher (>= 3 AEs). (E) Outcomes for OS and PFS in squamous subgroup. (F and G) Outcomes for OS and PFS in non-squamous subgroup. (H) Outcomes for PFS according to programmed-death ligand 1 (PD-L1) expression levels. HR=hazard ratio; OR=odds ratio; CI=confidence interval; PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed; TC=tumor cells; IC=tumor-infiltrating immune cells.

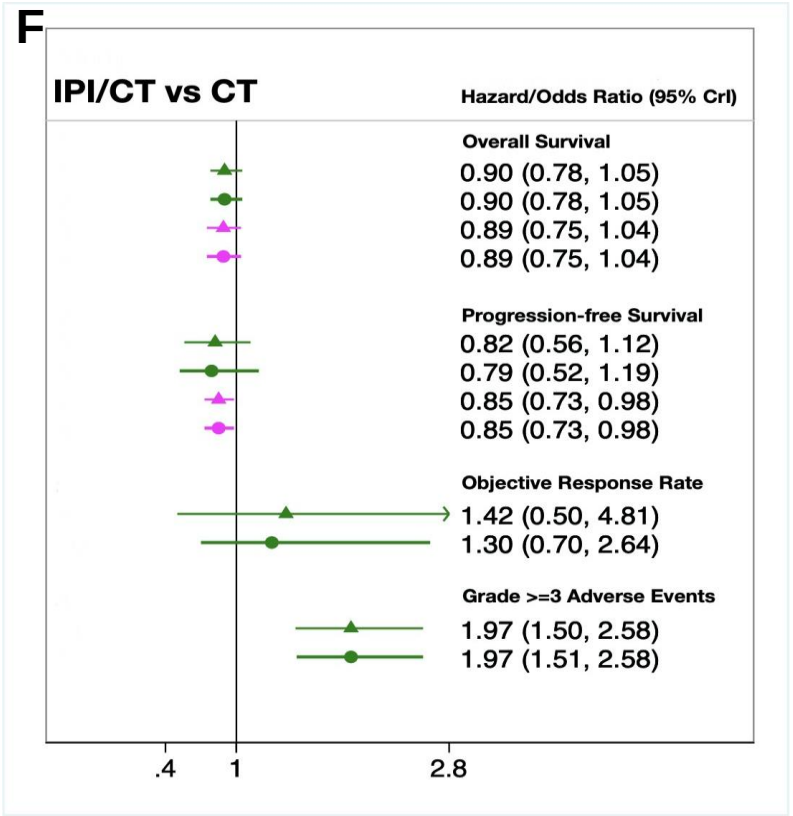

A
PEM vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.78 (0.69, 0.88)
0.78 (0.69, 0.88)
0.75 (0.61, 0.92)
0.75 (0.61, 0.92)
0.79 (0.68, 0.93)
0.79 (0.68, 0.93)
Progression-free Survival
0.75 (0.36, 1.51)
0.77 (0.51, 1.13)
Objective Response Rate
1.39 (0.69, 3.04)
1.40 (0.78, 2.60)
Grade >=3 Adverse Events
0.33 (0.26, 0.42)
0.33 (0.26, 0.41)
.2
1
1.5


B
NIV vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.92 (0.80, 1.05)
0.92 (0.80, 1.05)
Progression-free Survival
1.05 (0.88, 1.26)
1.04 (0.71, 1.53)
Objective Response Rate
1.23 (0.64, 2.51)
1.21 (0.69, 2.20)
Grade >=3 Adverse Events
0.31 (0.24, 0.40)
0.30 (0.24, 0.38)
.2
1
1.6

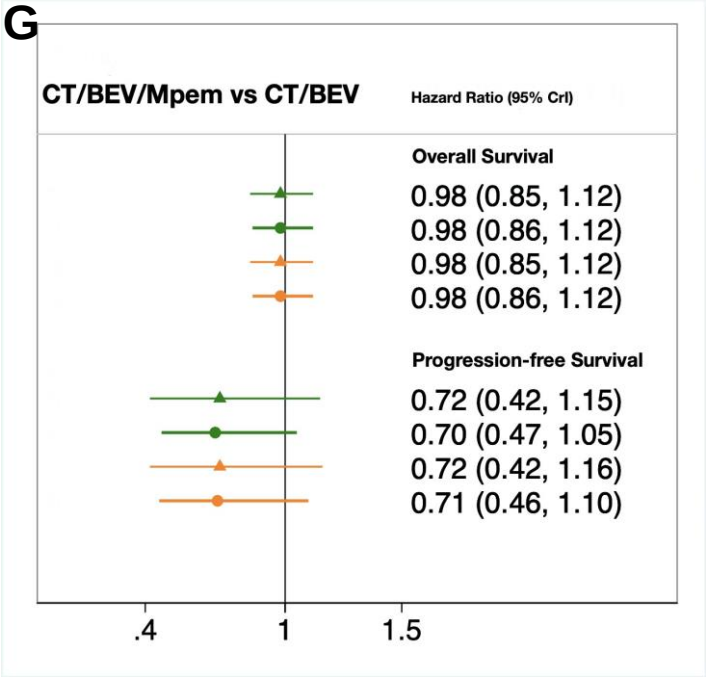

C
NIV/CT vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.80 (0.69, 0.92)
0.80 (0.69, 0.92)
Progression-free Survival
0.66 (0.45, 0.99)
0.68 (0.46, 1.01)
Objective Response Rate
2.29 (1.11, 4.52)
2.32 (1.30, 4.14)
Grade >=3 Adverse Events
1.75 (1.38, 2.24)
1.84 (1.45, 2.33)
0
1
2.5


D
PEM/CT vs CT/Mpem
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.56 (0.46, 0.68)
0.64 (0.55, 0.75)
0.56 (0.46, 0.69)
0.56 (0.46, 0.69)
Progression-free Survival
0.50 (0.29, 0.89)
0.57 (0.40, 0.82)
0.49 (0.28, 0.90)
0.50 (0.31, 0.80)
Objective Response Rate
2.92 (1.36, 6.44)
2.93 (1.76, 4.91)
Grade >=3 Adverse Events
1.35 (0.98, 1.88)
1.33 (1.01, 1.76)
.2
1
3.2

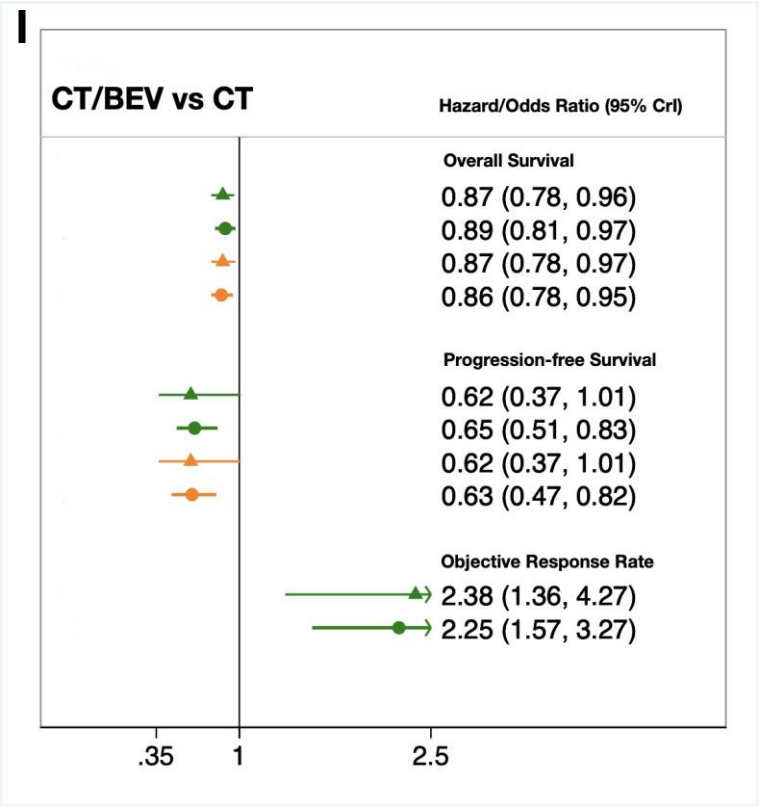

E
ATE/CT vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.84 (0.73, 0.97)
0.80 (0.72, 0.88)
Progression-free Survival
0.70 (0.48, 1.00)
0.62 (0.46, 0.83)
Objective Response Rate
1.75 (1.15, 2.72)
1.90 (1.25, 2.92)
Grade >=3 Adverse Events
1.75 (1.38, 2.20)
1.59 (1.30, 1.95)
.4
1
3


F
IPI/CT vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.90 (0.78, 1.05)
0.90 (0.78, 1.05)
0.89 (0.75, 1.04)
0.89 (0.75, 1.04)
Progression-free Survival
0.82 (0.56, 1.12)
0.79 (0.52, 1.19)
0.85 (0.73, 0.98)
0.85 (0.73, 0.98)
Objective Response Rate
1.42 (0.50, 4.81)
1.30 (0.70, 2.64)
Grade >=3 Adverse Events
1.97 (1.50, 2.58)
1.97 (1.51, 2.58)
.4
1
2.8

G


CT/BEV/Mpem vs CT/BEV
Hazard Ratio (95% CrI)
Overall Survival
0.98 (0.85, 1.12)
0.98 (0.86, 1.12)
0.98 (0.85, 1.12)
0.98 (0.86, 1.12)
Progression-free Survival
0.72 (0.42, 1.15)
0.70 (0.47, 1.05)
0.72 (0.42, 1.16)
0.71 (0.46, 1.10)
.4
1
1.5

I


CT/BEV vs CT
Hazard/Odds Ratio (95% CrI)
Overall Survival
0.87 (0.78, 0.96)
0.89 (0.81, 0.97)
0.87 (0.78, 0.97)
0.86 (0.78, 0.95)
Progression-free Survival
0.62 (0.37, 1.01)
0.65 (0.51, 0.83)
0.62 (0.37, 1.01)
0.63 (0.47, 0.82)
Objective Response Rate
2.38 (1.36, 4.27)
2.25 (1.57, 3.27)
.35
1
2.5

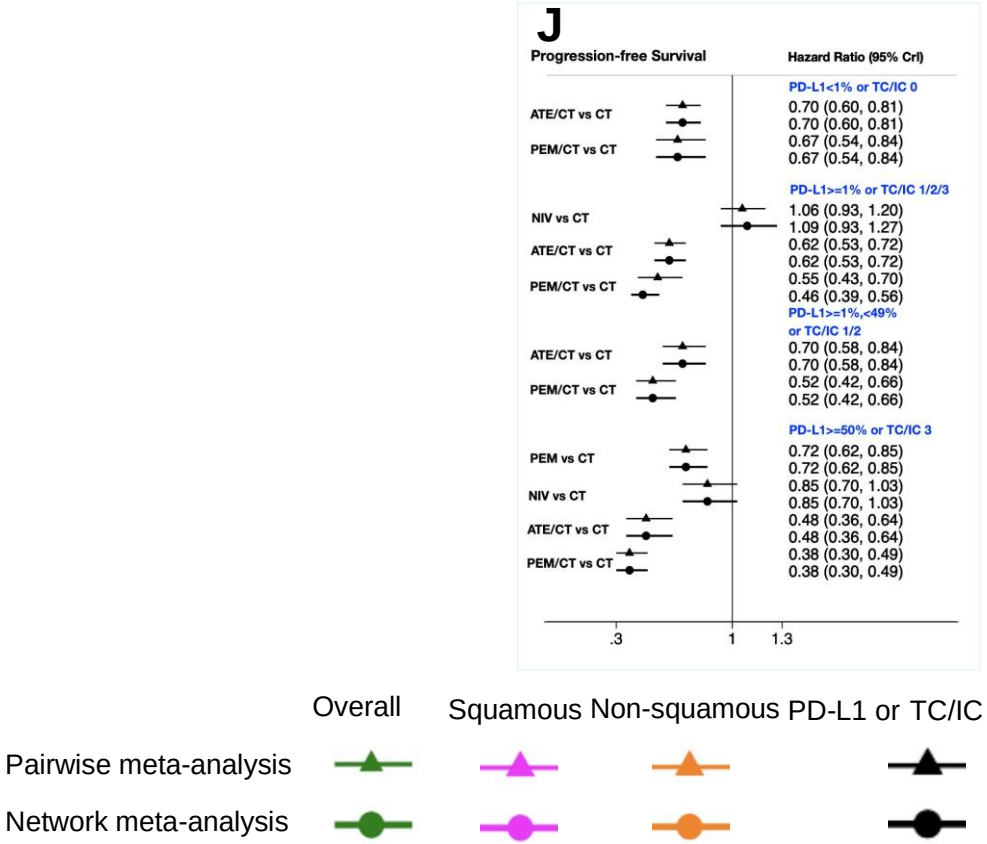
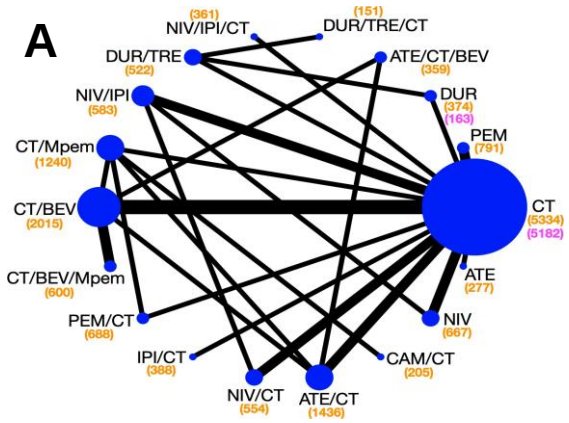
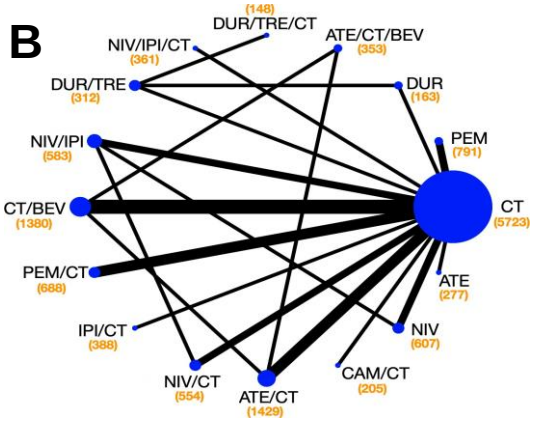
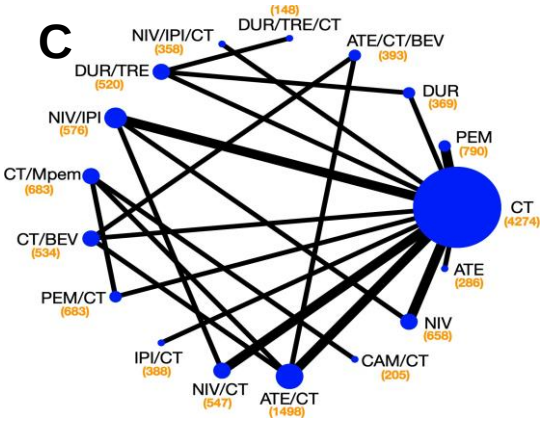


Figure S19. Forest Plots Depicting Results of Head-to-head Comparisons According to Bayesian Pairwise and Network Meta-analyses
Results of all comparisons in overall population (green), squamous (red), non-squamous (orange) and programmed-death ligand 1 (PD-L1) (black) subgroups were consistent between pairwise and network meta-analysis. CrI = credible interval; PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; IPI=ipilimumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed. TC=tumor cells; IC=tumor-infiltrating immune cells.







OS and PFS

ORR

>=3 AEs

D **PFS**

OS

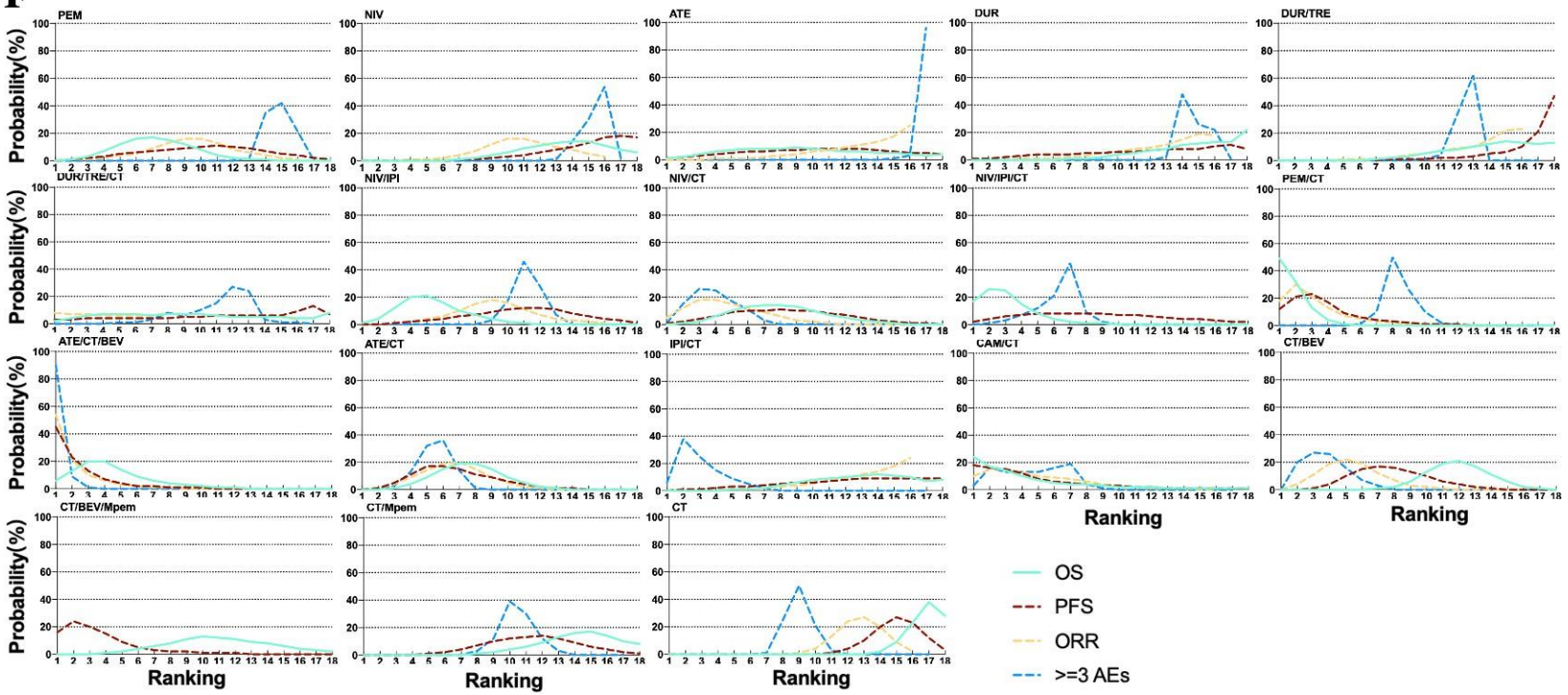
PEM	0.73 (0.39, 1.33)	0.99 (0.45, 2.09)	0.87 (0.37, 1.99)	0.61 (0.28, 1.28)	0.91 (0.33, 2.43)	0.97 (0.52, 1.76)	1.12 (0.59, 2.05)	1.12 (0.51, 2.36)	1.61 (0.83, 3.05)	1.96 (0.92, 4.11)	1.24 (0.70, 2.14)	0.88 (0.41, 1.83)	1.55 (0.64, 3.65)	1.16 (0.67, 1.97)	1.65 (0.82, 3.30)	0.94 (0.52, 1.66)	0.76 (0.48, 1.17)
0.85 (0.71, 1.02)	**NIV**	1.35 (0.64, 2.88)	1.20 (0.53, 2.74)	0.83 (0.40, 1.77)	1.24 (0.46, 3.35)	1.33 (0.79, 2.22)	1.53 (0.85, 2.76)	1.54 (0.72, 3.24)	**2.20 (1.17, 4.17)**	**2.69 (1.30, 5.65)**	1.70 (0.99, 2.93)	1.20 (0.57, 2.52)	2.12 (0.90, 5.02)	1.59 (0.95, 2.70)	**2.26 (1.15, 4.53)**	1.29 (0.74, 2.28)	1.04 (0.68, 1.60)
0.94 (0.72, 1.24)	1.11 (0.84, 1.47)	**ATE**	0.89 (0.35, 2.27)	0.62 (0.26, 1.48)	0.92 (0.31, 2.73)	0.98 (0.46, 2.08)	1.13 (0.53, 2.41)	1.13 (0.47, 2.72)	1.62 (0.75, 3.56)	1.98 (0.84, 4.75)	1.25 (0.62, 2.55)	0.88 (0.37, 2.12)	1.56 (0.60, 4.15)	1.17 (0.59, 2.36)	1.66 (0.74, 3.83)	0.96 (0.47, 1.97)	0.77 (0.41, 1.44)
0.82 (0.66, 1.00)	0.96 (0.77, 1.19)	0.87 (0.64, 1.17)	**DUR**	0.70 (0.34, 1.44)	1.04 (0.39, 2.75)	1.10 (0.49, 2.52)	1.28 (0.56, 2.92)	1.28 (0.50, 3.27)	1.83 (0.79, 4.32)	2.24 (0.89, 5.70)	1.41 (0.65, 3.10)	1.00 (0.39, 2.55)	1.77 (0.64, 4.96)	1.32 (0.62, 2.87)	1.88 (0.79, 4.63)	1.08 (0.49, 2.40)	0.87 (0.43, 1.77)
0.83 (0.68, 1.02)	0.98 (0.79, 1.21)	0.88 (0.66, 1.19)	1.02 (0.81, 1.29)	**DUR/TRE**	1.49 (0.78, 2.84)	1.59 (0.75, 3.35)	1.84 (0.86, 3.88)	1.84 (0.77, 4.39)	**2.64 (1.21, 5.75)**	**3.22 (1.37, 7.65)**	**2.03 (1.01, 4.11)**	1.44 (0.60, 3.42)	2.54 (0.97, 6.69)	1.90 (0.96, 3.79)	**2.70 (1.20, 6.19)**	1.55 (0.76, 3.18)	1.25 (0.67, 2.31)
0.95 (0.67, 1.33)	1.11 (0.79, 1.58)	1.00 (0.67, 1.51)	1.16 (0.81, 1.66)	1.14 (0.86, 1.50)	**DUR/TRE/CT**	1.07 (0.40, 2.84)	1.23 (0.46, 3.29)	1.23 (0.42, 3.63)	1.77 (0.64, 4.85)	2.16 (0.74, 6.34)	1.36 (0.53, 3.53)	0.96 (0.33, 2.84)	1.70 (0.53, 5.45)	1.27 (0.50, 3.28)	1.81 (0.65, 5.18)	1.04 (0.40, 2.73)	0.84 (0.34, 2.04)
1.06 (0.89, 1.27)	**1.25 (1.03, 1.51)**	1.13 (0.85, 1.50)	**1.30 (1.05, 1.62)**	**1.28 (1.03, 1.58)**	1.12 (0.79, 1.59)	**NIV/IPI**	1.16 (0.68, 1.96)	1.16 (0.55, 2.44)	1.66 (0.88, 3.13)	2.03 (0.98, 4.25)	1.28 (0.75, 2.20)	0.91 (0.43, 1.89)	1.60 (0.68, 3.77)	1.20 (0.72, 2.02)	1.70 (0.87, 3.40)	0.98 (0.56, 1.71)	0.79 (0.52, 1.20)
0.98 (0.81, 1.18)	1.15 (0.95, 1.41)	1.04 (0.78, 1.39)	1.20 (0.97, 1.50)	1.18 (0.95, 1.47)	1.04 (0.73, 1.47)	0.92 (0.76, 1.12)	**NIV/CT**	1.00 (0.47, 2.13)	1.44 (0.76, 2.75)	1.75 (0.84, 3.71)	1.11 (0.64, 1.93)	0.78 (0.37, 1.66)	1.38 (0.59, 3.3)	1.03 (0.62, 1.77)	1.47 (0.75, 2.97)	0.84 (0.48, 1.50)	0.68 (0.44, 1.05)
1.19 (0.95, 1.48)	**1.40 (1.11, 1.76)**	1.26 (0.92, 1.72)	**1.45 (1.13, 1.87)**	**1.43 (1.11, 1.83)**	1.25 (0.87, 1.82)	1.12 (0.89, 1.41)	1.21 (0.96, 1.53)	**NIV/IPI/CT**	1.43 (0.66, 3.14)	1.75 (0.74, 4.17)	1.11 (0.55, 2.24)	0.78 (0.33, 1.86)	1.38 (0.53, 3.66)	1.03 (0.52, 2.07)	1.47 (0.66, 3.37)	0.84 (0.41, 1.74)	0.68 (0.37, 1.26)
1.29 (1.06, 1.57)	**1.52 (1.23, 1.87)**	**1.37 (1.02, 1.84)**	**1.58 (1.25, 1.99)**	**1.55 (1.23, 1.94)**	1.36 (0.95, 1.94)	1.21 (0.98, 1.49)	**1.31 (1.06, 1.63)**	1.09 (0.85, 1.39)	**PEM/CT**	1.22 (0.58, 2.57)	0.77 (0.45, 1.33)	0.54 (0.25, 1.18)	0.96 (0.43, 2.16)	0.72 (0.43, 1.23)	1.02 (0.52, 2.07)	**0.59 (0.37, 0.95)**	**0.47 (0.29, 0.76)**
1.11 (0.88, 1.40)	**1.31 (1.03, 1.66)**	1.18 (0.86, 1.62)	**1.36 (1.05, 1.76)**	**1.33 (1.03, 1.72)**	1.17 (0.81, 1.71)	1.04 (0.82, 1.33)	1.13 (0.89, 1.45)	0.93 (0.71, 1.23)	0.86 (0.67, 1.10)	**ATE/CT/BEV**	0.63 (0.35, 1.12)	0.45 (0.19, 1.04)	0.79 (0.31, 1.97)	0.59 (0.33, 1.05)	0.84 (0.41, 1.74)	**0.48 (0.25, 0.92)**	**0.39 (0.21, 0.70)**
0.99 (0.84, 1.16)	1.16 (0.98, 1.38)	1.05 (0.80, 1.37)	1.21 (0.99, 1.47)	1.19 (0.97, 1.44)	1.04 (0.74, 1.46)	0.93 (0.78, 1.10)	1.01 (0.84, 1.20)	0.83 (0.67, 1.03)	**0.77 (0.64, 0.92)**	0.89 (0.72, 1.10)	**ATE/CT**	0.71 (0.35, 1.41)	1.25 (0.58, 2.68)	0.94 (0.64, 1.37)	1.33 (0.75, 2.40)	0.76 (0.51, 1.14)	**0.62 (0.44, 0.86)**
0.86 (0.70, 1.05)	1.01 (0.82, 1.25)	0.91 (0.68, 1.23)	1.05 (0.83, 1.33)	1.03 (0.82, 1.30)	0.91 (0.63, 1.30)	**0.81 (0.65, 1.00)**	0.88 (0.71, 1.09)	**0.72 (0.56, 0.93)**	**0.67 (0.53, 0.84)**	**0.77 (0.60, 1.00)**	0.87 (0.72, 1.06)	**IPI/CT**	1.77 (0.68, 4.64)	1.32 (0.67, 2.63)	1.88 (0.85, 4.28)	1.08 (0.53, 2.20)	0.87 (0.47, 1.60)
1.16 (0.80, 1.69)	1.37 (0.94, 2.00)	1.23 (0.80, 1.90)	1.42 (0.96, 2.10)	1.39 (0.94, 2.06)	1.23 (0.76, 1.98)	1.09 (0.75, 1.59)	1.18 (0.81, 1.73)	0.98 (0.65, 1.46)	0.90 (0.62, 1.30)	1.05 (0.70, 1.56)	1.18 (0.82, 1.68)	1.35 (0.91, 1.99)	**CAM/CT**	0.75 (0.35, 1.61)	1.06 (0.44, 2.58)	0.61 (0.32, 1.17)	0.49 (0.23, 1.03)
0.89 (0.76, 1.03)	1.04 (0.89, 1.23)	0.94 (0.72, 1.23)	1.09 (0.90, 1.32)	1.07 (0.88, 1.29)	0.94 (0.67, 1.31)	**0.83 (0.71, 0.98)**	0.90 (0.76, 1.07)	**0.75 (0.61, 0.92)**	**0.69 (0.58, 0.82)**	**0.80 (0.67, 0.95)**	0.90 (0.80, 1.01)	1.03 (0.85, 1.25)	0.76 (0.53, 1.09)	**CT/BEV**	1.42 (0.92, 2.22)	0.82 (0.55, 1.21)	**0.66 (0.48, 0.88)**
0.91 (0.74, 1.11)	1.07 (0.86, 1.32)	0.96 (0.72, 1.30)	1.11 (0.88, 1.41)	1.09 (0.86, 1.38)	0.96 (0.67, 1.38)	0.85 (0.69, 1.06)	0.93 (0.74, 1.15)	**0.77 (0.60, 0.98)**	**0.71 (0.56, 0.88)**	0.82 (0.66, 1.02)	0.92 (0.77, 1.10)	1.06 (0.84, 1.33)	0.78 (0.53, 1.15)	1.02 (0.89, 1.17)	**CT/BEV/ Mpem**	0.57 (0.31, 1.03)	**0.46 (0.27, 0.78)**
0.84 (0.70, 0.99)	0.98 (0.82, 1.18)	0.89 (0.67, 1.17)	1.02 (0.83, 1.26)	1.00 (0.82, 1.23)	0.88 (0.63, 1.24)	**0.79 (0.65, 0.94)**	0.85 (0.70, 1.03)	**0.70 (0.56, 0.88)**	**0.65 (0.55, 0.76)**	**0.75 (0.60, 0.94)**	**0.85 (0.74, 0.97)**	0.97 (0.79, 1.19)	**0.72 (0.52, 1.00)**	0.94 (0.82, 1.08)	0.92 (0.76, 1.12)	**CT/Mpem**	0.81 (0.56, 1.16)
0.78 (0.69, 0.88)	0.92 (0.80, 1.05)	0.83 (0.65, 1.07)	0.96 (0.81, 1.13)	0.94 (0.80, 1.11)	0.83 (0.60, 1.14)	**0.74 (0.64, 0.84)**	**0.80 (0.69, 0.92)**	**0.66 (0.55, 0.80)**	**0.61 (0.52, 0.71)**	**0.71 (0.58, 0.86)**	**0.79 (0.71, 0.88)**	0.91 (0.77, 1.07)	**0.67 (0.47, 0.96)**	**0.88 (0.80, 0.97)**	0.86 (0.73, 1.02)	0.94 (0.83, 1.06)	**CT**

E **>=3 AEs**

ORR

PEM	1.10 (0.78, 1.53)	**1.80 (1.12, 2.92)**	0.97 (0.63, 1.49)	**0.57 (0.38, 0.85)**	0.47 (0.21, 1.04)	**0.44 (0.31, 0.60)**	**0.18 (0.13, 0.25)**	**0.23 (0.16, 0.33)**	**0.31 (0.21, 0.45)**	**0.12 (0.08, 0.17)**	**0.21 (0.15, 0.28)**	**0.16 (0.11, 0.24)**	**0.20 (0.11, 0.34)**	**0.18 (0.12, 0.26)**	**0.39 (0.26, 0.57)**	**0.33 (0.26, 0.41)**
1.15 (0.52, 2.58)	**NIV**	**1.64 (1.01, 2.69)**	0.88 (0.57, 1.37)	**0.52 (0.35, 0.78)**	**0.43 (0.19, 0.96)**	**0.40 (0.30, 0.53)**	**0.16 (0.12, 0.23)**	**0.21 (0.14, 0.31)**	**0.28 (0.19, 0.41)**	**0.11 (0.07, 0.16)**	**0.19 (0.14, 0.26)**	**0.15 (0.10, 0.22)**	**0.18 (0.10, 0.31)**	**0.16 (0.11, 0.24)**	**0.36 (0.24, 0.53)**	**0.30 (0.24, 0.38)**
1.57 (0.59, 4.31)	1.36 (0.52, 3.71)	**ATE**	**0.54 (0.31, 0.94)**	**0.32 (0.18, 0.54)**	**0.26 (0.11, 0.62)**	**0.24 (0.15, 0.39)**	**0.10 (0.06, 0.16)**	**0.13 (0.08, 0.21)**	**0.17 (0.10, 0.29)**	**0.07 (0.04, 0.11)**	**0.11 (0.07, 0.18)**	**0.09 (0.05, 0.15)**	**0.11 (0.06, 0.21)**	**0.10 (0.06, 0.16)**	**0.22 (0.13, 0.37)**	**0.18 (0.12, 0.28)**
1.53 (0.56, 4.31)	1.32 (0.49, 3.73)	0.97 (0.30, 3.14)	**DUR**	**0.59 (0.40, 0.85)**	0.49 (0.22, 1.06)	**0.45 (0.29, 0.69)**	**0.19 (0.12, 0.29)**	**0.24 (0.15, 0.38)**	**0.32 (0.20, 0.50)**	**0.12 (0.07, 0.19)**	**0.21 (0.14, 0.32)**	**0.17 (0.11, 0.27)**	**0.20 (0.11, 0.38)**	**0.18 (0.11, 0.29)**	**0.40 (0.25, 0.65)**	**0.34 (0.24, 0.49)**
1.61 (0.58, 4.57)	1.40 (0.51, 3.90)	1.02 (0.31, 3.31)	1.05 (0.45, 2.49)	**DUR/TRE**	0.83 (0.42, 1.63)	0.76 (0.51, 1.14)	**0.32 (0.21, 0.47)**	**0.40 (0.26, 0.63)**	**0.54 (0.35, 0.84)**	**0.21 (0.13, 0.33)**	**0.36 (0.25, 0.53)**	**0.29 (0.19, 0.45)**	**0.34 (0.19, 0.63)**	**0.31 (0.20, 0.48)**	0.68 (0.44, 1.08)	**0.58 (0.42, 0.81)**
0.84 (0.21, 3.41)	0.73 (0.18, 2.95)	0.54 (0.12, 2.40)	0.55 (0.16, 1.95)	0.53 (0.21, 1.33)	**DUR/TRE/CT**	0.92 (0.42, 2.03)	**0.38 (0.17, 0.85)**	0.49 (0.22, 1.11)	0.65 (0.29, 1.47)	**0.25 (0.11, 0.57)**	**0.44 (0.20, 0.96)**	**0.35 (0.15, 0.78)**	0.41 (0.17, 1.04)	**0.38 (0.17, 0.84)**	0.82 (0.36, 1.89)	0.70 (0.33, 1.50)
0.97 (0.45, 2.18)	0.84 (0.44, 1.65)	0.62 (0.23, 1.64)	0.63 (0.23, 1.76)	0.60 (0.22, 1.66)	1.15 (0.29, 4.52)	**NIV/IPI**	**0.41 (0.31, 0.56)**	**0.53 (0.36, 0.77)**	0.71 (0.49, 1.03)	**0.27 (0.18, 0.40)**	**0.47 (0.35, 0.64)**	**0.38 (0.26, 0.55)**	**0.45 (0.26, 0.78)**	**0.41 (0.28, 0.59)**	0.90 (0.61, 1.32)	**0.76 (0.60, 0.96)**
0.60 (0.28, 1.38)	0.52 (0.25, 1.14)	0.38 (0.15, 1.03)	0.39 (0.14, 1.09)	0.37 (0.14, 1.05)	0.71 (0.18, 2.88)	0.62 (0.32, 1.21)	**NIV/CT**	1.28 (0.87, 1.87)	**1.71 (1.17, 2.49)**	**0.65 (0.43, 0.97)**	1.14 (0.84, 1.56)	0.91 (0.62, 1.33)	1.08 (0.62, 1.91)	0.98 (0.68, 1.43)	**2.16 (1.46, 3.19)**	**1.83 (1.45, 2.32)**
0.74 (0.29, 2.01)	0.65 (0.25, 1.73)	0.47 (0.15, 1.46)	0.49 (0.15, 1.55)	0.46 (0.14, 1.49)	0.88 (0.20, 3.93)	0.77 (0.29, 2.00)	1.24 (0.47, 3.22)	**NIV/IPI/CT**	1.33 (0.87, 2.04)	**0.51 (0.33, 0.79)**	0.90 (0.62, 1.29)	0.71 (0.47, 1.08)	0.85 (0.47, 1.54)	0.77 (0.51, 1.17)	**1.69 (1.10, 2.62)**	**1.44 (1.07, 1.94)**
0.48 (0.22, 1.11)	**0.42 (0.19, 0.94)**	**0.31 (0.11, 0.83)**	**0.32 (0.11, 0.89)**	**0.30 (0.11, 0.84)**	0.57 (0.14, 2.31)	0.50 (0.23, 1.10)	0.80 (0.36, 1.78)	0.65 (0.24, 1.72)	**PEM/CT**	**0.38 (0.25, 0.57)**	**0.67 (0.49, 0.92)**	**0.53 (0.35, 0.81)**	0.64 (0.38, 1.05)	**0.58 (0.39, 0.85)**	1.27 (0.94, 1.71)	1.08 (0.80, 1.45)
0.38 (0.15, 1.00)	**0.33 (0.13, 0.85)**	**0.24 (0.08, 0.72)**	**0.25 (0.08, 0.78)**	**0.24 (0.08, 0.74)**	0.45 (0.10, 1.99)	**0.39 (0.16, 0.99)**	0.63 (0.25, 1.60)	0.51 (0.17, 1.53)	0.79 (0.31, 2.06)	**ATE/CT/BEV**	**1.77 (1.34, 2.33)**	1.40 (0.90, 2.18)	1.68 (0.95, 2.94)	**1.52 (1.15, 2.00)**	**3.34 (2.25, 4.96)**	**2.83 (2.04, 3.94)**
0.74 (0.38, 1.54)	0.65 (0.33, 1.32)	0.47 (0.19, 1.18)	0.49 (0.19, 1.27)	0.46 (0.18, 1.19)	0.89 (0.23, 3.35)	0.77 (0.39, 1.53)	1.24 (0.62, 2.49)	1.00 (0.42, 2.45)	1.55 (0.76, 3.13)	1.96 (0.94, 4.04)	**ATE/CT**	0.80 (0.56, 1.13)	0.95 (0.58, 1.55)	0.86 (0.67, 1.10)	**1.89 (1.42, 2.52)**	**1.60 (1.31, 1.97)**
1.57 (0.61, 4.31)	1.37 (0.54, 3.62)	1.00 (0.33, 3.09)	1.03 (0.33, 3.32)	0.98 (0.31, 3.12)	1.88 (0.42, 8.30)	1.63 (0.63, 4.21)	**2.61 (1.00, 6.82)**	2.11 (0.69, 6.51)	**3.26 (1.24, 8.69)**	**4.13 (1.41, 12.26)**	2.12 (0.88, 5.09)	**IPI/CT**	1.19 (0.66, 2.15)	1.08 (0.72, 1.64)	**2.38 (1.55, 3.65)**	**2.02 (1.51, 2.71)**
0.59 (0.23, 1.66)	0.52 (0.19, 1.42)	0.38 (0.12, 1.20)	0.39 (0.12, 1.27)	0.37 (0.11, 1.21)	0.71 (0.16, 3.17)	0.62 (0.23, 1.63)	0.99 (0.37, 2.66)	0.80 (0.26, 2.48)	1.23 (0.45, 3.38)	1.56 (0.52, 4.79)	0.80 (0.32, 1.99)	0.38 (0.12, 1.17)	**CAM/CT**	0.91 (0.53, 1.57)	**1.99 (1.33, 2.99)**	**1.69 (1.02, 2.81)**
0.66 (0.34, 1.34)	0.57 (0.29, 1.14)	0.42 (0.17, 1.02)	0.43 (0.17, 1.10)	0.41 (0.16, 1.04)	0.78 (0.21, 2.97)	0.68 (0.35, 1.32)	1.09 (0.55, 2.14)	0.88 (0.37, 2.13)	1.36 (0.69, 2.72)	1.73 (0.83, 3.56)	0.88 (0.54, 1.43)	**0.42 (0.17, 1.00)**	1.11 (0.45, 2.72)	**CT/BEV**	**2.20 (1.52, 3.19)**	**1.86 (1.40, 2.49)**
															CT/Mpem	0.85 (0.62, 1.16)
1.39 (0.81, 2.55)	1.21 (0.71, 2.17)	0.89 (0.40, 2.01)	0.91 (0.39, 2.15)	0.87 (0.37, 2.05)	1.65 (0.47, 5.95)	1.44 (0.84, 2.47)	**2.31 (1.32, 4.06)**	1.87 (0.85, 4.13)	**2.89 (1.63, 5.15)**	**3.67 (1.73, 7.79)**	**1.87 (1.24, 2.83)**	0.88 (0.40, 1.93)	**2.34 (1.03, 5.34)**	**2.12 (1.45, 3.10)**		**CT**

F


PEM
NIV
ATE
DUR
DUR/TRE
DUR/TRE/CT
NIV/IPI
NIV/CT
NIV/IPI/CT
PEM/CT
ATE/CT/BEV
ATE/CT
IPI/CT
CAM/CT
CT/BEV
CT/BEV/Mpem
CT/Mpem
CT
Probability(%)
Ranking
OS
PFS
ORR
>=3 AEs

G **PFS**

OS

PEM	**0.42 (0.18, 0.97)**			0.56 (0.23, 1.33)	0.69 (0.31, 1.51)		0.61 (0.29, 1.29)	**0.47 (0.22, 0.97)**	**0.40 (0.19, 0.84)**	**0.35 (0.17, 0.72)**
0.91 (0.57, 1.45)	**NIV**			1.32 (0.69, 2.52)	1.63 (0.96, 2.76)		1.46 (0.91, 2.32)	1.11 (0.70, 1.74)	0.95 (0.61, 1.49)	0.83 (0.54, 1.27)
1.34 (0.53, 3.33)	1.46 (0.55, 3.90)	**ATE**								
0.84 (0.52, 1.36)	0.92 (0.50, 1.68)	0.63 (0.23, 1.70)	**DUR**							
1.18 (0.86, 1.63)	1.30 (0.80, 2.10)	0.89 (0.35, 2.24)	1.41 (0.85, 2.33)	**NIV/IPI**	1.23 (0.69, 2.21)		1.10 (0.65, 1.87)	0.84 (0.50, 1.41)	0.72 (0.43, 1.21)	0.63 (0.39, 1.03)
1.08 (0.73, 1.60)	1.19 (0.70, 2.02)	0.81 (0.31, 2.10)	1.29 (0.74, 2.24)	0.92 (0.61, 1.38)	**NIV/CT**		0.89 (0.62, 1.30)	**0.68 (0.48, 0.97)**	**0.59 (0.41, 0.83)**	**0.51 (0.37, 0.70)**
1.21 (0.82, 1.77)	1.32 (0.78, 2.23)	0.90 (0.35, 2.33)	1.43 (0.83, 2.47)	1.02 (0.68, 1.53)	1.11 (0.70, 1.77)	**NIV/IPI/CT**				
1.05 (0.79, 1.41)	1.16 (0.72, 1.84)	0.79 (0.32, 1.97)	1.25 (0.77, 2.04)	0.89 (0.65, 1.23)	0.97 (0.66, 1.44)	0.87 (0.59, 1.28)	**PEM/CT**	**0.76 (0.59, 0.98)**	**0.66 (0.51, 0.84)**	**0.57 (0.47, 0.69)**
0.85 (0.65, 1.12)	0.93 (0.59, 1.46)	0.64 (0.26, 1.58)	1.01 (0.63, 1.62)	**0.72 (0.53, 0.98)**	0.78 (0.54, 1.15)	0.70 (0.49, 1.02)	0.81 (0.61, 1.06)	**ATE/CT**	0.86 (0.69, 1.07)	**0.75 (0.64, 0.88)**
0.82 (0.63, 1.07)	0.90 (0.58, 1.41)	0.62 (0.25, 1.52)	0.98 (0.61, 1.56)	**0.69 (0.52, 0.93)**	0.76 (0.52, 1.10)	**0.68 (0.47, 0.98)**	0.78 (0.60, 1.02)	0.97 (0.76, 1.24)	**IPI/CT**	0.87 (0.75, 1.01)
0.75 (0.61, 0.92)	0.82 (0.54, 1.24)	0.56 (0.23, 1.36)	0.89 (0.57, 1.38)	**0.63 (0.49, 0.81)**	**0.69 (0.50, 0.96)**	**0.62 (0.45, 0.86)**	**0.71 (0.58, 0.87)**	0.88 (0.73, 1.06)	0.91 (0.77, 1.07)	**CT**

squamous

H

PFS

OS

PEM	0.43 (0.14, 1.27)			1.00 (0.33, 3.08)	0.82 (0.28, 2.43)		1.61 (0.50, 5.33)	1.54 (0.53, 4.60)	1.01 (0.41, 2.57)	1.27 (0.39, 4.23)	0.88 (0.37, 2.12)	1.25 (0.46, 3.53)	0.77 (0.31, 1.97)	0.55 (0.25, 1.21)
0.68 (0.50, 0.92)	**NIV**			2.35 (0.78, 7.02)	1.93 (0.67, 5.55)		**3.78 (1.19, 12.33)**	**3.62 (1.27, 10.55)**	2.37 (0.97, 5.88)	2.97 (0.92, 9.74)	2.06 (0.90, 4.85)	**2.93 (1.11, 8.11)**	1.81 (0.74, 4.50)	1.29 (0.61, 2.74)
1.28 (0.80, 2.04)	**1.88 (1.14, 3.13)**	**ATE**												
1.13 (0.80, 1.61)	**1.67 (1.11, 2.51)**	0.89 (0.52, 1.52)	**DUR**											
1.02 (0.81, 1.28)	**1.50 (1.11, 2.03)**	0.80 (0.50, 1.27)	0.90 (0.63, 1.28)	**NIV/IPI**	0.82 (0.28, 2.45)		1.61 (0.49, 5.40)	1.54 (0.52, 4.65)	1.01 (0.40, 2.61)	1.27 (0.38, 4.30)	0.88 (0.37, 2.16)	1.25 (0.46, 3.58)	0.77 (0.30, 2.00)	0.55 (0.25, 1.23)
0.92 (0.70, 1.21)	1.36 (0.97, 1.91)	0.72 (0.44, 1.18)	0.81 (0.55, 1.20)	0.91 (0.69, 1.20)	**NIV/CT**		1.96 (0.62, 6.33)	1.88 (0.67, 5.43)	1.23 (0.51, 3.03)	1.54 (0.48, 5.05)	1.07 (0.47, 2.50)	1.52 (0.58, 4.18)	0.94 (0.39, 2.32)	0.67 (0.32, 1.41)
1.15 (0.87, 1.52)	**1.69 (1.20, 2.39)**	0.90 (0.55, 1.47)	1.01 (0.69, 1.50)	1.13 (0.85, 1.50)	1.25 (0.90, 1.72)	**NIV/IPI/CT**								
1.66 (1.22, 2.26)	**2.45 (1.70, 3.54)**	1.30 (0.78, 2.17)	1.47 (0.97, 2.21)	**1.63 (1.20, 2.23)**	**1.80 (1.28, 2.55)**	**1.45 (1.02, 2.05)**	**PEM/CT**	0.96 (0.33, 2.83)	0.63 (0.25, 1.56)	0.79 (0.27, 2.28)	0.55 (0.22, 1.34)	0.78 (0.28, 2.21)	0.48 (0.23, 1.01)	**0.34 (0.14, 0.83)**
1.16 (0.90, 1.50)	**1.71 (1.24, 2.37)**	0.91 (0.56, 1.47)	1.03 (0.71, 1.49)	1.14 (0.88, 1.48)	1.26 (0.93, 1.70)	1.01 (0.75, 1.37)	**0.70 (0.51, 0.96)**	**ATE/CT/BEV**	0.66 (0.33, 1.31)	0.82 (0.27, 2.45)	0.57 (0.29, 1.14)	0.81 (0.34, 1.96)	0.50 (0.23, 1.10)	**0.36 (0.17, 0.74)**
1.09 (0.88, 1.34)	**1.60 (1.20, 2.14)**	0.85 (0.54, 1.34)	0.96 (0.68, 1.35)	1.07 (0.86, 1.32)	1.18 (0.91, 1.52)	0.94 (0.72, 1.23)	**0.65 (0.50, 0.85)**	0.93 (0.75, 1.16)	**ATE/CT**	1.25 (0.50, 3.17)	0.87 (0.54, 1.43)	1.24 (0.61, 2.57)	0.77 (0.45, 1.30)	**0.54 (0.33, 0.88)**
1.29 (0.87, 1.92)	**1.91 (1.23, 2.97)**	1.01 (0.57, 1.78)	1.14 (0.71, 1.84)	1.27 (0.86, 1.89)	1.40 (0.92, 2.14)	1.13 (0.74, 1.72)	0.78 (0.52, 1.16)	1.11 (0.75, 1.66)	1.19 (0.83, 1.72)	**CAM/CT**	0.69 (0.28, 1.73)	0.99 (0.35, 2.86)	0.61 (0.28, 1.31)	0.43 (0.17, 1.07)
0.93 (0.77, 1.12)	**1.37 (1.04, 1.80)**	0.73 (0.46, 1.14)	0.82 (0.59, 1.14)	0.91 (0.75, 1.11)	1.01 (0.79, 1.29)	0.81 (0.63, 1.04)	**0.56 (0.43, 0.73)**	**0.80 (0.67, 0.95)**	**0.86 (0.75, 0.97)**	0.72 (0.50, 1.03)	**CT/BEV**	1.42 (0.84, 2.43)	0.88 (0.53, 1.44)	**0.63 (0.42, 0.90)**
0.95 (0.76, 1.20)	**1.40 (1.03, 1.91)**	0.74 (0.47, 1.19)	0.84 (0.59, 1.20)	0.93 (0.74, 1.18)	1.03 (0.78, 1.36)	0.83 (0.62, 1.10)	**0.57 (0.42, 0.77)**	0.82 (0.66, 1.02)	0.88 (0.73, 1.06)	0.73 (0.50, 1.08)	1.02 (0.89, 1.17)	**CT/BEV/ Mpem**	0.62 (0.30, 1.26)	**0.44 (0.23, 0.82)**
0.93 (0.75, 1.15)	**1.37 (1.02, 1.84)**	0.73 (0.46, 1.15)	0.82 (0.58, 1.16)	0.92 (0.74, 1.14)	1.01 (0.78, 1.32)	0.81 (0.62, 1.06)	**0.56 (0.45, 0.70)**	0.80 (0.64, 1.01)	**0.86 (0.74, 1.00)**	**0.72 (0.52, 1.00)**	1.00 (0.87, 1.16)	0.98 (0.80, 1.19)	**CT/Mpem**	0.71 (0.43, 1.16)
0.79 (0.68, 0.93)	1.17 (0.91, 1.51)	**0.62 (0.40, 0.96)**	**0.70 (0.51, 0.96)**	**0.78 (0.66, 0.92)**	0.86 (0.69, 1.08)	**0.69 (0.55, 0.87)**	**0.48 (0.37, 0.62)**	**0.68 (0.56, 0.83)**	**0.73 (0.64, 0.84)**	**0.61 (0.43, 0.88)**	**0.85 (0.77, 0.94)**	**0.83 (0.71, 0.99)**	0.85 (0.74, 0.98)	**CT**

non-squamous

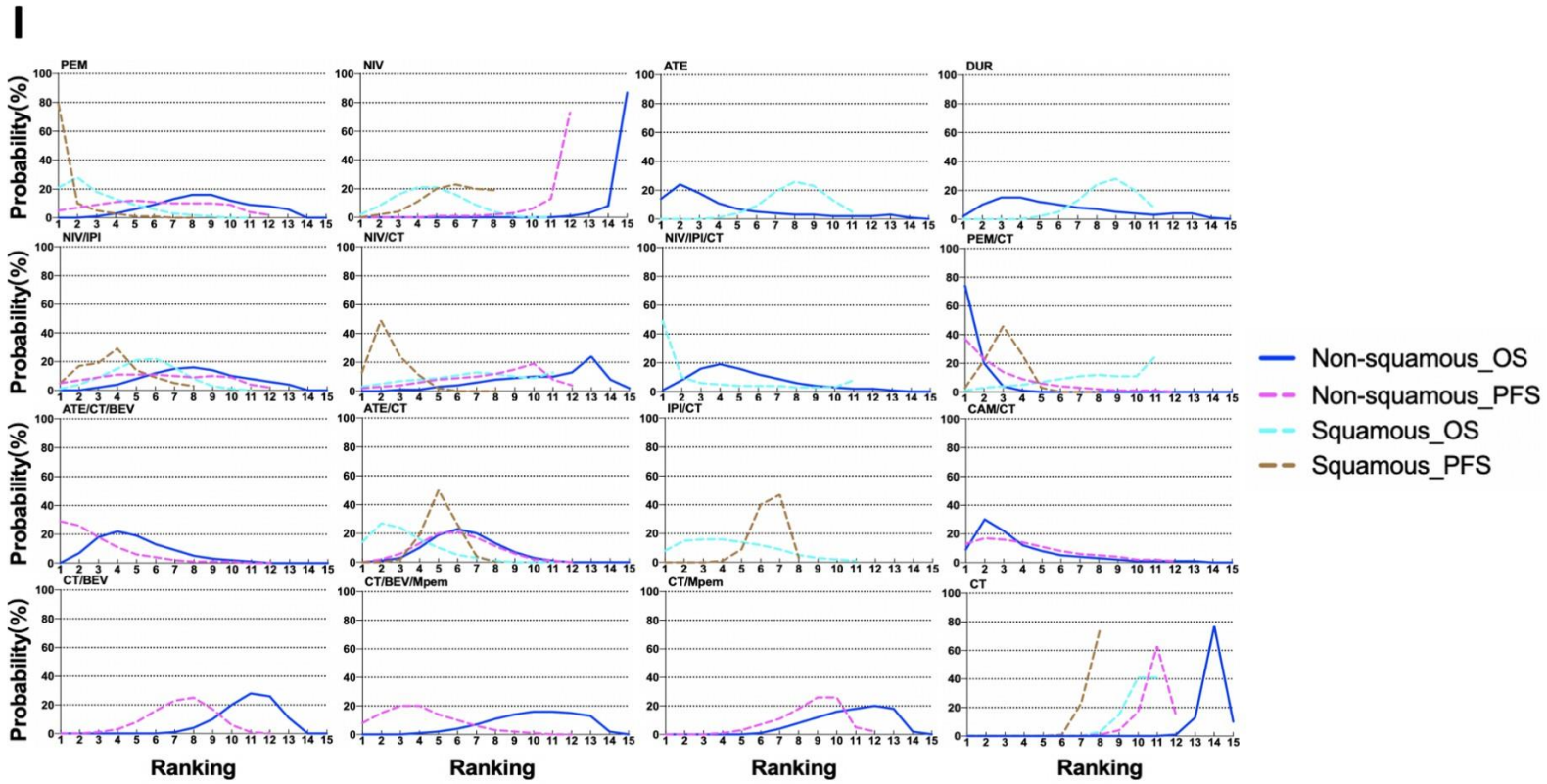
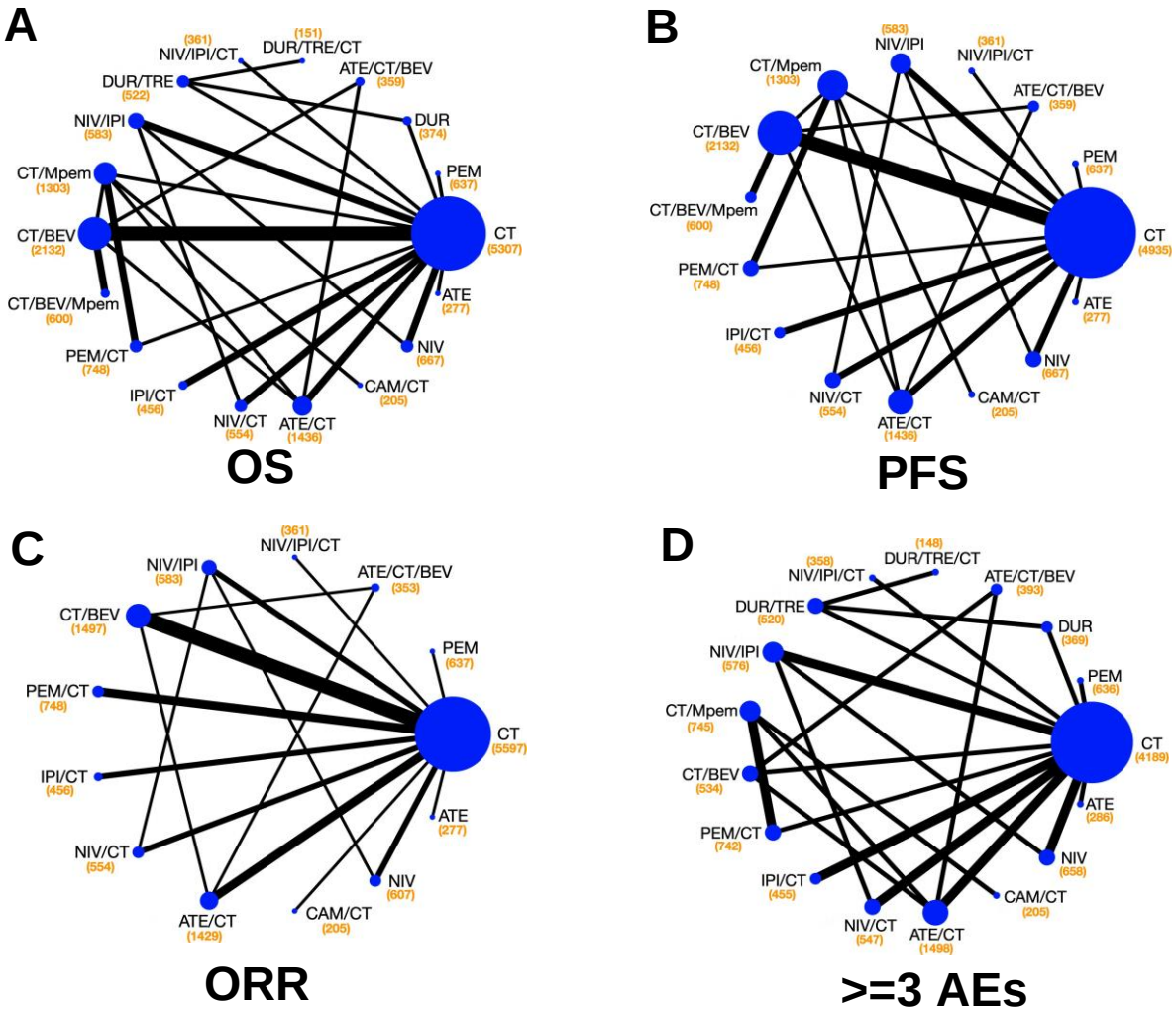


Figure S20. Network Meta-analysis of the First Sensitive Analysis Excluding Phase II Trials

(A) Network diagrams for overall survival (OS) and progression-free survival (PFS), objective response rate (ORR) (B), adverse events of grade 3 or higher (>= 3 AEs) (C). Each circular node represents a type of treatment. Each line represents a type of head-to-head comparison. The size of the nodes and the thickness of the lines are weighted according to the number of studies evaluating each treatment and direct comparison, respectively. The total number of patients receiving a treatment was shown in brackets. The total number in each node for OS is the same as that for PFS except for durvalumab (DUR) and chemotherapy (CT) (marked in red for PFS). (D) Pooled hazard ratios (HR) (95% CrIs (credible intervals)) for OS and PFS. (E) Pooled odds ratio (OR) (95% CrIs) for >= 3 AEs and ORR. Data in each cell are HR or OR (95% credible intervals) for the comparison of row-defining treatment versus column-defining treatment. HR less than 1 and OR more than 1 favour upper-row treatment. Significant results are highlighted in red and bold. (F) Ranking curves indicate the probability of each comparable treatment being ranked from first to last on OS, PFS, ORR, and >=3AEs. (G) Pooled HR (95% CrIs) for OS and PFS in squamous subgroup. (H) Pooled HR (95% CrIs) for OS and PFS in non-squamous subgroup. (I) Ranking curves indicate the probability of each comparable treatment being ranked from first to last on OS and PFS in squamous and non-squamous subgroups. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.


A
(361)
NIV/IPI/CT
(151)
DUR/TRE/CT
ATE/CT/BEV
(359)
DUR/TRE
(522)
DUR
(374)
NIV/IPI
(583)
PEM
(637)
CT/Mpem
(1303)
CT
(5307)
CT/BEV
(2132)
ATE
(277)
CT/BEV/Mpem
(600)
NIV
(667)
PEM/CT
(748)
CAM/CT
(205)
IPI/CT
(456)
NIV/CT
(554)
ATE/CT
(1436)
OS
B
(583)
NIV/IPI
(361)
NIV/IPI/CT
CT/Mpem
(1303)
ATE/CT/BEV
(359)
CT/BEV
(2132)
PEM
(637)
CT/BEV/Mpem
(600)
CT
(4935)
PEM/CT
(748)
ATE
(277)
IPI/CT
(456)
NIV
(667)
NIV/CT
(554)
CAM/CT
(205)
ATE/CT
(1436)
PFS
C
(361)
NIV/IPI/CT
NIV/IPI
(583)
ATE/CT/BEV
(353)
CT/BEV
(1497)
PEM
(637)
PEM/CT
(748)
CT
(5597)
IPI/CT
(456)
ATE
(277)
NIV/CT
(554)
NIV
(607)
CAM/CT
(205)
ATE/CT
(1429)
ORR
D
(148)
DUR/TRE/CT
(358)
NIV/IPI/CT
ATE/CT/BEV
(393)
DUR/TRE
(520)
DUR
(369)
NIV/IPI
(576)
PEM
(636)
CT/Mpem
(745)
CT
(4189)
CT/BEV
(534)
ATE
(286)
PEM/CT
(742)
NIV
(658)
IPI/CT
(455)
CAM/CT
(205)
NIV/CT
(547)
ATE/CT
(1498)
>=3 AEs

E **PFS**

OS

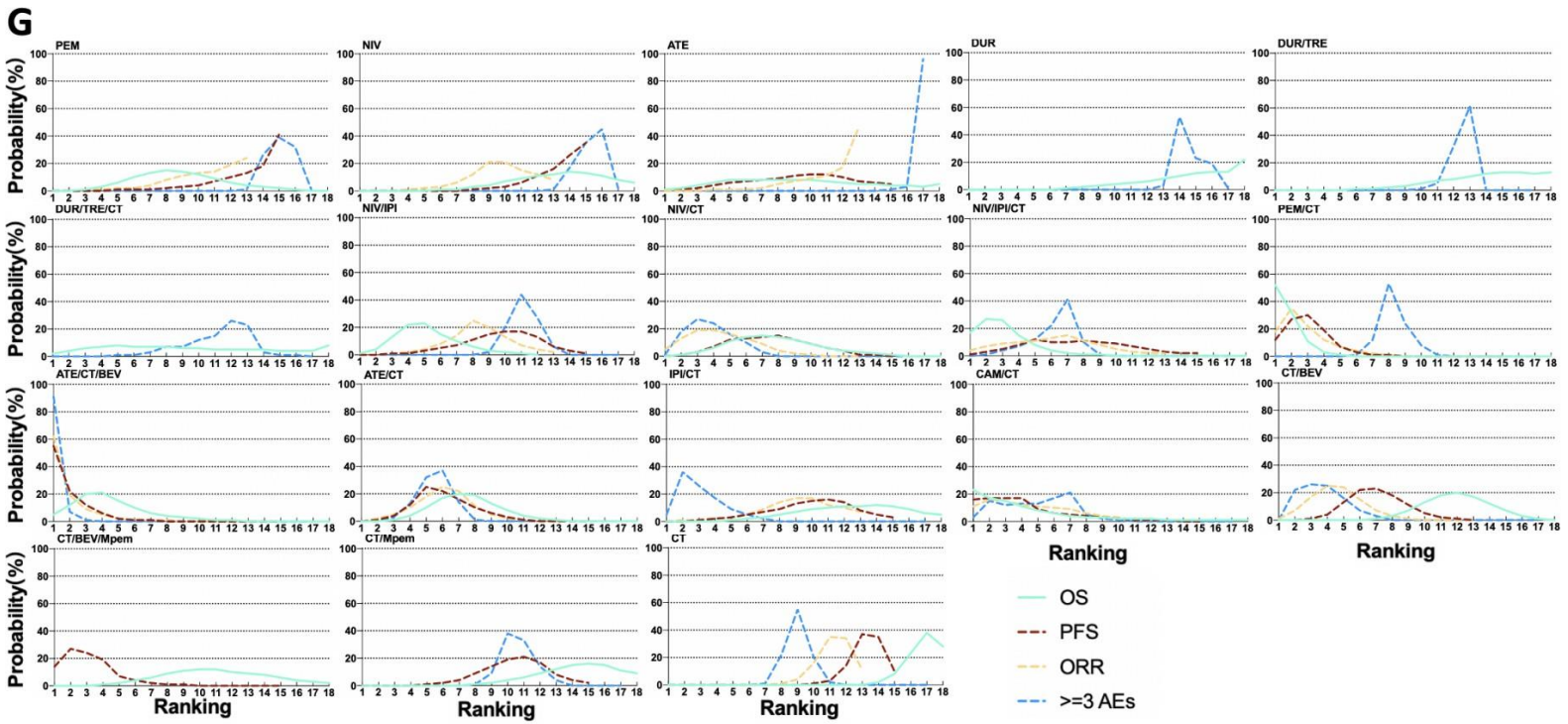
PEM	1.01 (0.57, 1.79)	1.36 (0.69, 2.68)				1.33 (0.75, 2.36)	1.55 (0.87, 2.76)	1.55 (0.79, 3.02)	**2.25 (1.26, 4.04)**	**2.70 (1.40, 5.27)**	**1.70 (1.00, 2.91)**	1.32 (0.73, 2.39)	**2.11 (1.00, 4.53)**	1.60 (0.96, 2.70)	**2.25 (1.23, 4.26)**	1.29 (0.75, 2.23)	1.05 (0.66, 1.67)
0.89 (0.74, 1.08)	**NIV**	1.36 (0.75, 2.45)				1.32 (0.88, 1.99)	1.54 (0.96, 2.46)	1.54 (0.85, 2.77)	**2.23 (1.38, 3.64)**	**2.69 (1.52, 4.81)**	**1.69 (1.11, 2.60)**	1.31 (0.80, 2.16)	**2.10 (1.06, 4.18)**	**1.59 (1.07, 2.39)**	**2.23 (1.33, 3.86)**	1.28 (0.83, 2.00)	1.04 (0.75, 1.46)
0.99 (0.74, 1.31)	1.11 (0.84, 1.47)	**ATE**				0.98 (0.54, 1.76)	1.14 (0.62, 2.06)	1.13 (0.57, 2.25)	1.65 (0.91, 3.02)	**1.98 (1.01, 3.94)**	1.25 (0.72, 2.18)	0.96 (0.53, 1.79)	1.55 (0.72, 3.37)	1.17 (0.69, 2.02)	1.65 (0.88, 3.17)	0.94 (0.54, 1.67)	0.77 (0.47, 1.26)
0.85 (0.69, 1.06)	0.96 (0.77, 1.19)	0.87 (0.64, 1.17)	**DUR**														
0.87 (0.71, 1.08)	0.98 (0.79, 1.21)	0.88 (0.66, 1.19)	1.02 (0.81, 1.29)	**DUR/TRE**													
0.99 (0.70, 1.40)	1.11 (0.79, 1.58)	1.00 (0.67, 1.51)	1.16 (0.81, 1.67)	1.14 (0.86, 1.49)	**DUR/TRE/CT**												
1.11 (0.92, 1.35)	**1.25 (1.04, 1.51)**	1.13 (0.85, 1.50)	**1.30 (1.05, 1.62)**	**1.28 (1.03, 1.58)**	1.12 (0.79, 1.59)	**NIV/IPI**	1.17 (0.75, 1.79)	1.16 (0.65, 2.09)	**1.69 (1.05, 2.74)**	**2.03 (1.15, 3.63)**	1.28 (0.84, 1.96)	0.99 (0.61, 1.63)	1.58 (0.80, 3.15)	1.20 (0.81, 1.80)	**1.69 (1.01, 2.91)**	0.97 (0.63, 1.50)	0.79 (0.57, 1.10)
1.03 (0.84, 1.25)	1.15 (0.95, 1.41)	1.04 (0.78, 1.39)	1.20 (0.97, 1.50)	1.18 (0.95, 1.47)	1.04 (0.73, 1.47)	0.92 (0.76, 1.12)	**NIV/CT**	0.99 (0.55, 1.81)	1.45 (0.90, 2.38)	1.74 (0.98, 3.15)	1.09 (0.72, 1.70)	0.85 (0.52, 1.42)	1.36 (0.69, 2.73)	1.03 (0.69, 1.57)	1.45 (0.86, 2.53)	0.83 (0.54, 1.31)	**0.68 (0.48, 0.96)**
1.24 (0.99, 1.57)	**1.40 (1.11, 1.76)**	1.26 (0.92, 1.72)	**1.46 (1.13, 1.87)**	**1.42 (1.11, 1.83)**	1.25 (0.86, 1.82)	1.12 (0.89, 1.41)	1.21 (0.95, 1.53)	**NIV/IPI/CT**	1.45 (0.81, 2.66)	1.75 (0.90, 3.45)	1.10 (0.64, 1.91)	0.85 (0.47, 1.57)	1.36 (0.64, 2.96)	1.03 (0.61, 1.77)	1.45 (0.78, 2.8)	0.83 (0.48, 1.47)	0.68 (0.42, 1.10)
1.36 (1.11, 1.67)	**1.53 (1.24, 1.87)**	**1.38 (1.03, 1.84)**	**1.59 (1.26, 2.00)**	**1.56 (1.24, 1.95)**	1.37 (0.96, 1.95)	1.22 (0.99, 1.50)	**1.32 (1.07, 1.63)**	1.09 (0.86, 1.40)	**PEM/CT**	1.20 (0.68, 2.11)	0.76 (0.51, 1.13)	**0.59 (0.36, 0.97)**	0.94 (0.51, 1.74)	0.71 (0.49, 1.05)	1.00 (0.60, 1.70)	**0.57 (0.42, 0.79)**	**0.47 (0.33, 0.66)**
1.16 (0.91, 1.47)	**1.30 (1.02, 1.65)**	1.17 (0.85, 1.61)	**1.35 (1.04, 1.75)**	**1.32 (1.02, 1.71)**	1.16 (0.80, 1.70)	1.04 (0.82, 1.31)	1.12 (0.88, 1.43)	0.93 (0.71, 1.22)	0.85 (0.67, 1.09)	**ATE/CT/BEV**	**0.63 (0.40, 1.00)**	**0.49 (0.27, 0.88)**	0.78 (0.38, 1.62)	**0.59 (0.38, 0.93)**	0.83 (0.48, 1.48)	**0.48 (0.29, 0.79)**	**0.39 (0.24, 0.62)**
1.03 (0.87, 1.23)	1.16 (0.98, 1.38)	1.04 (0.80, 1.37)	1.21 (0.99, 1.47)	1.18 (0.97, 1.44)	1.04 (0.74, 1.46)	0.93 (0.78, 1.10)	1.00 (0.84, 1.20)	0.83 (0.67, 1.03)	**0.76 (0.64, 0.91)**	0.89 (0.73, 1.10)	**ATE/CT**	0.77 (0.50, 1.21)	1.24 (0.67, 2.29)	0.94 (0.71, 1.26)	1.32 (0.85, 2.11)	0.76 (0.55, 1.04)	**0.62 (0.47, 0.80)**
0.91 (0.74, 1.11)	1.02 (0.83, 1.25)	0.92 (0.69, 1.23)	1.06 (0.85, 1.33)	1.04 (0.83, 1.30)	0.91 (0.64, 1.30)	**0.81 (0.66, 1.00)**	0.88 (0.72, 1.09)	**0.73 (0.57, 0.93)**	**0.67 (0.54, 0.83)**	0.79 (0.61, 1.01)	0.88 (0.73, 1.06)	**IPI/CT**	1.60 (0.80, 3.22)	1.21 (0.79, 1.86)	1.71 (0.99, 2.99)	0.98 (0.62, 1.55)	0.80 (0.55, 1.14)
1.21 (0.83, 1.77)	1.36 (0.93, 1.98)	1.23 (0.80, 1.89)	1.42 (0.96, 2.09)	1.39 (0.94, 2.05)	1.22 (0.76, 1.96)	1.09 (0.74, 1.59)	1.18 (0.80, 1.72)	0.97 (0.65, 1.45)	0.89 (0.62, 1.28)	1.05 (0.70, 1.56)	1.17 (0.82, 1.68)	1.33 (0.91, 1.96)	**CAM/CT**	0.76 (0.41, 1.39)	1.07 (0.53, 2.17)	0.61 (0.36, 1.03)	**0.50 (0.27, 0.90)**
0.92 (0.79, 1.09)	1.04 (0.88, 1.22)	0.94 (0.72, 1.22)	1.08 (0.89, 1.31)	1.06 (0.88, 1.28)	0.93 (0.67, 1.30)	**0.83 (0.71, 0.98)**	0.90 (0.76, 1.07)	**0.74 (0.60, 0.92)**	**0.68 (0.57, 0.81)**	**0.80 (0.67, 0.95)**	0.90 (0.80, 1.00)	1.02 (0.85, 1.22)	0.76 (0.53, 1.09)	**CT/BEV**	**1.41 (1.00, 2.01)**	0.81 (0.59, 1.09)	**0.66 (0.52, 0.81)**
0.95 (0.77, 1.17)	1.06 (0.86, 1.31)	0.96 (0.71, 1.29)	1.11 (0.88, 1.40)	1.08 (0.86, 1.37)	0.95 (0.67, 1.37)	0.85 (0.69, 1.05)	0.92 (0.74, 1.14)	**0.76 (0.59, 0.98)**	**0.70 (0.56, 0.87)**	0.82 (0.66, 1.02)	0.92 (0.77, 1.10)	1.04 (0.83, 1.30)	0.78 (0.53, 1.15)	1.02 (0.89, 1.17)	**CT/BEV/ Mpem**	**0.57 (0.36, 0.90)**	**0.47 (0.30, 0.70)**
0.87 (0.72, 1.05)	0.98 (0.81, 1.17)	0.88 (0.67, 1.16)	1.02 (0.83, 1.25)	1.00 (0.81, 1.23)	0.88 (0.62, 1.24)	**0.78 (0.65, 0.94)**	0.85 (0.70, 1.02)	**0.70 (0.56, 0.88)**	**0.64 (0.55, 0.75)**	**0.75 (0.60, 0.94)**	**0.84 (0.73, 0.97)**	0.96 (0.79, 1.17)	**0.72 (0.52, 1.00)**	0.94 (0.82, 1.08)	0.92 (0.76, 1.12)	**CT/Mpem**	0.82 (0.61, 1.08)
0.82 (0.72, 0.94)	0.92 (0.80, 1.05)	0.83 (0.65, 1.07)	0.96 (0.81, 1.13)	0.94 (0.80, 1.11)	0.83 (0.60, 1.14)	**0.74 (0.64, 0.84)**	**0.80 (0.69, 0.92)**	**0.66 (0.55, 0.80)**	**0.60 (0.52, 0.71)**	**0.71 (0.58, 0.86)**	**0.80 (0.72, 0.88)**	0.90 (0.78, 1.05)	**0.68 (0.48, 0.97)**	**0.89 (0.81, 0.97)**	0.87 (0.74, 1.02)	0.94 (0.83, 1.07)	**CT**

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PEM	1.04 (0.73, 1.49)	**1.71 (1.04, 2.81)**	0.92 (0.59, 1.44)	**0.54 (0.36, 0.82)**	**0.45 (0.20, 1.00)**	**0.41 (0.29, 0.58)**	**0.17 (0.12, 0.24)**	**0.22 (0.15, 0.32)**	**0.29 (0.19, 0.42)**	**0.11 (0.07, 0.17)**	**0.20 (0.14, 0.27)**	**0.16 (0.11, 0.23)**	**0.19 (0.11, 0.34)**	**0.17 (0.12, 0.25)**	**0.38 (0.25, 0.57)**	**0.31 (0.24, 0.41)**
0.86 (0.32, 2.19)	**NIV**	**1.64 (1.01, 2.69)**	0.88 (0.57, 1.38)	**0.52 (0.35, 0.78)**	**0.43 (0.19, 0.97)**	**0.40 (0.30, 0.53)**	**0.16 (0.12, 0.23)**	**0.21 (0.14, 0.31)**	**0.27 (0.19, 0.40)**	**0.11 (0.07, 0.16)**	**0.19 (0.14, 0.26)**	**0.15 (0.11, 0.22)**	**0.18 (0.10, 0.32)**	**0.16 (0.11, 0.24)**	**0.37 (0.25, 0.54)**	**0.30 (0.24, 0.39)**
1.17 (0.38, 3.58)	1.36 (0.52, 3.72)	**ATE**	**0.54 (0.31, 0.94)**	**0.32 (0.18, 0.54)**	**0.26 (0.11, 0.62)**	**0.24 (0.15, 0.39)**	**0.10 (0.06, 0.16)**	**0.13 (0.08, 0.21)**	**0.17 (0.10, 0.28)**	**0.07 (0.04, 0.11)**	**0.12 (0.07, 0.18)**	**0.09 (0.06, 0.15)**	**0.11 (0.06, 0.22)**	**0.10 (0.06, 0.17)**	**0.22 (0.13, 0.38)**	**0.18 (0.12, 0.28)**
			DUR	**0.59 (0.40, 0.85)**	0.49 (0.22, 1.07)	**0.45 (0.29, 0.69)**	**0.19 (0.12, 0.29)**	**0.24 (0.15, 0.38)**	**0.31 (0.19, 0.49)**	**0.12 (0.07, 0.20)**	**0.21 (0.14, 0.32)**	**0.17 (0.11, 0.27)**	**0.21 (0.11, 0.38)**	**0.18 (0.12, 0.29)**	**0.41 (0.25, 0.66)**	**0.34 (0.24, 0.49)**
				DUR/TRE	0.83 (0.41, 1.64)	0.76 (0.51, 1.14)	**0.32 (0.21, 0.47)**	**0.40 (0.26, 0.63)**	**0.53 (0.34, 0.82)**	**0.21 (0.13, 0.33)**	**0.36 (0.25, 0.53)**	**0.29 (0.19, 0.45)**	**0.35 (0.19, 0.64)**	**0.31 (0.20, 0.48)**	0.70 (0.45, 1.10)	**0.58 (0.42, 0.81)**
					DUR/TRE/CT	0.92 (0.41, 2.04)	**0.38 (0.17, 0.85)**	0.49 (0.21, 1.10)	0.64 (0.28, 1.44)	**0.25 (0.11, 0.57)**	**0.44 (0.20, 0.97)**	**0.36 (0.16, 0.80)**	0.43 (0.17, 1.07)	**0.38 (0.17, 0.85)**	0.85 (0.37, 1.94)	0.70 (0.33, 1.50)
0.72 (0.28, 1.86)	0.84 (0.44, 1.65)	0.62 (0.23, 1.64)				**NIV/IPI**	**0.41 (0.31, 0.56)**	**0.53 (0.36, 0.77)**	0.69 (0.48, 1.01)	**0.27 (0.18, 0.40)**	**0.48 (0.35, 0.65)**	**0.39 (0.27, 0.55)**	**0.46 (0.26, 0.80)**	**0.41 (0.28, 0.59)**	0.92 (0.63, 1.36)	**0.76 (0.61, 0.96)**
0.45 (0.17, 1.18)	0.52 (0.25, 1.15)	0.38 (0.14, 1.03)				0.62 (0.32, 1.23)	**NIV/CT**	1.27 (0.87, 1.87)	**1.67 (1.14, 2.44)**	**0.65 (0.44, 0.98)**	1.15 (0.85, 1.57)	0.93 (0.65, 1.34)	1.11 (0.64, 1.95)	0.99 (0.68, 1.44)	**2.22 (1.51, 3.28)**	**1.84 (1.45, 2.32)**
0.55 (0.18, 1.67)	0.64 (0.25, 1.74)	0.47 (0.15, 1.47)				0.77 (0.29, 2.02)	1.24 (0.46, 3.27)	**NIV/IPI/CT**	1.31 (0.86, 1.99)	**0.51 (0.33, 0.79)**	0.91 (0.63, 1.3)	0.73 (0.49, 1.09)	0.87 (0.48, 1.58)	0.78 (0.51, 1.18)	**1.74 (1.13, 2.68)**	**1.44 (1.07, 1.95)**
0.36 (0.14, 0.90)	**0.41 (0.20, 0.89)**	**0.30 (0.12, 0.77)**				0.49 (0.23, 1.03)	0.79 (0.37, 1.65)	0.64 (0.25, 1.63)	**PEM/CT**	**0.39 (0.26, 0.59)**	**0.69 (0.50, 0.94)**	**0.56 (0.37, 0.83)**	0.67 (0.41, 1.09)	**0.59 (0.40, 0.88)**	**1.33 (1.01, 1.77)**	1.10 (0.82, 1.48)
0.28 (0.09, 0.81)	**0.32 (0.13, 0.82)**	**0.23 (0.08, 0.71)**				**0.38 (0.15, 0.96)**	0.61 (0.24, 1.54)	0.50 (0.16, 1.47)	0.77 (0.32, 1.92)	**ATE/CT/BEV**	**1.76 (1.34, 2.33)**	1.42 (0.93, 2.18)	1.71 (0.98, 3.01)	**1.52 (1.15, 2.00)**	**3.40 (2.30, 5.05)**	**2.81 (2.03, 3.91)**
0.55 (0.23, 1.31)	0.64 (0.32, 1.29)	0.47 (0.19, 1.15)				0.76 (0.38, 1.48)	1.22 (0.61, 2.40)	0.99 (0.40, 2.40)	1.54 (0.81, 2.96)	2.00 (0.97, 4.14)	**ATE/CT**	0.81 (0.58, 1.13)	0.97 (0.59, 1.59)	0.86 (0.67, 1.10)	**1.93 (1.45, 2.57)**	**1.59 (1.30, 1.95)**
0.82 (0.28, 2.08)	0.95 (0.39, 2.12)	0.70 (0.23, 1.83)				1.13 (0.46, 2.47)	1.83 (0.73, 4.01)	1.48 (0.50, 3.86)	2.30 (0.96, 4.89)	**2.99 (1.03, 7.58)**	1.49 (0.66, 3.03)	**IPI/CT**	1.20 (0.67, 2.13)	1.07 (0.72, 1.59)	**2.39 (1.58, 3.63)**	**1.97 (1.51, 2.58)**
0.44 (0.14, 1.39)	0.52 (0.19, 1.44)	0.38 (0.12, 1.22)				0.62 (0.23, 1.67)	0.99 (0.36, 2.71)	0.80 (0.26, 2.57)	1.25 (0.48, 3.35)	1.62 (0.54, 5.05)	0.81 (0.32, 2.07)	0.54 (0.20, 1.66)	**CAM/CT**	0.89 (0.51, 1.55)	**1.99 (1.33, 3.00)**	1.65 (0.99, 2.74)
0.46 (0.20, 1.07)	0.54 (0.28, 1.06)	**0.39 (0.16, 0.94)**				0.64 (0.33, 1.22)	1.03 (0.53, 1.97)	0.84 (0.35, 1.98)	1.30 (0.70, 2.42)	1.69 (0.81, 3.51)	0.84 (0.52, 1.35)	0.56 (0.29, 1.24)	1.04 (0.42, 2.55)	**CT/BEV**	**2.24 (1.55, 3.24)**	**1.85 (1.38, 2.48)**
															CT/Mpem	0.83 (0.60, 1.13)
1.04 (0.48, 2.27)	1.21 (0.70, 2.16)	0.89 (0.39, 1.99)				1.44 (0.83, 2.47)	**2.32 (1.32, 4.03)**	1.88 (0.85, 4.16)	**2.92 (1.78, 4.86)**	**3.79 (1.79, 8.08)**	**1.90 (1.27, 2.87)**	1.27 (0.71, 2.58)	**2.34 (1.01, 5.33)**	**2.25 (1.59, 3.24)**		**CT**


G
PEM
NIV
ATE
DUR
DUR/TRE
DUR/TRE/CT
NIV/IPI
NIV/CT
NIV/IPI/CT
PEM/CT
ATE/CT/BEV
ATE/CT
IPI/CT
CAM/CT
CT/BEV
CT/BEV/Mpem
CT/Mpem
CT
Probability(%)
Ranking
OS
PFS
ORR
>=3 AEs

H

PFS

PEM								
0.91 (0.57, 1.46)	**NIV**		1.63 (0.96, 2.76)		1.46 (0.91, 2.32)	1.11 (0.70, 1.74)	0.98 (0.63, 1.53)	0.83 (0.54, 1.27)
1.19 (0.86, 1.65)	1.30 (0.80, 2.10)	**NIV/IPI**						
1.09 (0.73, 1.62)	1.19 (0.70, 2.02)	0.92 (0.61, 1.38)	**NIV/CT**		0.89 (0.62, 1.30)	**0.68 (0.48, 0.97)**	**0.60 (0.42, 0.85)**	**0.51 (0.37, 0.70)**
1.21 (0.82, 1.79)	1.32 (0.78, 2.24)	1.02 (0.68, 1.53)	1.11 (0.70, 1.77)	**NIV/IPI/CT**				
1.06 (0.78, 1.43)	1.16 (0.72, 1.84)	0.89 (0.64, 1.23)	0.97 (0.66, 1.44)	0.87 (0.59, 1.28)	**PEM/CT**	**0.76 (0.59, 0.98)**	**0.67 (0.53, 0.86)**	**0.57 (0.47, 0.69)**
0.85 (0.64, 1.13)	0.93 (0.59, 1.47)	**0.72 (0.53, 0.97)**	0.78 (0.54, 1.14)	0.70 (0.49, 1.02)	0.81 (0.61, 1.06)	**ATE/CT**	0.89 (0.71, 1.10)	**0.75 (0.64, 0.88)**
0.85 (0.65, 1.11)	0.93 (0.59, 1.45)	**0.71 (0.53, 0.96)**	0.78 (0.54, 1.13)	**0.70 (0.49, 1.00)**	0.80 (0.62, 1.04)	0.99 (0.78, 1.27)	**IPI/CT**	**0.85 (0.73, 0.98)**
0.75 (0.60, 0.93)	0.82 (0.54, 1.24)	**0.63 (0.49, 0.81)**	**0.69 (0.50, 0.96)**	**0.62 (0.45, 0.86)**	**0.71 (0.58, 0.87)**	0.88 (0.73, 1.06)	0.89 (0.75, 1.04)	**CT**

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PEM													
0.73 (0.54, 1.00)	**NIV**		1.92 (0.80, 4.61)		**3.64 (1.52, 8.79)**	**3.61 (1.54, 8.68)**	**2.36 (1.14, 5.00)**	1.59 (0.62, 4.11)	**2.96 (1.13, 7.94)**	**2.06 (1.05, 4.13)**	**2.92 (1.32, 6.69)**	1.81 (0.87, 3.84)	1.29 (0.69, 2.41)
1.10 (0.87, 1.41)	**1.50 (1.11, 2.03)**	**NIV/IPI**											
1.00 (0.75, 1.33)	1.36 (0.97, 1.91)	0.91 (0.69, 1.20)	**NIV/CT**		1.89 (0.79, 4.53)	1.88 (0.80, 4.47)	1.23 (0.60, 2.57)	0.83 (0.32, 2.12)	1.54 (0.59, 4.09)	1.07 (0.55, 2.12)	1.51 (0.69, 3.44)	0.94 (0.46, 1.97)	0.67 (0.36, 1.23)
1.25 (0.93, 1.67)	**1.70 (1.20, 2.39)**	1.13 (0.85, 1.50)	1.25 (0.90, 1.72)	**NIV/IPI/CT**									
1.80 (1.32, 2.44)	**2.45 (1.71, 3.50)**	**1.63 (1.21, 2.20)**	**1.80 (1.28, 2.51)**	**1.44 (1.03, 2.02)**	**PEM/CT**	0.99 (0.45, 2.23)	0.65 (0.34, 1.24)	0.44 (0.17, 1.12)	0.81 (0.37, 1.80)	0.57 (0.31, 1.06)	0.80 (0.38, 1.73)	**0.50 (0.31, 0.80)**	**0.35 (0.19, 0.66)**
1.25 (0.96, 1.63)	**1.70 (1.23, 2.35)**	1.13 (0.88, 1.47)	1.25 (0.93, 1.69)	1.00 (0.74, 1.36)	**0.70 (0.51, 0.94)**	**ATE/CT/BEV**	0.66 (0.37, 1.16)	0.44 (0.17, 1.11)	0.82 (0.33, 2.03)	**0.57 (0.32, 1.00)**	0.81 (0.40, 1.66)	**0.50 (0.26, 0.96)**	**0.36 (0.19, 0.65)**
1.17 (0.94, 1.46)	**1.59 (1.19, 2.12)**	1.06 (0.86, 1.31)	1.17 (0.90, 1.52)	0.94 (0.72, 1.22)	**0.65 (0.50, 0.84)**	0.94 (0.75, 1.16)	**ATE/CT**	0.67 (0.29, 1.52)	1.25 (0.58, 2.71)	0.87 (0.59, 1.30)	1.23 (0.69, 2.25)	0.76 (0.50, 1.18)	**0.55 (0.36, 0.81)**
0.74 (0.45, 1.20)	1.00 (0.59, 1.70)	0.67 (0.41, 1.09)	0.74 (0.44, 1.23)	**0.59 (0.35, 0.99)**	**0.41 (0.24, 0.69)**	**0.59 (0.36, 0.97)**	0.63 (0.39, 1.01)	**IPI/CT**	1.86 (0.66, 5.32)	1.30 (0.60, 2.82)	1.83 (0.77, 4.51)	1.14 (0.50, 2.60)	0.81 (0.40, 1.66)
1.40 (0.93, 2.09)	**1.90 (1.22, 2.96)**	1.27 (0.85, 1.88)	1.40 (0.92, 2.14)	1.12 (0.73, 1.72)	0.78 (0.53, 1.15)	1.12 (0.75, 1.67)	1.19 (0.83, 1.72)	**1.90 (1.06, 3.41)**	**CAM/CT**	0.70 (0.33, 1.48)	0.98 (0.42, 2.37)	0.61 (0.32, 1.15)	**0.44 (0.20, 0.92)**
1.00 (0.82, 1.22)	**1.36 (1.04, 1.79)**	0.91 (0.75, 1.10)	1.00 (0.78, 1.28)	0.80 (0.63, 1.03)	**0.56 (0.43, 0.72)**	**0.80 (0.67, 0.95)**	**0.85 (0.75, 0.97)**	1.36 (0.85, 2.18)	0.72 (0.50, 1.03)	**CT/BEV**	1.41 (0.92, 2.21)	0.88 (0.59, 1.31)	**0.63 (0.47, 0.82)**
1.02 (0.80, 1.31)	**1.39 (1.03, 1.89)**	0.93 (0.73, 1.17)	1.02 (0.77, 1.35)	0.82 (0.62, 1.09)	**0.57 (0.43, 0.76)**	0.82 (0.66, 1.02)	0.87 (0.73, 1.05)	1.39 (0.85, 2.27)	0.73 (0.50, 1.08)	1.02 (0.89, 1.17)	**CT/BEV/ Mpem**	0.62 (0.34, 1.11)	**0.44 (0.26, 0.73)**
1.01 (0.80, 1.26)	**1.37 (1.02, 1.83)**	0.91 (0.73, 1.13)	1.01 (0.77, 1.31)	0.81 (0.62, 1.06)	**0.56 (0.46, 0.69)**	0.80 (0.64, 1.01)	**0.86 (0.74, 1.00)**	1.37 (0.85, 2.22)	**0.72 (0.52, 1.00)**	1.01 (0.87, 1.16)	0.98 (0.81, 1.20)	**CT/Mpem**	0.71 (0.47, 1.06)
0.86 (0.72, 1.03)	1.17 (0.91, 1.51)	**0.78 (0.66, 0.92)**	0.86 (0.69, 1.08)	**0.69 (0.55, 0.87)**	**0.48 (0.37, 0.61)**	**0.69 (0.56, 0.84)**	**0.73 (0.64, 0.84)**	1.17 (0.74, 1.85)	**0.62 (0.43, 0.88)**	**0.86 (0.78, 0.95)**	**0.84 (0.71, 0.99)**	**0.85 (0.74, 0.99)**	**CT**

OS

non-squamous

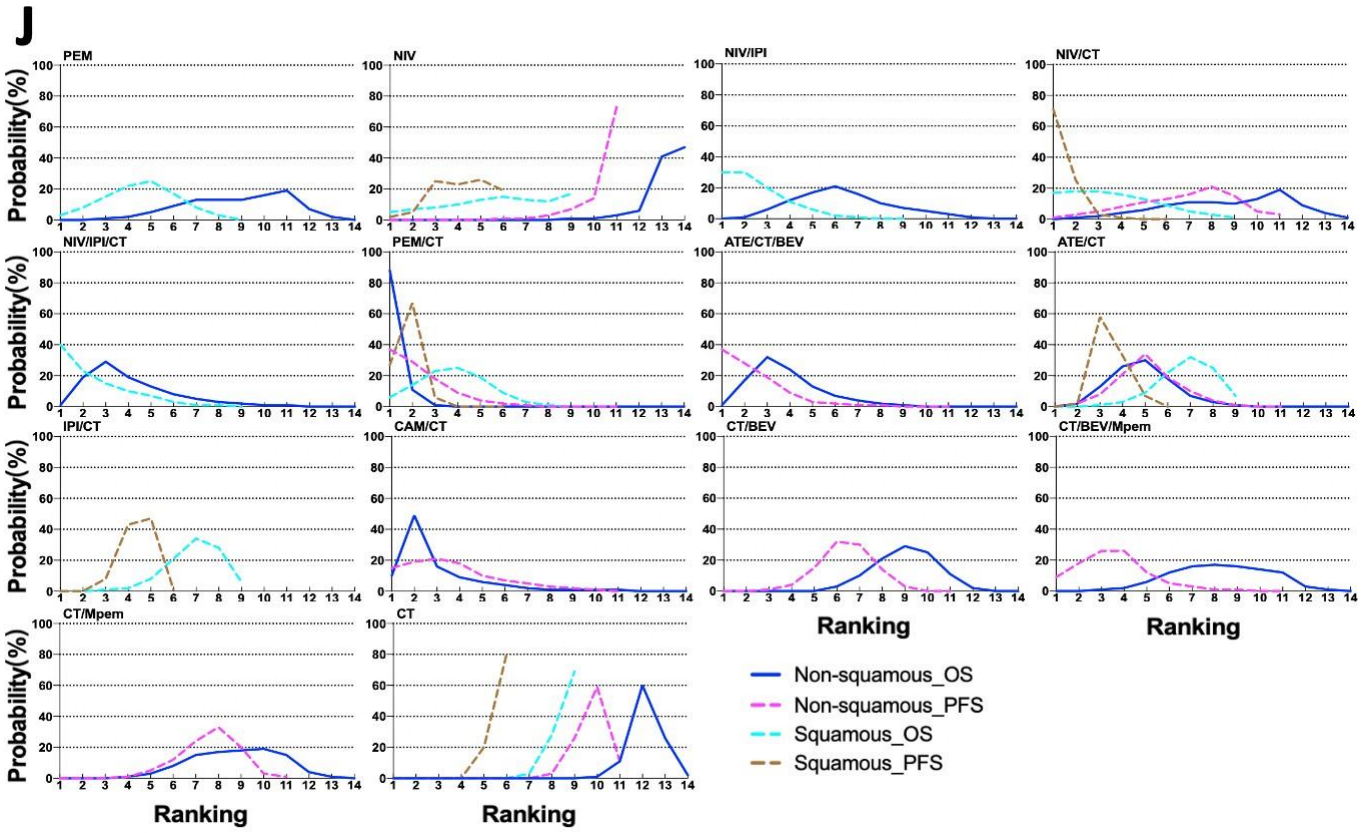


Figure S21. Network Meta-analysis of the Second Sensitive Analysis Excluding Highly Selected Patients

(A) Network diagrams for overall survival (OS), progression-free survival (PFS) (B), objective response rate (ORR) (C), and adverse events of grade 3 or higher (>= 3 AEs) (D). Each circular node represents a type of treatment. Each line represents a type of head-to-head comparison. The size of the nodes and the thickness of the lines are weighted according to the number of studies evaluating each treatment and direct comparison, respectively. The total number of patients receiving a treatment was shown in brackets. (E) Pooled hazard ratios (HR) (95% CrIs (credible intervals)) for OS and PFS. (F) Pooled odds ratio (OR) (95% CrIs) for >= 3 AEs and ORR. Data in each cell are HR or OR (95% CrIs) for the comparison of row-defining treatment versus column-defining treatment. HR less than 1 and OR more than 1 favour upper-row treatment. Significant results are highlighted in red and bold. (G) Ranking curves indicate the probability of each comparable treatment being ranked from first to last on OS, PFS, ORR, and >=3AEs. (H) Pooled HR (95% CrIs) for OS and PFS in squamous subgroup. (I) Pooled HR (95% CrIs) for OS and PFS in non-squamous subgroup. (J) Ranking curves indicate the probability of each comparable treatment being ranked from first to last on OS and PFS in squamous and non-squamous subgroups. PEM=pembrolizumab; ATE=atezolizumab; NIV=nivolumab; DUR=durvalumab; TRE=tremelimumab; IPI=ipilimumab; CAM=camrelizumab; BEV=bevacizumab; CT=chemotherapy; CT/Mpem=CT followed by maintenance of pemetrexed.